

Chapter 1

Introduction to Artificial Intelligence

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Overview

- 1 Introduction to Artificial Intelligence
- 2 Definition of Artificial Intelligence
- 3 Characteristics of Artificial Intelligence
- 4 Brief history of Artificial Intelligence
- 5 Application of Artificial Intelligence
- 6 Some Terminology of Artificial Intelligence

Definition of Artificial Intelligence

Meaning of the word: ``Intelligence''

- The ability to reason
- The ability to understand
- The ability to create
- The ability to learn from past experience
- The ability to plan and execute complex task

What Behaviors are Intelligent?

Everyday tasks:

- recognize a friend, recognize who is calling, translate from one language to another, interpret a photograph, talk, cook a dinner

Formal tasks:

- prove a logic theorem, geometry, calculus, play chess, checkers, or Go

Expert tasks:

- engineering design, medical designers, financial analysis

Artificial Intelligence

Based on the above definition of intelligence, 'artificial intelligence' is the branch of the science and engineering necessary to create artifacts that can

- **acquire knowledge**, i.e., can learn and extract knowledge; and
- **reason with knowledge** (leading to doing tasks such as planning, explaining, diagnosing, acting rationally, etc.),
- it is the branch of science concerned with making computers behave like humans
- AI is the science of making intelligent machines, especially intelligent computer programs

Artificial Intelligence

- According to **Barr and Feigenbaum**, “Artificial Intelligence is the part of computer science concerned with designing intelligent computer systems, that is, systems that exhibit the characteristics we associate with intelligence in human behavior.”
- According to Elaine Rich, “AI is the study of how to make computers do things at which, at the moment, people are better”.

Artificial Intelligence

An AI System should have:

- Capability to provide reason about something
- Capability of Natural Language Processing
- Capability of learning from past experience
- Capability of self correction

Approach/ View of AI falls under four categories

Thinking humanly

Thinking rationally

Acting humanly

Acting rationally

	Humanly	Rationally
Thinking	Thinking humanly – cognitive modeling. Systems should solve problems the same way humans do.	Thinking rationally – the use of logic. Need to worry about modeling uncertainty and dealing with complexity.
Acting	Acting humanly – the Turing Test approach.	Acting rationally – the study of rational agents: agents that maximize the expected value of their performance measure given what they currently know.

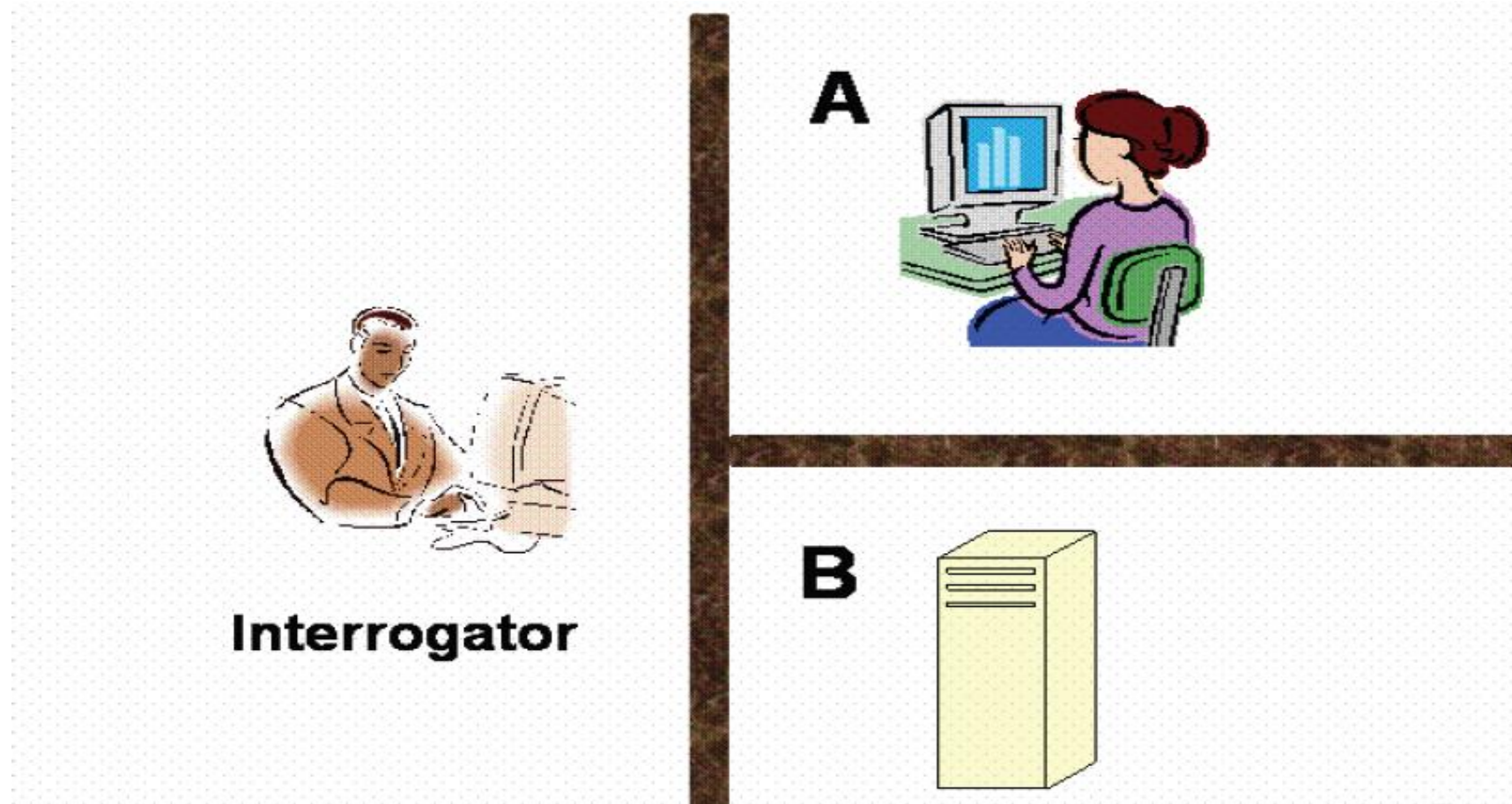
Thinking Humanly: The cognitive modelling approach

- If we are going to say that a given program thinks like a human, we must have some way of determining how human thinks.
- We need to get inside the actual working of human minds.
- Two way of doing this is:
 - Predicting and testing human behavior (cognitive science)
 - Identification of neurological data (cognitive neuro science)
- If the program's input/output and timing behavior matches human behavior, that is evidence that some of the program's mechanisms may also be operating in humans.
- Activities that we associate with human thinking are decision-making, problem solving, learning, etc.

Acting Humanly: The Turing Test Approach

- The Turing test is a method for determining whether or not a computer is capable of thinking like a human.
- The test is named after Alan Turing, an English mathematician who pioneered artificial intelligence during the 1940s and 1950s, and who is credited with devising the original version of test.
- According to this kind of test, a computer is deemed to have artificial intelligence if it can mimic human responses under specific conditions.

The Turing Test Approach



The Turing Test Approach

Consider the following scenario:

- There are two rooms, A and B.
- One of the rooms contain a computer. The other contains a human.
- The interrogator is outside and does not know which one is a computer.
- He can ask questions through a teletype and receives answers from both A and B.
- The interrogator needs to identify whether A or B are humans.
- To pass the Turing test, the machine has to fool the interrogator into believing that it is human.
- The unofficial “Total Turing Test” adds that the interrogation should also involve visual activity and physical interaction.

The Turing Test Approach

Turing test are continually evolving to maintain relevance during technological advancements.

- **The Reverse Turing Test** aims to have a human trick a computer into having the computer believe it is not interrogating a human.
- **The Total Turing Test** incorporates perceptual abilities and the person being question's ability to manipulate objects.
- **The Marcus Test** has test subjects view media and respond to questions about the content consumed.
- **The Minimum Intelligent Signal** test asks test subjects only binary questions (i.e. only true/false or yes/no answers are allowed).

The Turing Test Approach

To pass a Turing test, a computer must have following capabilities:

- Natural Language Processing:
 - Must be able to communicate in English successfully
- Knowledge representation:
 - To store what it knows and hears.
- Automated reasoning:
 - Answer the Questions based on the stored information.
- Machine learning:
 - Must be able to adapt in new circumstances

Thinking Rationally: The “laws of thought approach”

- The study of mental faculties through the use of computational models.
- The study of computations that make it possible to perceive, reason and act.
- The Greek philosopher Aristotle was one of the first to attempt to codify “right thinking”
- His famous syllogisms provided patterns for argument structures that always gave correct conclusions given correct premises.
- For example, “Socrates is a man; all men are mortal; therefore Socrates is mortal.” These laws of thought were supposed to govern the operation of the mind, and initiated the field of logic.

Acting Rationally: The rational agent approach

- Acting rationally means acting so as to achieve one's goals, given one's beliefs.
- An agent is just something that perceives and acts.
- In this approach, AI is viewed as the study and construction of rational agents.

AI and related fields/ Foundations of AI

Philosophy: Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality.

Mathematics: Formal representation and proof algorithms, computation, (un)decidability, (in)tractability, probability.

Economics: utility, decision theory, rational economic agents

Neuroscience: neurons as information processing units

Psychology: how do people behave, perceive, process Cognitive Science information, represent knowledge

Linguistics: knowledge representation, grammar

Brief History of AI

- The term “Artificial Intelligence” was used for the first time in 1956 by an American scientist John McCarthy who is referred to as the Father of AI.
- McCarthy also come up with a programming language called LISP (i.e. List-Processing), which is still used to program computer in AI that allow the computer to learn.
- Further, the major achievements can be listed as below:

Brief History of AI

1943	First electronic computer “Colossus” was developed.
1949	First commercial stored program computer was developed.
1950	- Alan Turing proposes the Turing test as a measure of machine intelligence. - Claude Shannon published a detail analysis of chess playing as search. - Isaac Asimov published his three laws of Robotics.
1951	The first working AI programs were written to run on the Ferranti Mark machine of the University of Manchester; a checkers-playing program written by Christopher Stavechey and a chess-playing program is written by Dietrich Prinz
1955	The first Dartmouth college summer AI conference is organized by John McCarthy, Marvin Minsky, Nathan Rochester of IBM and Claude Shannon.
1956	- The name artificial intelligence is used for the 1st time as the topic of the second Dartmouth Conference, organized by John McCarthy.

Brief History of AI

	- The first demonstration of the Logic Theorist (LT) written by Allen Newell, J.C. Shaw and Merbart Simon pus is called the first AI program
1957	The general problem Solver (GPS) demonstrated by Newell, Shaw and Simon
1958	John McCarthy at MIT invented the Lisp Programming Language.
1959	- John McCarthy and Marvin Minsky founded the MIT AI Lab. - First industrial robot company, animation was established.

Brief History of AI

1961	First Robot was introduced in GM Lab
1965	First Chat Bot ELIZA was developed
1972	Prolog Programming language was introduced
Mid 1980's	Neural Network was widely used
1997	First time AI system controlled a spacecraft named "Deep Space II"
1999	Sony company developed robot and MIT AI Lab developed emotional robot
2007	Checkers was solved by the team at University of Alberta
2009	Google developed self driving car
2011	IBM robot beats cheese player, SIRI and google now was introduced
2016	Google <u>Deepmind</u> beats <u>AlphaGo</u> Champion, first humanoid robot acquiring citizenship " <u>Shopia</u> " was introduced by Hanson Robotics



Brief History of AI

2016	The AI audio generator WaveNet is launched
2017	The birth of deepfakes
2017	The first 'Transformer' lays the technological foundation for large language models (LLMs)
2018	OpenAI releases GPT-1
2020	Generative AI goes mainstream (GPT-3.5)
2023	The release of Chat GPT-4

Application of AI

- Artificial intelligence has been used in a wide range of fields including medical diagnosis, stock trading, robot control, law, remote sensing, scientific discovery and toys.
- Many thousands of AI applications are deeply embedded in the infrastructure of every industry.
- In the late 90s and early 21st century, AI technology became widely used as elements of larger systems, but the field is rarely credited for these successes.

Application of AI

1. Game Playing

- Machines can play master level chess. There is some AI in them, but they play well against people mainly through brute force method, looking at hundreds of thousands of positions.
- Few of the major achievements in game playing are:
 - In 2016, Google's Deepmind AI named "ALphaGo Zero" defeated Lee Sedol by beating him in 4 games out of 5.
 - Lee Sedol was South Korea's champion in Go.
 - Lee Sedol eventually retired from the game when in 2017 this AI next version was released.
 - In 2017, the next version of AI named "Alpha Go" beat World Champion Ke Jie in 3 matches out of 3.
 - In latest version, this AI beats the human opponents at the ratio of 100 out of 0.
 - Similarly, in 2017 AI Poker Bot beats World's Best Player.

2. Speech Recognition

- It is possible to instruct some computers using speech. In 1990s, computer speech recognition reached a practical level for limited purposes
- Voice recognition system such as SIRI and CORTANA.

Application of AI

3. Understanding Natural Language

- To perform many natural language processing tasks such as machine translation, summarization, information extraction, word sense disambiguation need the AI in machine.

4. Computer Vision

- Computer vision is concerned with the theory behind artificial system that extract information from images. The image data can take many forms such as videos sequences views from multiple cameras and data from a medical scanner.

Application of AI

5. Expert System

- Expert system needs the AI to perform its task.
- One of the first expert system was MYCIN in 1974 which diagnosis bacterial infections of the blood and suggests treatments.
- IBM's "Watson" which is an question answer system.

6. Finance

- Financial institutions have long used artificial neural network systems to detect charges or claims outside of the norm, flagging these for human investigation.
- Use of AI in banking can be traced back to 1987 when Security Pacific National Bank in USA set-up a Fraud Prevention Task force to counter the unauthorized use of debit cards.

Application of AI

7. Hospitals and medicine

- Artificial neural networks are used as clinical decision support systems for medical diagnosis.
- Other tasks in medicine that can potentially be performed by artificial intelligence include:
 - Computer-aided interpretation of medical images. Such systems help scan digital images, *e.g.* from computed tomography, for typical appearances and to highlight conspicuous sections, such as possible diseases. A typical application is the detection of a tumor.
 - Heart sound analysis
 - Companion robots for the care of the elderly.

Few achievements in this fields are:

- Path AI and Philips helps in diagnosis of breast cancer with accuracy of 99%.
- Google AI predicts inpatient death risks with 95% accuracy.

Application of AI

8. Music

- The evolution of music has always been affected by technology.
- With AI, scientists are trying to make the computer emulate the activities of the skillful musician.
- Composition, performance, music theory, sound processing are some of the major areas on which research in Music and Artificial Intelligence are focusing.

9. Aviation

- The Air Operations Division (AOD) uses AI for the rule based expert systems.
- The AOD has use for artificial intelligence for replacement operators for fighting and training simulators, mission management aids, support systems for tactical decision making, and post processing of the simulator data into symbolic summaries.

Application of AI

10. Identification Technologies

- *ID cards*
 - e.g., ATM cards
 - can be a nuisance and security risk:
 - cards can be lost, stolen, passwords forgotten, etc
- *Biometric Identification*
 - walk up to a locked door
 - camera
 - fingerprint device
 - microphone
 - computer uses your biometric signature for identification face, eyes, fingerprints, voice pattern

Ethical issues in Artificial Intelligence

- Unemployment
 - What happens after the end of jobs?
- Humanity
 - How do machines affect our behavior and interaction?
- Security
 - How do we keep AI safe from adversaries?
- Singularity
 - How do we stay in control of a complex intelligence?

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- E. Rich and Knight, *Artificial Intelligence*, McGraw Hill, 1991.
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Chapter 2

Intelligent Agent

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Overview

1. Introduction to Intelligent Agent
2. Properties of Agent
3. Sensor/Actuator/Effectors and Actions
4. Types of Agents
5. Properties of Task Environment

Intelligent Agents

Agent

- An AI system is composed of an agent and its environment. The agents act in their environment. The environment may contain other agents.
- An agent is anything that can be viewed as:
 - perceiving its environment through sensors and
 - acting upon that environment through actuators

Agents

Human agent:

- eyes, ears, and other organs for sensors;
- hands, legs, mouth, and other body parts for actuators

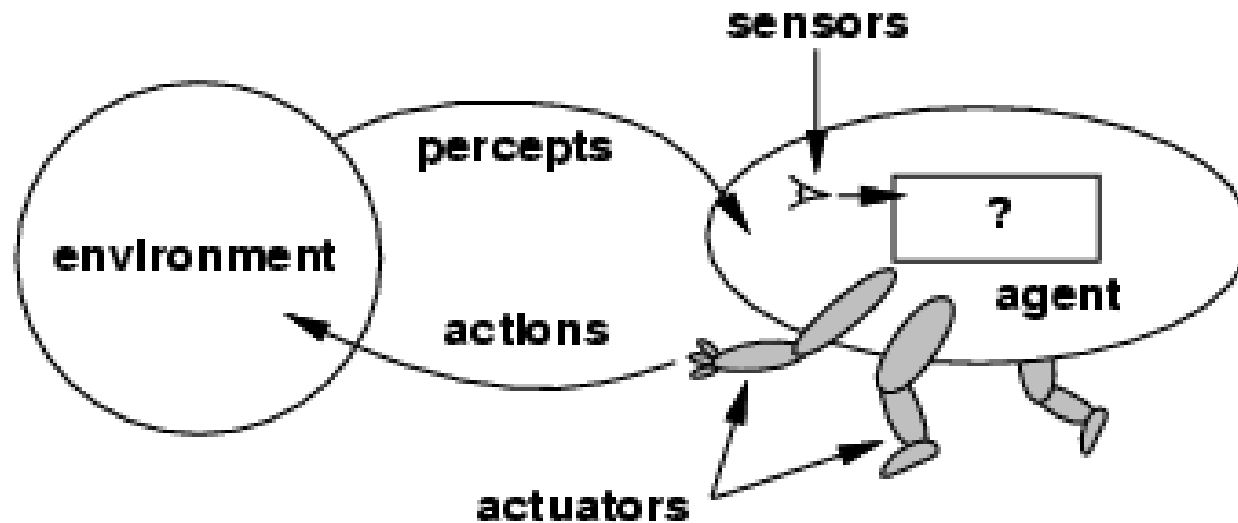
Robotic agent:

- cameras and infrared range finders for sensors;
- Various motors for actuators

Software agent:

- Keyboard, mouse, microphone for sensors;
- Monitor, printer for actuators

Agents and Environments



Performance Measure: determines how successful the agent is.

Behavior: activities the agent performs to achieve the goal

Percepts: formation of concepts based on the inputs.

Percept Sequence: set of all the perceptions till date

Agent Function: mapping of perception

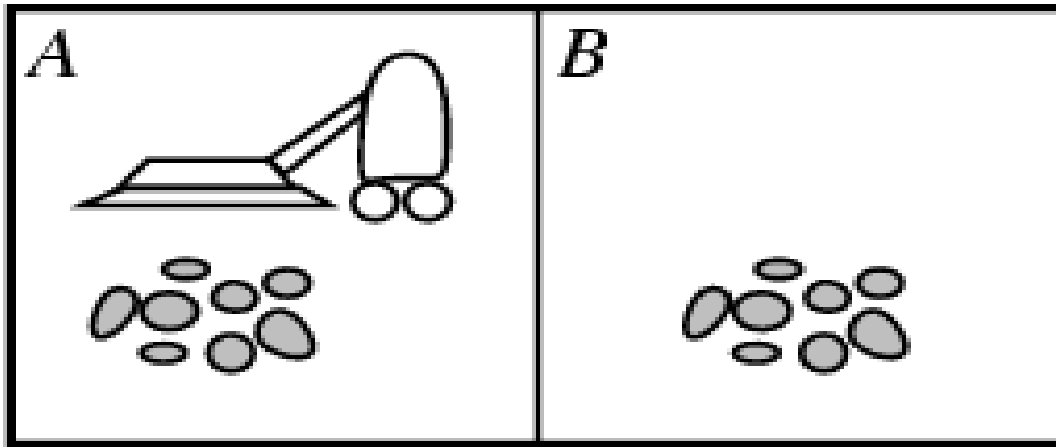
Agents and environments

- The agent function maps from percept histories to actions:

$$[f: P^* \rightarrow A]$$

- The agent program runs on the physical architecture to produce f ,
agent = architecture + program
- Architecture = the machinery that an agent executes on.
- Agent Program = an implementation of an agent function

Vacuum Cleaner World



Percepts: location and state of the environment, e.g., [A,Dirty], [B,Clean]

Actions: *Left, Right, Suck, NoOp*

Rational agents

Rational Agent:

- For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, based on the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Performance measure:

- An objective criterion for success of an agent's behavior. E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

Rational agents

- Rationality is distinct from omniscience (all-knowing with infinite knowledge)
- Agents can perform actions in order to modify future percepts so as to obtain useful information (information gathering, exploration)
- An agent is autonomous if its behavior is determined by its own percepts & experience (with ability to learn and adapt) without depending solely on build-in knowledge

Task Environment

PEAS

- PEAS define task environments about formulating the performance of intelligent agents. It stands for:
 - Performance measure
 - Environment
 - Actuators
 - Sensors

Task Environment

Q. Point out the task of designing an automated taxi driver according to PEAS description.

A. Agent= taxi-driver

- Performance measure: Safe, fast, legal, comfortable trip, maximize profits
- Environment: Roads, other traffic, pedestrians, customers
- Actuators: Steering wheel, accelerator, brake, signal, horn
- Sensors: Cameras, sonar, speedometer, GPS, odometer, engine sensors

Task Environment

Q. Point out the task of designing a Medical diagnosis system according to PEAS description.

A. Agent= Medical diagnosis system

- Performance measure: reliable, minimize costs, lawsuits
- Environment: Patient, hospital, staff
- Actuators: Screen display (questions, tests, diagnoses, treatments, referrals)
- Sensors: Keyboard (entry of symptoms, findings, patient's answers)

Task Environment

Q. Point out the task of designing a Part picking Robot according to PEAS description.

- A. Agent = Part-picking robot
- Performance measure: Percentage of parts in correct bins
- Environment: Conveyor belt with parts, bins
- Actuators: Jointed arm and hand
- Sensors: Camera, joint angle sensor

Environment types

- Fully observable (vs. partially observable)
- Deterministic (vs. stochastic)
- Static (vs. dynamic)
- Discrete (vs. continuous)
- Single agent (vs. multiagent)

Fully Observable vs Partially Observable

- Is everything an agent requires to choose its actions available to it via its sensors? Perfect or Full information.
 - If so, the environment is fully accessible. If not, parts of the environment are inaccessible.
 - Agent must make informed guesses about world.
 - In decision theory: perfect information vs. imperfect information.
 - Fully Observable: Part picking robot
 - Partially Observable: Taxi driver

Deterministic vs Stochastic

- Deterministic environments:
 - Depend only on current state and agent's action?
- Non-deterministic environments:
 - Have aspects beyond the control of the agent ▪
- Utility functions have to guess at changes in world
- Deterministic: Cross word
- Non-deterministic: Taxi driver

Static vs Dynamic

- Static environments don't change:
 - While the agent is deliberating over what to do .
- Dynamic environments do change:
 - So, agent should/could consult the world when choosing actions.
 - Alternatively: anticipate the change during deliberation OR make decision very fast

Static: Cross word

Dynamic: Taxi driver

Discrete vs Continuous

- A limited number of distinct, clearly defined percepts and actions vs. a different range of values (continuous)
- Discrete: Cross word
- Continuous: Taxi driver

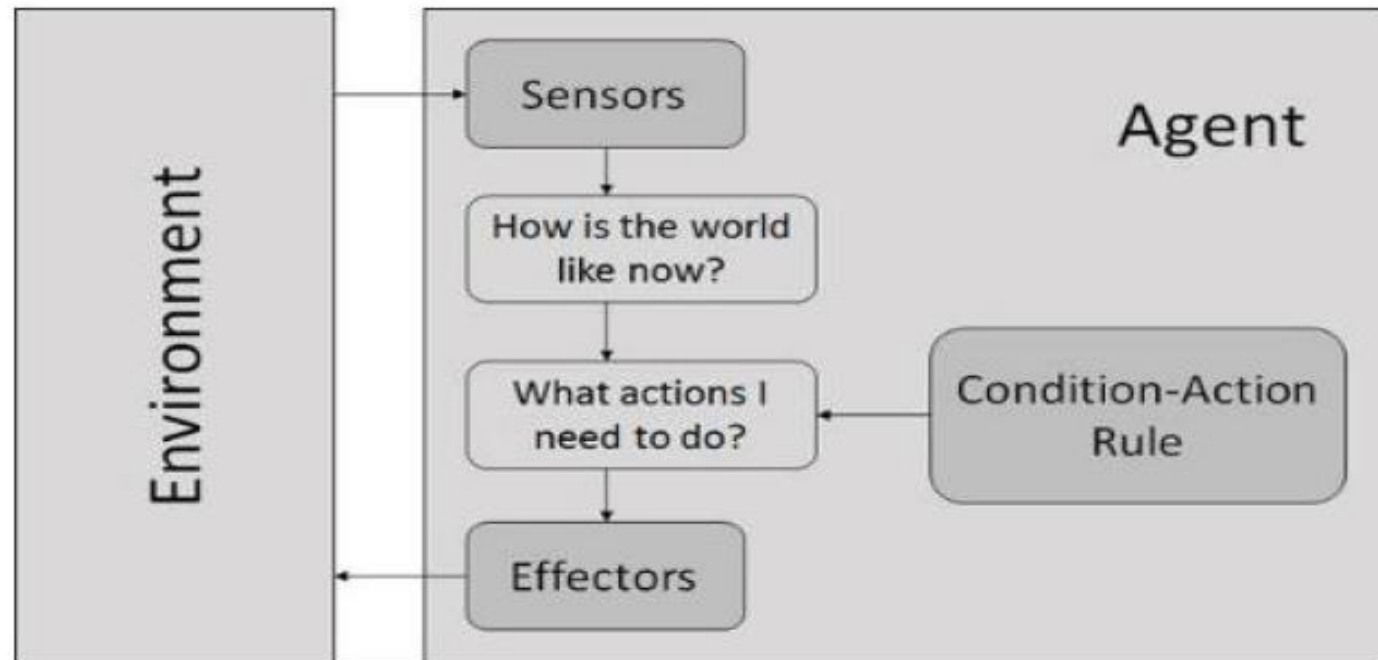
Single agent vs Multi agent

- An agent operating by itself in an environment or there are many agents working together.
- Single agent: Cross word
- Multi agent: Taxi driver

Types of Agents

Simple Reflex Agent

- They choose actions only based on the current percept or situation.
- This is useful when a quick automated system is required. Human have very similar reaction to fire.

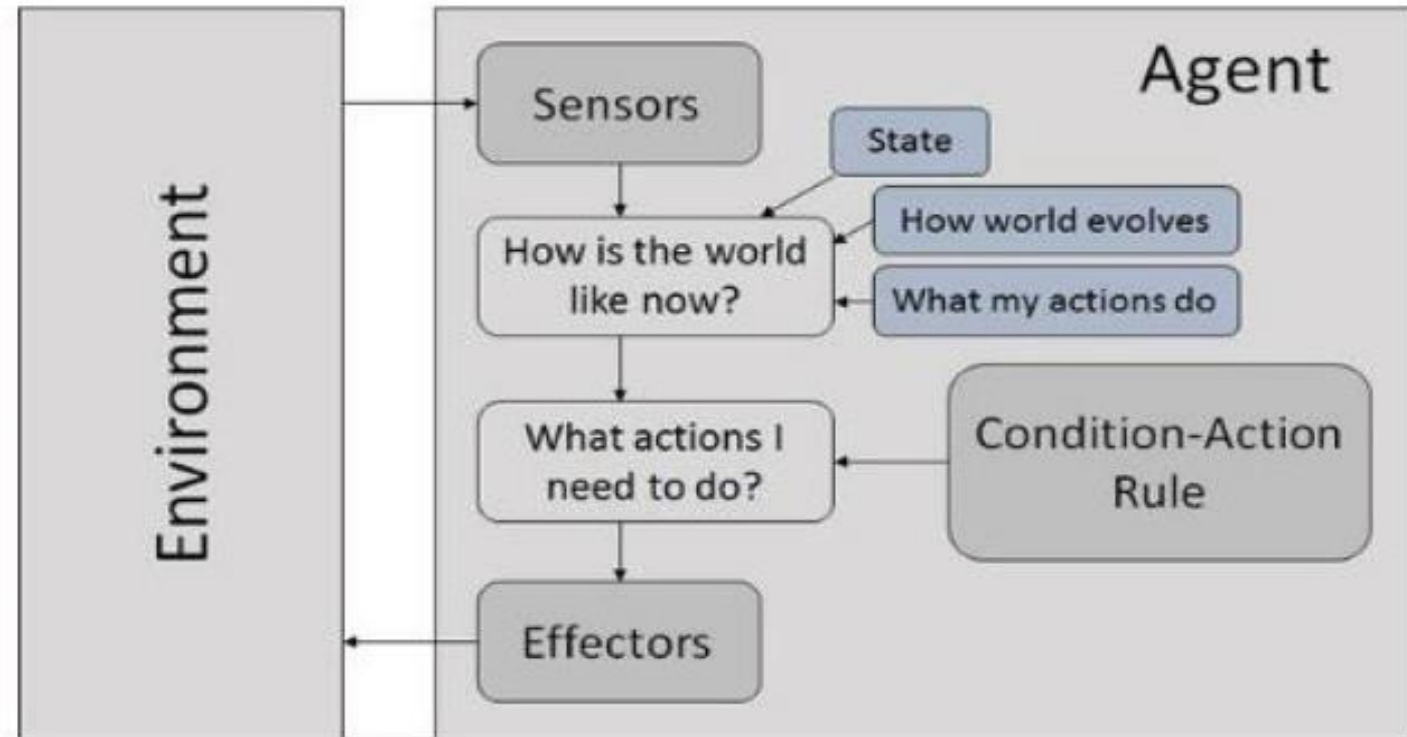


(Condition- Action Rule/IF-THEN Rule IF hands on fire, THEN pull away)

Types of Agent

Model Based Agent

- They choose action only based on their model and maintain their internal state where model means the information about perception function and Internal States are

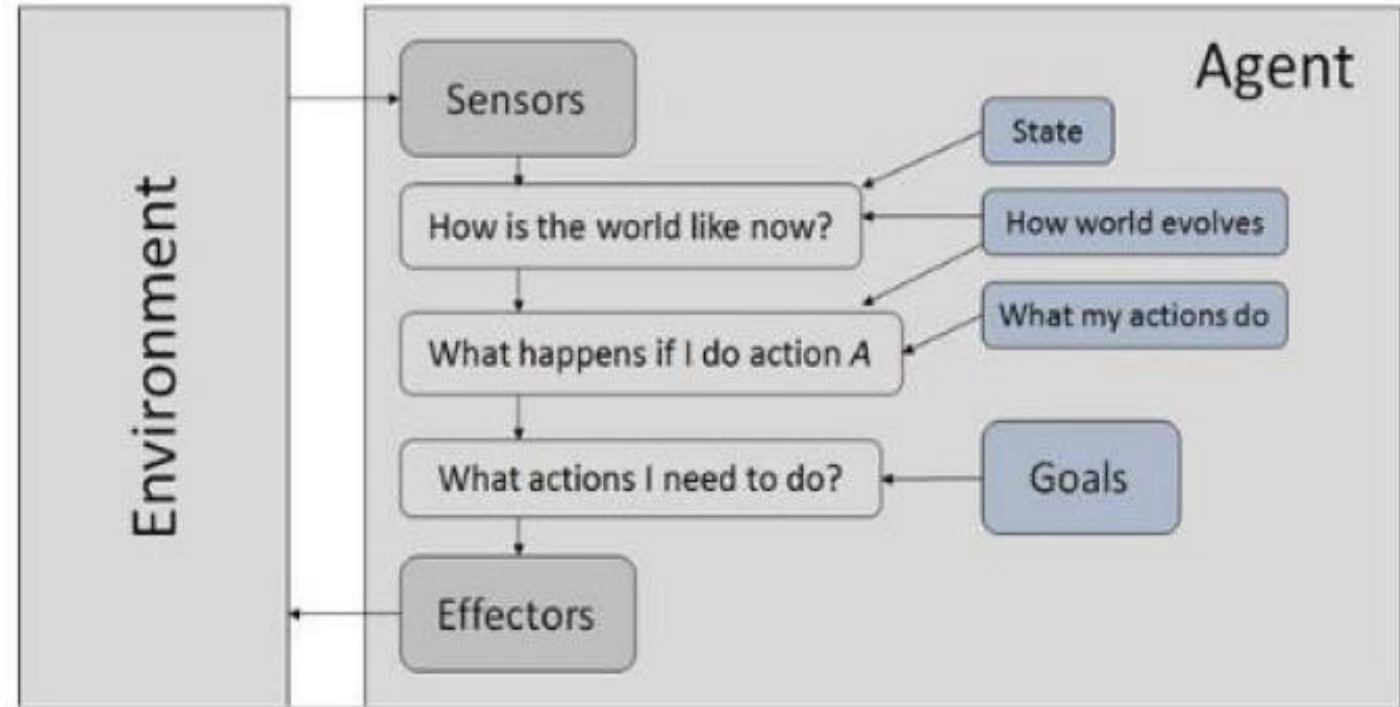


the representation of unobserved aspects of current state depending on percept history.

Types of Agent

Goal Based Agent

- They choose their actions in order to achieve goals. Goal-based approach is more flexible than reflex agent since the knowledge/information

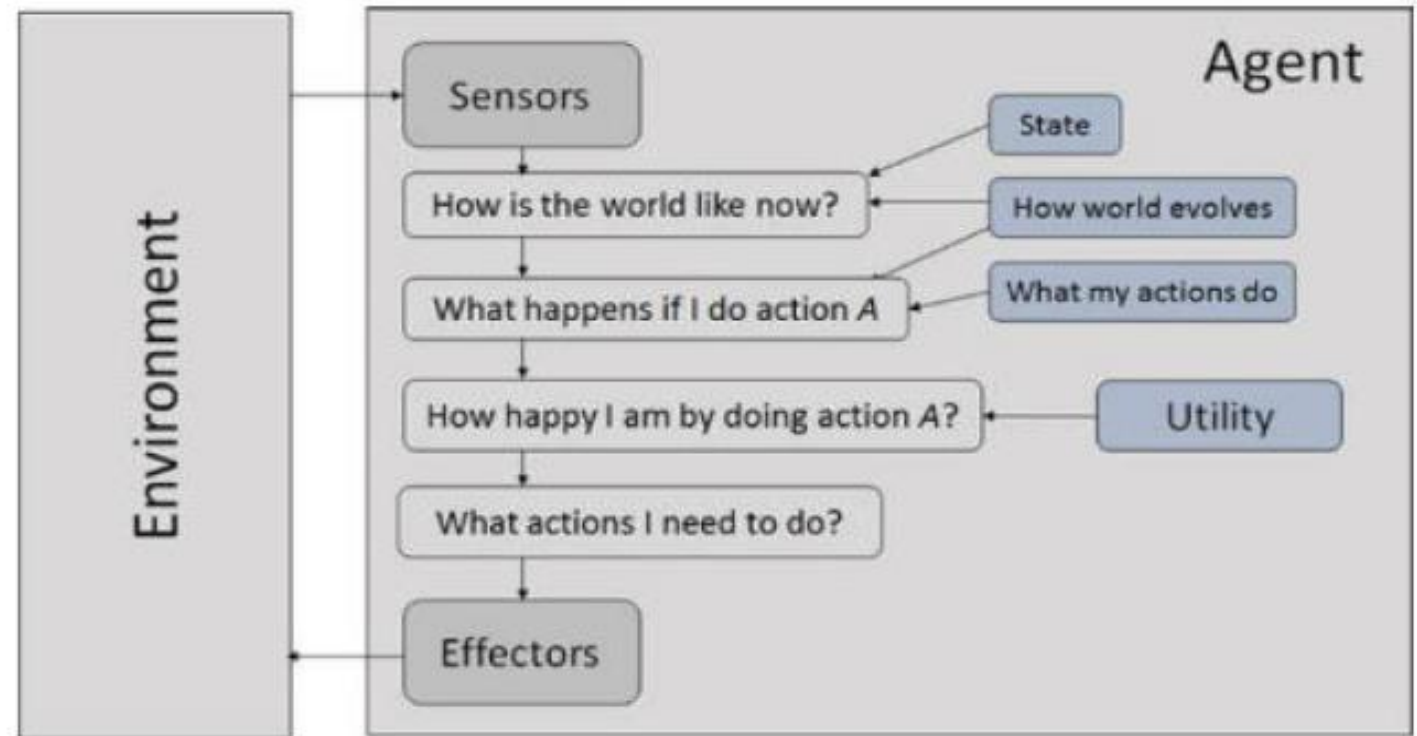


is modeled in such a way that they are easy for modifications.

Types of Agent

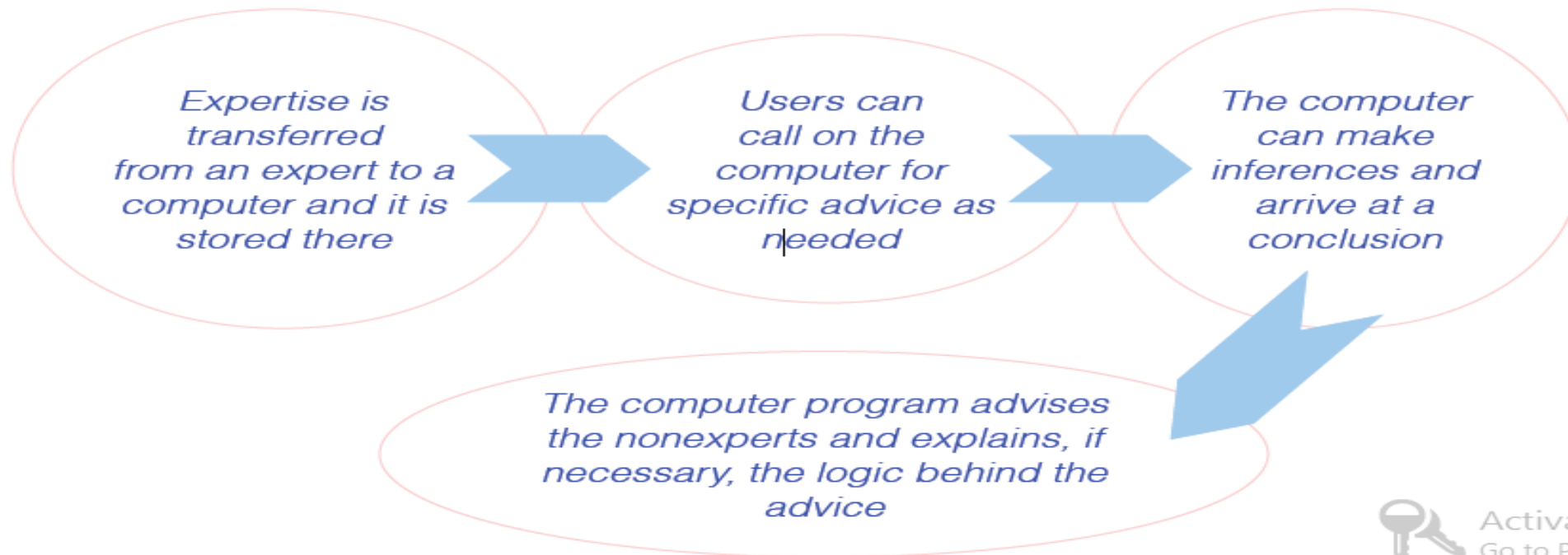
Utility Based Agent

- They choose actions based on a utility for each state.
- Utility function maps each state after each action to achieve the goal. This is useful when we either have many actions for same goal or we have many goals for same actions.



Intelligent Agent Example: Expert System

- The expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extraordinary human intelligence and expertise.



Characteristics of Expert Systems

- High performance
- Reliable
- Highly responsive

When to Use an Expert System

- Provide a high potential payoff or significantly reduced downside risk
- Capture and preserve irreplaceable human expertise.
- Provide expertise needed at a number of locations at the same time or in a hostile environment that is dangerous to human health.
- Provide expertise that is expensive or rare.
- Develop a solution faster than human experts can
- Provide expertise needed for training and development to share the wisdom of human experts with a large number of people.

Examples of Expert System

- MYCIN would attempt to diagnose patients based on reported symptoms and medical test results.
- The program could request further information concerning the patient, as well as suggest additional laboratory tests, to arrive at a probable diagnosis, after which it would recommend a course of treatment.
- If requested, MYCIN would explain the reasoning that led to its diagnosis and recommendation.

Examples of Expert System

- CALEX is a blackboard based integrated system for agricultural management, developed at University of California.
- CALEX can be used by growers, pest control advisors, consultants and other managers.
- DENDRAL's primary aim was to help organic chemists in identifying unknown organic molecules, by analyzing their mass spectra and using knowledge of chemistry.

Components of Expert Systems

- The components of Expert System include :
 - Knowledge Base
 - Inference Engine
 - User Interface

Components of Expert Systems

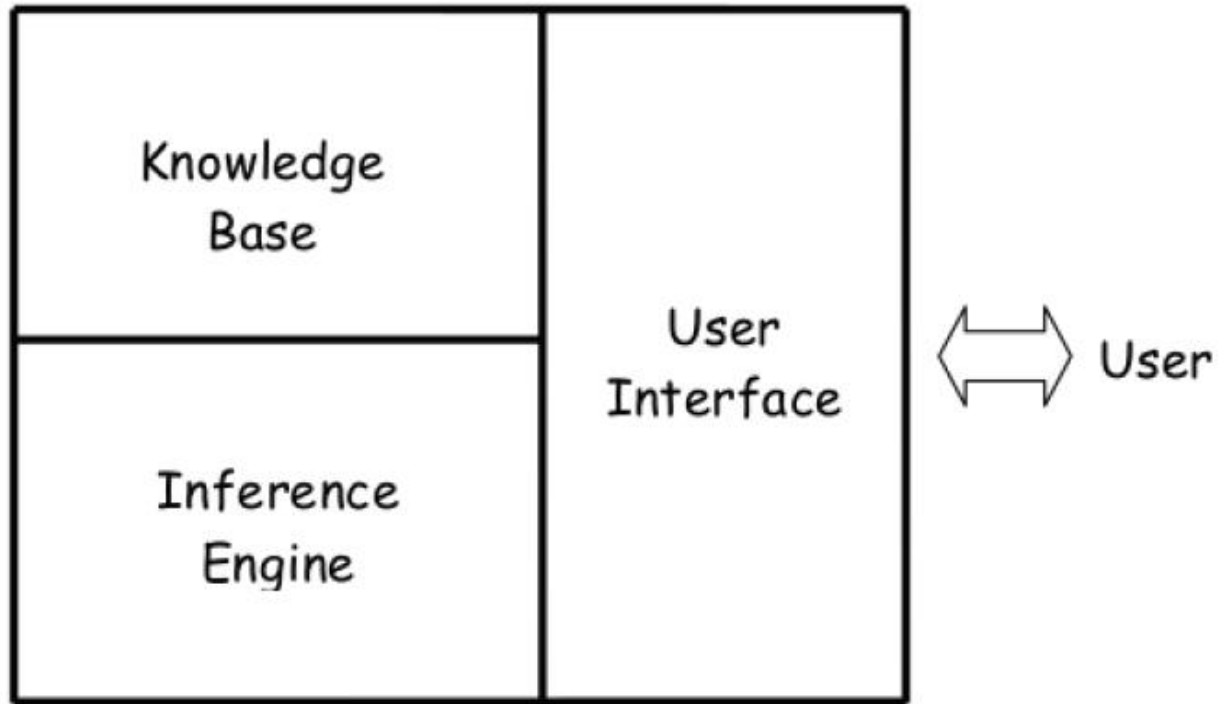


Fig: Block Diagram of expert system

Knowledge Base

- Knowledge is required to exhibit intelligence.
- The success of any ES depends upon the collection of highly accurate and precise knowledge.

Components of Knowledge Base

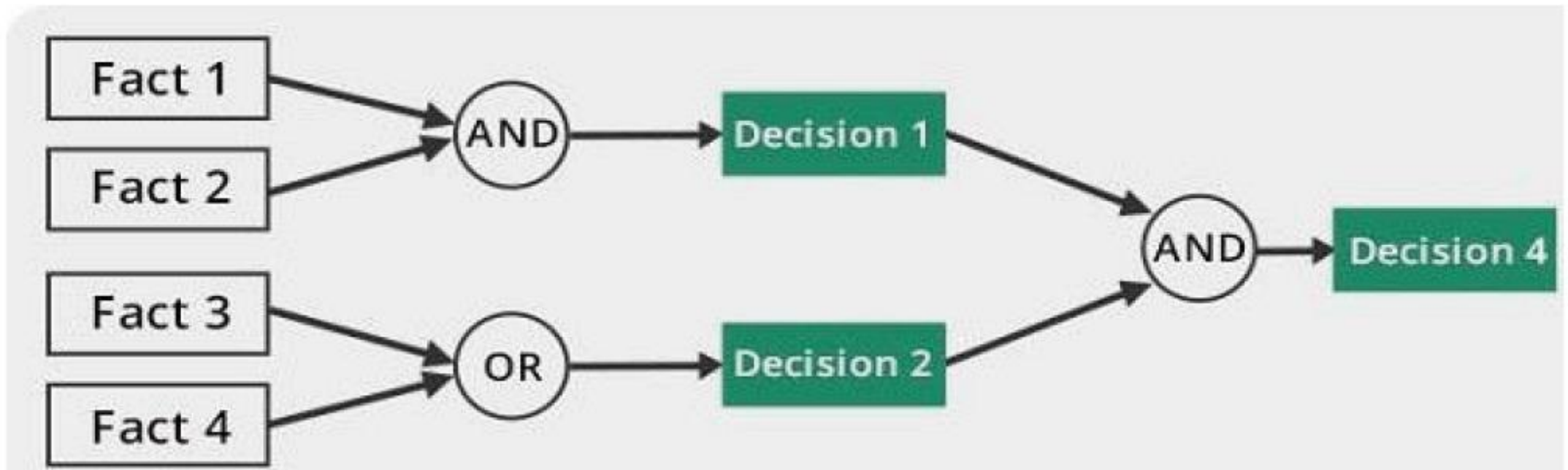
- The knowledge base of an ES is a store of both, factual and heuristic knowledge.
 - Factual Knowledge – It is the information widely accepted by the Knowledge Engineers and scholars in the task domain.
 - Heuristic Knowledge – It is about practice, accurate judgment, one's ability of evaluation, and guessing.

Inference Engine

- In case of knowledge-based Expert System, the Inference Engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.
- In case of rule based Expert System, it –
 - Applies rules repeatedly to the facts, which are obtained from earlier rule application.
 - Adds new knowledge into the knowledge base if required.
 - To recommend a solution, the Inference Engine uses the following strategies:
 - Forward Chaining
 - Backward Chaining

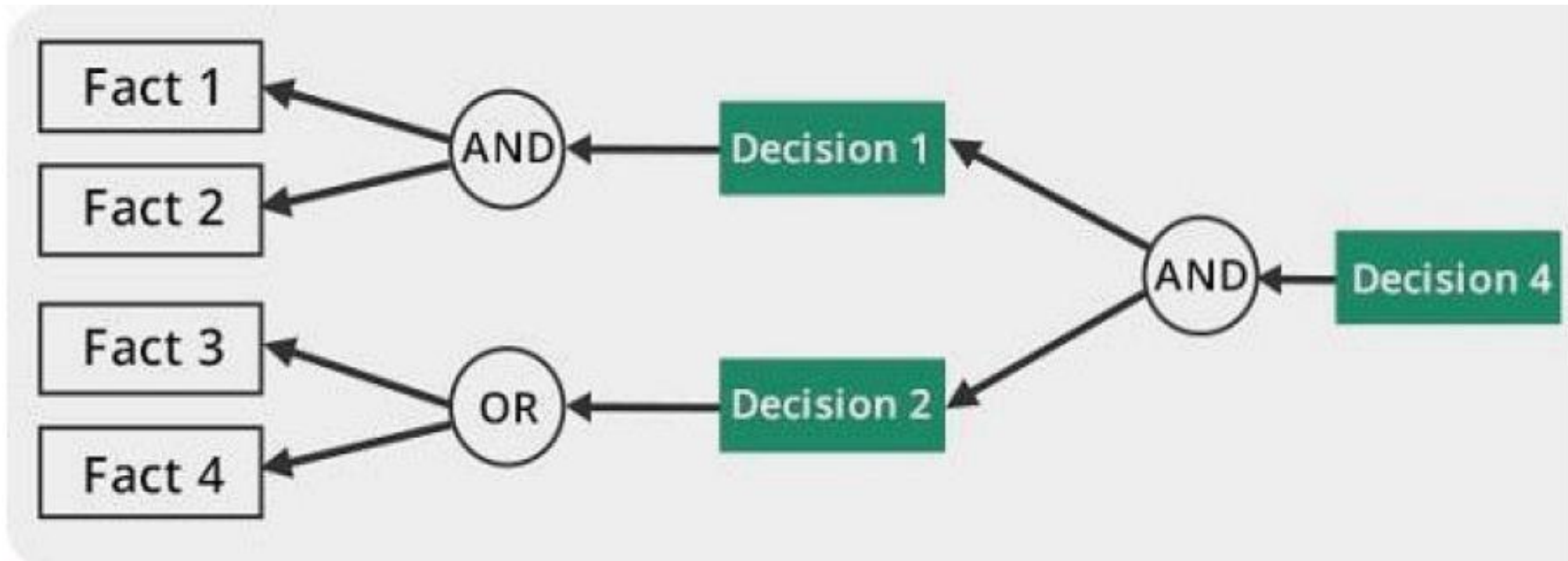
Forward Chaining

- It is a strategy of an expert system to answer the question, “What can happen next?”



Backward Chaining

- With this strategy, an expert system finds out the answer to the question, “Why this happened?”.



User Interface

- User interface provides interaction between user of the ES and the ES itself.
- It is generally Natural Language Processing so as to be used by the user who is well-versed in the task domain.
- The user of the ES need not be necessarily an expert in Artificial Intelligence.

Advantages of Expert System

- These systems are highly reproducible.
- They can be used for risky places where the human presence is not safe.
- Error possibilities are less if the KB contains correct knowledge.
- The performance of these systems remains steady as it is not affected by emotions, tension, or fatigue.
- They provide a very high speed to respond to a particular query

Disadvantages of Expert System

- The response of the expert system may get wrong if the knowledge base contains the wrong information.
- Like a human being, it cannot produce a creative output for different scenarios.
- Its maintenance and development costs are very high.
- Knowledge acquisition for designing is much difficult.
- For each domain, we require a specific ES, which is one of the big limitations.

Knowledge

- “The fact or condition of *knowing something* with familiarity gained through experience or association.” (Webster’s Dictionary, 1988)(Knowing something via seeing, hearing, touching, feeling, and tasting)
- “The fact or condition of *being aware of something*” . (Ex. Sun is hot, balls are round, sky is blue,...)

Knowledge Storing

- Natural language for people:
 - Symbols for computer: a number or character string that represents an object or idea (Internal representation of the knowledge).
 - The core concepts: mapping from facts to an internal computer representation and also to a form that people can understand.

Knowledge Representation

- Simple facts or complex relationships
- Mathematical formulas
- Associations between related concepts
- Inheritance hierarchies between classes of objects

Knowledge

- Note: *knowledge* $\neq$ *data*.

Example: determining voltage from current and resistance

Data	Knowledge
V=12, I=6, R=2 V=28, I=4, R=7 V=9, I=3, R=3 V=12, I=3, R=4 V=15, I=3, R=5	$V = I * R$

- Knowledge is *more compact* and *faster to manipulate*.
- Knowledge is *more general* (that is, may be applied to situations we have *not* been programmed for).
- Important feature of intelligence: *creating knowledge* from data.

References

- E. Rich and Knight, *Artificial Intelligence*, McGraw Hill, 1991.
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Chapter 3

Problem Solving

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Overview

1. Introduction to Problems
2. Problem Solving by Search
3. Example Problems
4. Search Algorithms

Problem

- A problem is a situation (difficult/easy) experienced by an agent.
- Problem is solved by a sequence of actions that reduce the difference between the initial situation and the goal.
- Problem solving methods are divided into 2 categories:
 1. Special Purpose Method.
 - ✓ A special-purpose method is tailor-made for a particular problem and often exploits very specific features of the situation in which the problem is embedded.
 2. General Purpose Method
 - ✓ A general-purpose method is applicable to a wide variety of problems.
- A problem is characterized by a set of goals, a set of objects and a set of operations.

Problem Solving

Four general steps in problem solving:

- Goal formulation
 - What are the successful world states
- Problem formulation
 - What actions and states to consider given the goal
- Search
 - Determine the possible sequence of actions that lead to the states of known values and then choosing the best sequence.
- Execute
 - Give the solution perform the actions.

Defining Problem as a State Space Search

A set of initial state, actions and the goal state for a given problem is known as state space for that problem. A state space represents a problem in terms of states and operators that changes the states. A state space essentially consists of a set of nodes representing each state of the problem, arcs between nodes representing the legal moves from one state to another, an initial state and a goal state

The major components of state space representation are:

- Initial state
- Goal state, and
- Operator or legal moves.

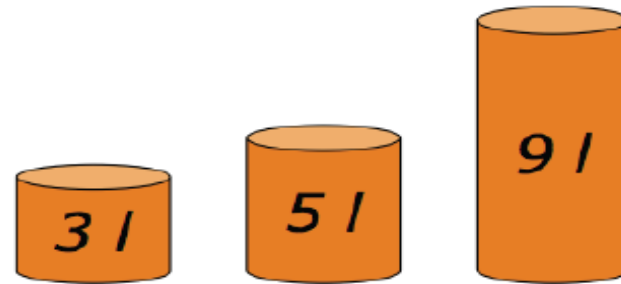
Defining Problem as a State Space Search

- The major components of state space representation:
 1. Initial State :- is the start state
 2. Goal State:- is the conditions that has to be fulfilled.
 3. Operators:-are used to change the state. Operators do actions that can transform one state into another.

Q. Explain problem as a state space with an example.

Example 1: Measuring Problem

Example: Measuring Problem



Problem: Using these three buckets,
measure 7 liters of water.

Example 1: Measuring Problem

Example: Measuring Problem

Measure 7 liters of water using a 3-liter, a 5-liter, and a 9-liter buckets.

- **Formulate goal:** Have 7 liters of water in 9-liter bucket
- **Formulate problem:**
 - States: amount of water in the buckets
 - Operators: Fill bucket from source, empty bucket
- **Find solution:** sequence of operators that bring you from current state to the goal state

Example 1: Measuring Problem

Example: Measuring Problem

- Solution 1:**

a	b	c	
0	0	0	start
3	0	0	
0	0	3	
3	0	3	
0	0	6	
3	0	6	
0	3	6	
3	3	6	
1	5	6	
0	5	7	goal

- Solution 2:**

a	b	c	
0	0	0	start
0	5	0	
3	2	0	
3	0	2	
3	5	2	
3	0	7	goal

Example 2: The 8-puzzle

- The puzzle consists of an 8-square frames and an empty slot. The tiles are numbered from 1-8. It is possible to move the tiles in the square field by moving tile into the empty slot. The objective is to get square in a numeric order.

1		2
4	5	3
7	8	6

Example 2: The 8-puzzle

- **Initial State**

1		2
4	5	3
7	8	6

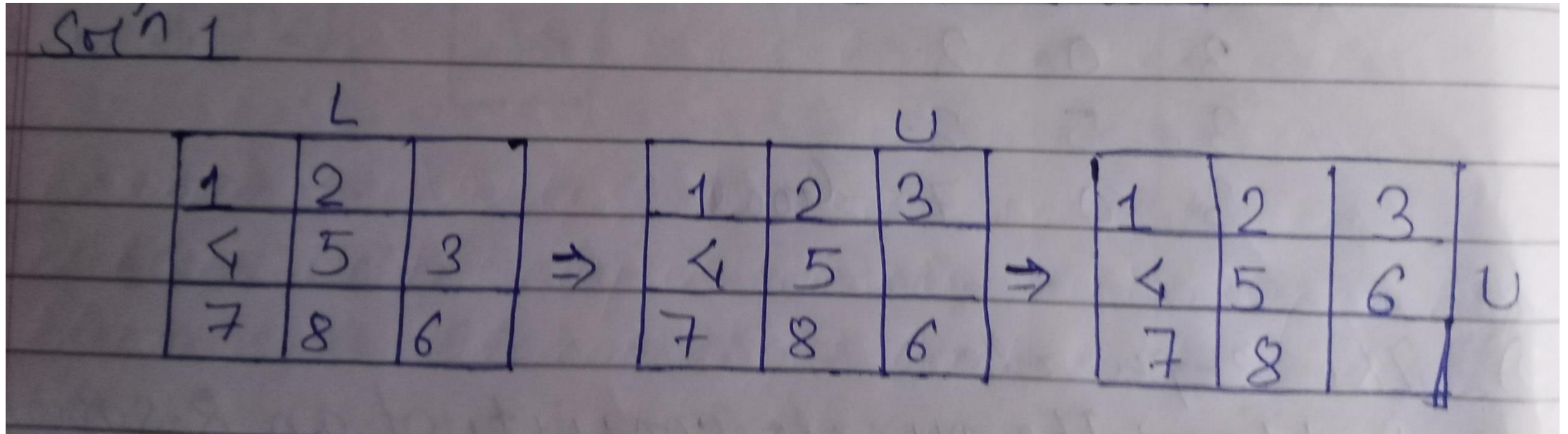
- **Operators**

- UP: If the empty slot is not touching the up-frame, move it up.
- DOWN: If the empty slot is not touching down-frame, move it down.
- LEFT: If the empty slot is not touching left-frame, move it left.
- RIGHT: If the empty slot is not touching right-frame, move it right.

- **Final State**

1	2	3
4	5	6
7	8	

Example 2: The 8-puzzle



Problem Formulation

Problem formulation:

A problem is defined by:

- An initial state: State from which agent start
- Successor function: Description of possible actions available to the agent.
- Goal test: Determine whether the given state is goal state or not
- Path cost: Sum of cost of each path from initial state to the given state.

A solution is a sequence of actions from initial to goal state. Optimal solution has the lowest path cost.

Problem Types

1. Single state problem = deterministic, accessible

- Agent knows everything about the world (the exact state)
- Can calculate optimal action sequence to reach the goal
- E.g. playing chess, any action will result in an exact state.

2. Multiple state problem = deterministic, inaccessible

- Agent doesn't know the exact state (could be in any of the possible states)
- Assume states while working towards the goal state.
- E.g. walking in a dark room

Problem Types

3. Contingency Problem = non-deterministic, inaccessible

- Must use sensors during execution
- Solution is a tree or policy
- Often interleave search and execution
- E.g. a new skater in an arena i.e. sliding problem and many skaters around

4. Exploration Problem = unknown state space

- Discover and learn about environment while taking actions.
- E.g. Maze

Constraint Satisfaction Problem

- Constraint satisfaction problems (CSPs) are mathematical questions defined as a set of objects whose state must satisfy a number of constraints or limitations.
- The process of finding a solution to set of variables V_i ($V_1, V_2, \dots, V_n$) and a set of constraint C_i ($C_1, C_2, \dots, C_m$)
- There is no any specified rule to define the procedure to solve the CSP.
- E.g. : **river crossing problem, 8-queens problem, Crypto Arithmetic Problems**

Example 1: River Crossing Problem

Problem: In a distinct land, bigamy is common. There are six people who want to cross a river in this land. This group consists of two men, each two wives. No man can tolerate any of his wives being in the company of another man unless at least he or his next wife is present in the boat or in next land. There is a boat that holds two people to be used for crossing the river. How is the trip possible?

Example 1: River Crossing Problem

- **Initial State**

$W \{ H1, W1, W1' \text{ and } H2, W2, W2' \}, E \{ \phi \}$

- **Operation**

- No man can tolerate any of his wives being in the company of another man.
- Crossing the river

- **Final State**

$W \{ \phi \}, E \{ H1, W1, W1' \text{ and } H2, W2, W2' \}$

Example 1: River Crossing Problem

- Solution**

S. N	West (W)	River		East (E)
0.	H1,W1,W1', H2,W2,W2'			(ϕ)
1.	H1,H2,W2,W2'	W1, W1'		W1,W1'
2.	H1,W1,H2,W2,W2'	W1,		W1'
3.	H1,W1,H2,	W2, W2'		W1',W2,W2'
4.	H1,W1,H2,W2	W2		W1',W2'
5.	H1, H2,	W1, W2		W1,W1',W2,W2'
6.	H1, H2, W2	W2		W1,W1', W2'
7.	H1,	H2, W2		W1,W1', H2,W2, W2'
8.	H1,H2,	H2,		W2
9.	(ϕ)	H1, H2,		H1, W1,W1', H2,W2, W2'

Example 2: Crypto Arithmetic Problems

- Crypto-Arithmetic Problems are substitution problems where digits representing a mathematical operation are replaced by unique digits.
- E. g:

	A	B	C
+	D	E	F
	G	H	I



	0	2	4
+	5	8	9
	6	1	3

Example 2: Crypto Arithmetic Problems

Solution:

1. $A \neq B \neq C \neq D \neq E \neq F \neq G \neq H \neq I$
2. $C + F = I$
 $C + F = 10 + I$ (I as carry)
3. $B + E = H$
 $B + E = 10 + H$
 $B + E + 1 = H$
 $B + E + 1 = 10 + H$ (H as carry)
4. $A + D = G$
 $A + D + 1 = G$

Example 2: Crypto Arithmetic Problems

Step 1:

Domain of C = {1, 2, 3, 4, 5, 6, 7, 8, 9}

Domain of F = {1, 2, 3, 4, 5, 6, 7, 8, 9}

So,

Domain of I = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}

Select C = 4 & F = 9 Then I = 3 (Carry = 1)

So,

	A	B	4
+	D	E	9
	G	H	3

Example 2: Crypto Arithmetic Problems

Step 2:

Domain of B = {1, 2, 5, 6, 7, 8}

Domain of E = {1, 2, 5, 6, 7, 8}

So,

Domain of H = {0, 1, 2, 5, 6, 7, 8}

Select B = 2 & E = 8

Then H = 10+1 {previous carry = 1 + (Carry = 1)}

So,

	A	2	4
+	D	8	9
	G	1	3

Example 2: Crypto Arithmetic Problems

Step 3:

Domain of A = {0, 5, 6, 7}

Domain of D = {0, 5, 6, 7}

So, Domain of G = {5, 6}

Select A = 0 & D = 5 Then G = 6 (with addition of Carry)

So,

Hence, the required solutions are:

A = 0, B = 2, C = 4, D = 5, E = 8, F = 9, G = 6, H = 1, I = 3.

Example 2: Crypto Arithmetic Problems

LOGIC + LOGIC = PROLOG

90452 + 90452 = 180904

Example 2: Crypto Arithmetic Problems

$$\begin{array}{r} \\ \\ + \\ \hline = M \end{array}$$

The solution to this puzzle is $O = 0$, $M = 1$, $Y = 2$, $E = 5$, $N = 6$, $D = 7$, $R = 8$, and $S = 9$.

Search Algorithms

- A search algorithm takes a search problem as input and returns a solution, or an indication of failure.
- Each node in the search tree corresponds to a state in the state space and the edges in the search tree correspond to actions.
- The root of the tree corresponds to the initial state of the problem.
- The leaf node represent the goal of the problem.

Example: Tic-tac-toe

- There are two players denoted by X and O.
- They are alternatively writing their letter in one of the 9 cells of a 3 by 3 board.
- The winner is the one who succeeds in writing three letters in line.
- The game begins with an empty board. It ends in a win for one player and a loss for the other, or possibly in a draw.
- A complete tree is a representation of all the possible plays of the game.
- The root node is the initial state, in which it is the first player's turn to move (the player X).
- The successors of the initial state are the states the player can reach in one move, their successors are the states resulting from the other player's possible replies, and so on.

References

- E. Rich and Knight, *Artificial Intelligence*, McGraw Hill, 1991.
- D. W. Patterson, *Artificial Intelligence and Expert Systems*, Prentice Hall, 2001.

Chapter 4

Searching Techniques

Er. Sharmila Bista
MECE

Overview

1 Uninformed Search Techniques

- Depth first search
- Breadth first search
- Uniform cost search

2 Informed Search Techniques

- Hill Climbing Search
- Best First Search
- Greedy Search
- A* search
- Adversarial search techniques – minimax and alpha beta techniques

Searching

- Search is a universal problem-solving technique.
- Searching is the process of finding the required states or nodes.
- Searching is to be performed through the state space.
- Search process is carried out by constructing a search tree.

SEARCH TERMINOLOGY

- Problem Space: Environment in which the search takes place.
- Problem Instance: It is Initial state + Goal state
- Problem Space Graph: Graph that represents problem state. States are shown by nodes and operators are shown by edges.
- Depth of a problem: Length of a shortest path or shortest sequence of operators from Initial State to goal state.
- Space Complexity: The maximum number of nodes that are stored in memory.
- Time Complexity: The maximum number of nodes that are created.
- Admissibility: A property of an algorithm to always find an optimal solution.
- Branching Factor: The average number of child nodes in the problem space graph.

CLASSIFICATION

- **Uninformed Search (Blind Search/Brute force search)**
 - The search algorithms that do not use any extra information regarding the problem. Uninformed search strategies use no information about the likely “direction” of the goal node(s).
 - Depth First Search
 - Breath First Search
 - Depth Limit Search
- **Informed search or Heuristic search**
 - Informed search have problem specific knowledge apart from problem definition. Informed search strategies use information about the domain to (try to) (usually) head in the general direction of the goal node(s).
 - Hill climbing Search
 - Best first Search
 - Greedy Search
 - A* Search
 - Adversarial Search Techniques

Uninformed Search Methods

Breadth-First Search (BFS)

- Proceeds level by level down the search tree.
- Starting from the root node (initial state) explores all children of the root node, left to right.
- If no solution is found, expands the first (leftmost) child of the root node, then expands the second node and so on.

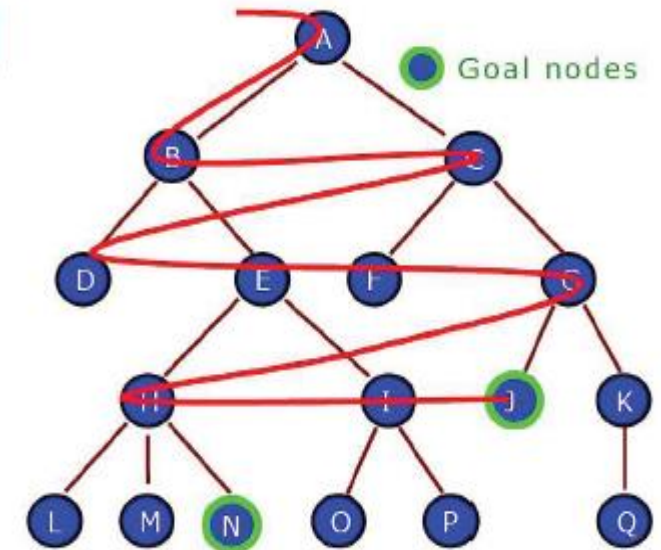
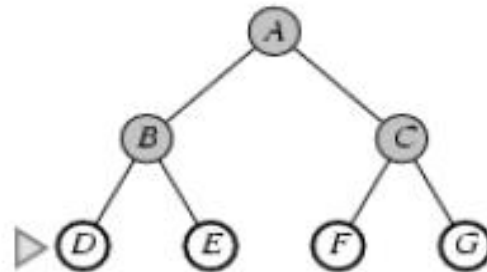
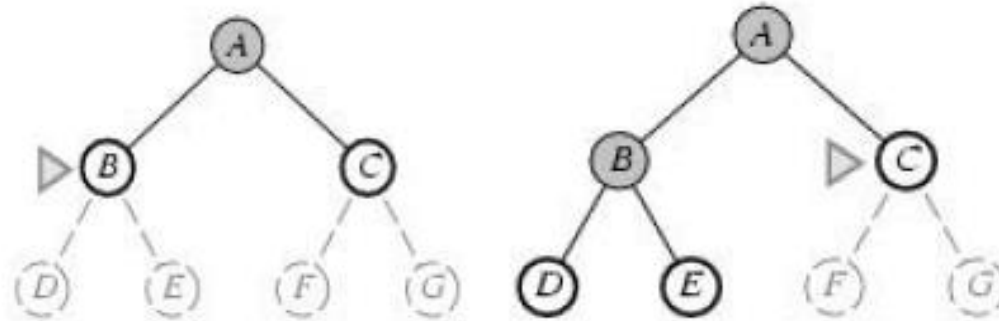
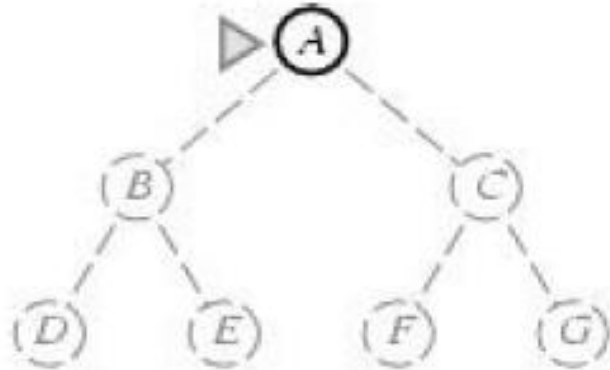


Fig. Breadth-first search (BFS)

BFS example (Find path from A to D)



Breadth-First Search (BFS)

- Completeness: Complete
- Optimality: It is guaranteed to find the shortest path.
- Space Complexity: $(b + b^2 + b^3 + \dots + b^d + (b^{d+1} - b)) = \mathbf{O(b^{d+1})}$
- Time Complexity: $\mathbf{O(b^{d+1})}$

Depth First Search

- Proceeds down a single branch of the tree at a time.
- Expands the root node, then the leftmost child of the root node.
- Always expands a node at the deepest level of the tree.
- Only when the search hits a dead end (a partial solution which can't be extended), the search backtrack and expand nodes at higher levels.

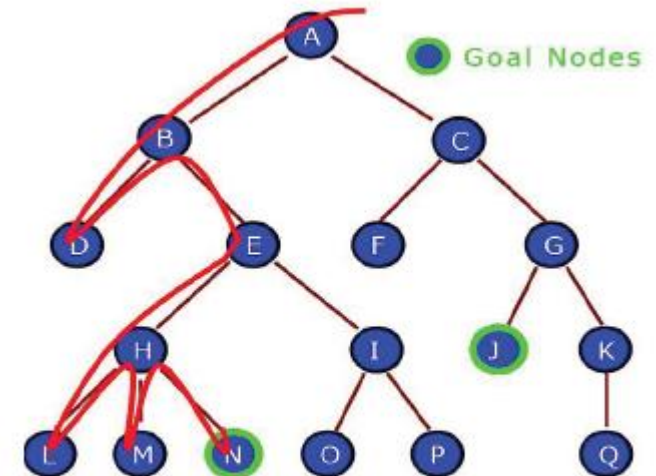
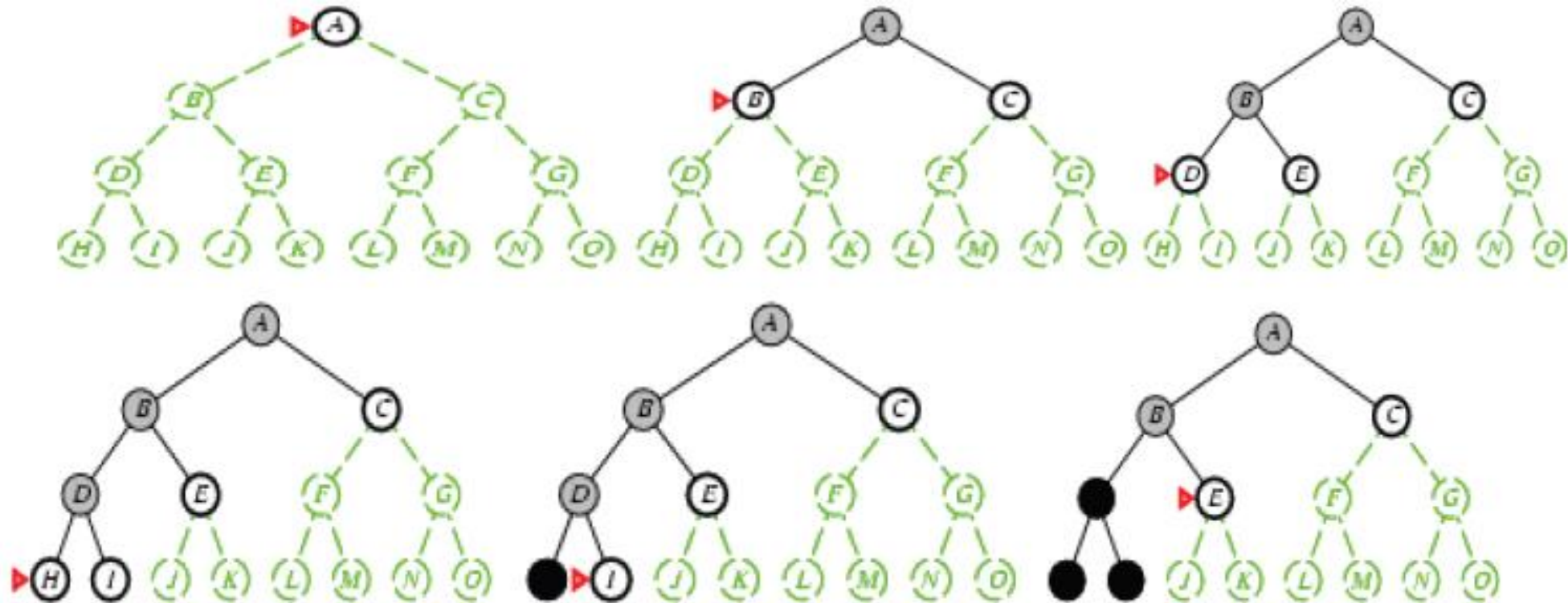


Fig. Depth-first search (DFS)

DFS example (find path from A to E)



DEPTH FIRST SEARCH (DFS)

- Completeness: Incomplete as it may get stuck going down an infinite branch that doesn't leads to solution.
- Optimality: The first solution found by the DFS may not be shortest.
- Space complexity: **b** as branching factor and maximum depth **m**,
Space complexity = **$O(bm)$**
- Time Complexity: **$O(b^m)$**

Depth Limit Search

- Depth-first search will not find a goal if it searches down a path that has infinite length. So, in general, depth-first search is not guaranteed to find a solution, so it is not complete.
- This problem is eliminated by limiting the depth of the search to some value L . However, this introduces another way of preventing depth-first search from finding the goal: if the goal is deeper than L it will not be found.
- Perform depth first search but only to a pre-specified depth limit L .
- No node on a path that is more than L steps from the initial state.

Depth Limit Search

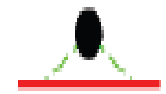
- Completeness: Incomplete as solution may be beyond specified depth level.
- Optimality: not optimal
- Space complexity: b as branching factor and L as tree depth level,
 $O(b.L)$
- Time Complexity: **$O(b^L)$**

Iterative Deepening Search (IDS)

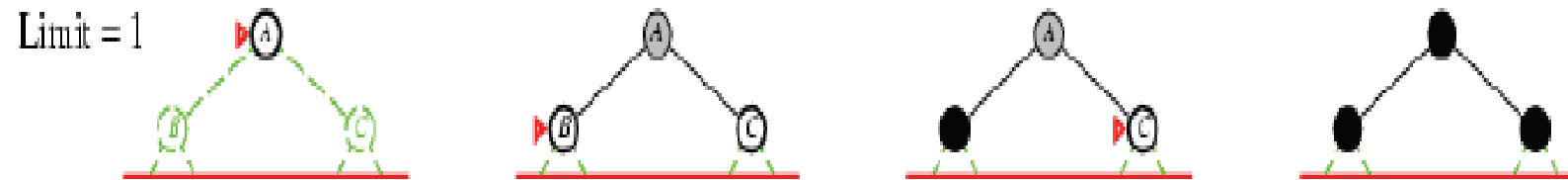
- Iterative deepening search is a strategy that sidesteps the issue of choosing the best depth limit by trying all possible depth limits: first depth 0, then depth 1, then depth 2, and so on.
- In effect, iterative deepening combines the benefits of depth-first and breadth-first search.
- It is optimal and complete, like breadth-first search, but has only the modest memory requirements of depth-first search.

Iterative deepening search $L=0$

Limit = 0

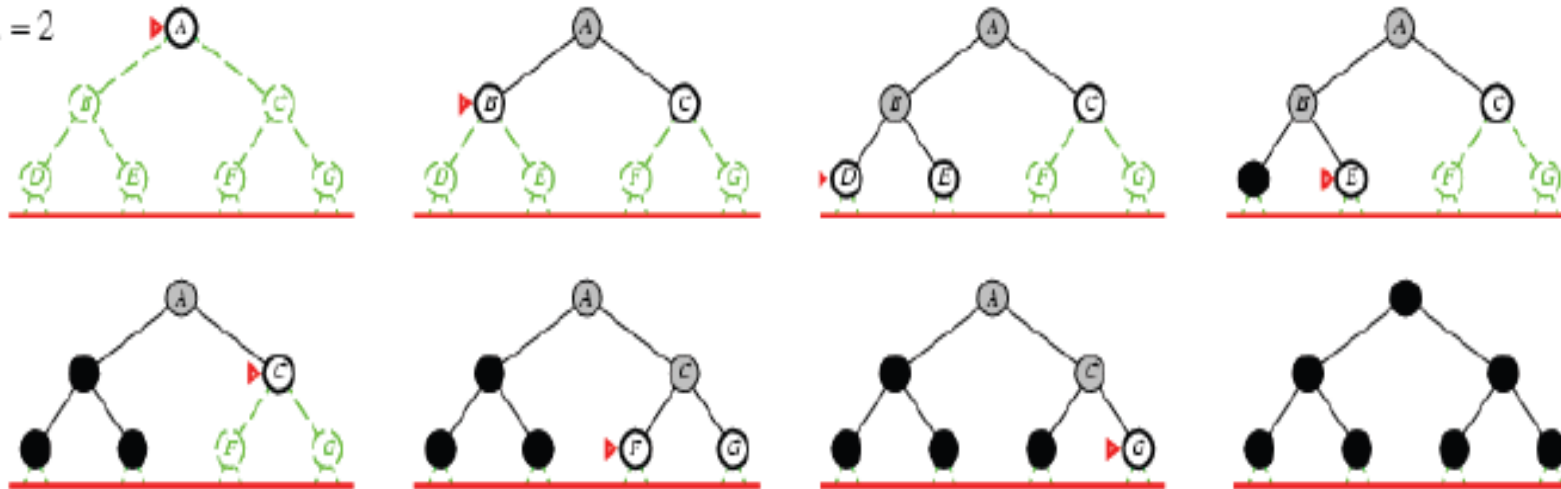


Iterative deepening search L=1



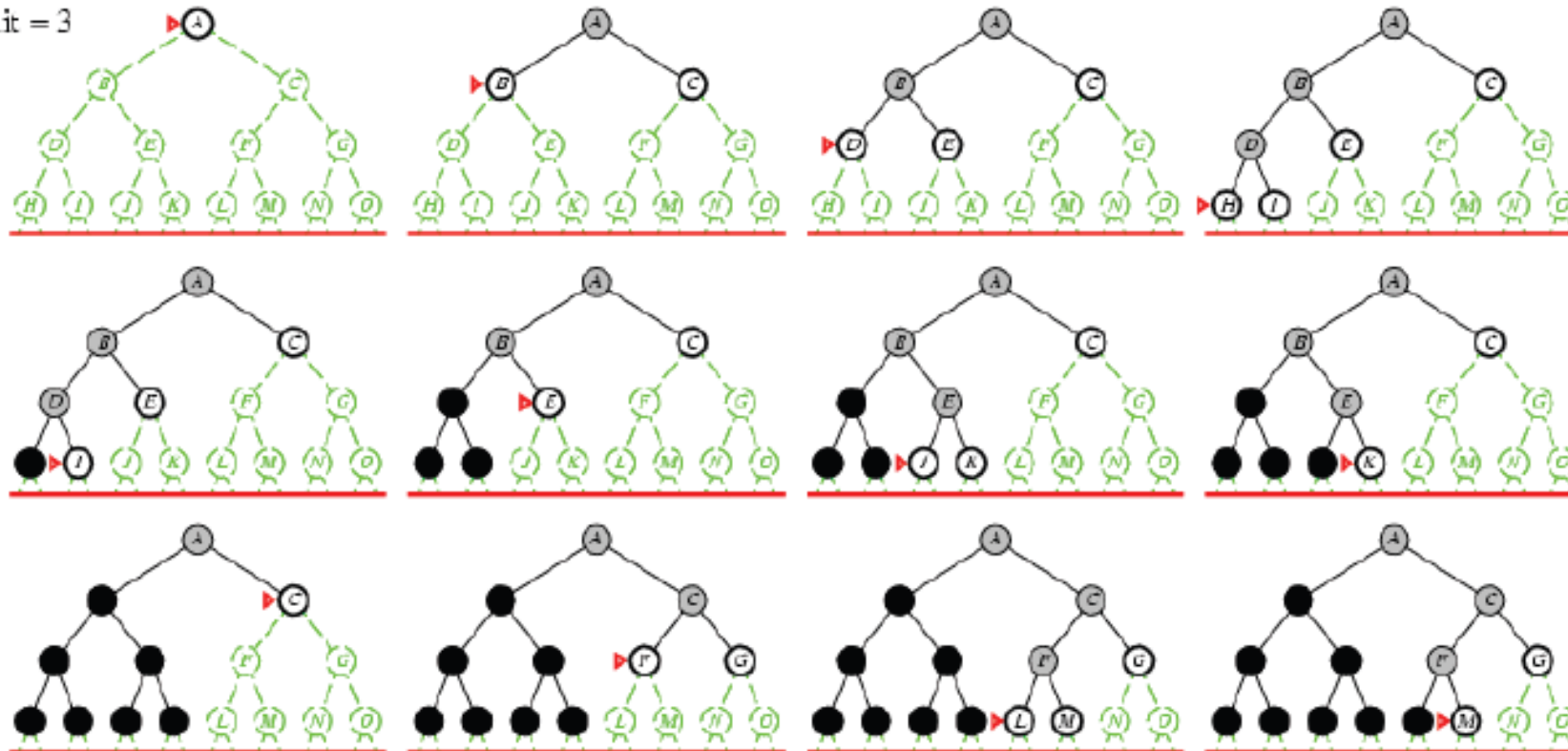
Iterative deepening search $L=2$

Limit = 2



Iterative Deepening Search L=3

Limit = 3



Iterative Deepening Search

- Completeness: Yes, when the branching factor is finite.
- Optimality: Optimal
- Space complexity: b as branching factor and d as depth level,
 $O(b \cdot d)$
- Time Complexity: **$O(b^d)$**

Uniform-Cost Search

- Uniform-cost is guided by path cost rather than path length like in BFS, the algorithm starts by expanding the root, then expanding the node with the lowest cost from the root, the search continues in this manner for all nodes.
- If all step costs are equal, this is identical to breadth-first search.
- Uniform-cost search does not care about the number of steps a path has, but only about their cost. Therefore, algorithm expands nodes in order of increasing path cost.

UCS example (Find path from A to E)

Expand A to B, C, and D.

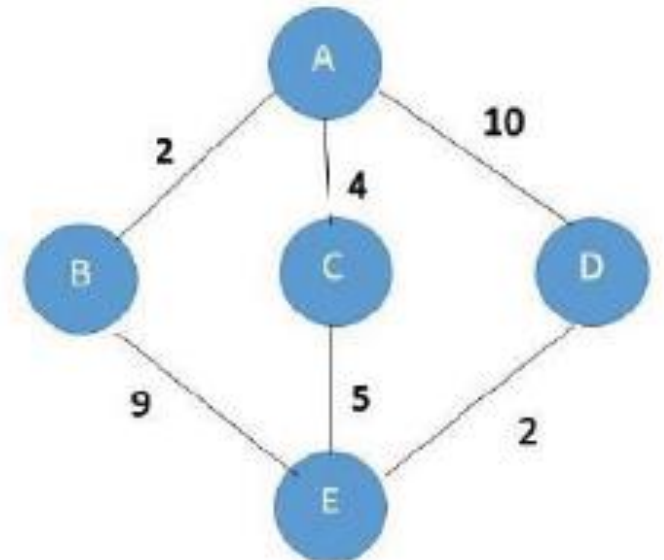
- The path to B is the cheapest one with path cost 2.

Expand B to E

- Total path cost = $2 + 9 = 11$
- This might not be the optimal solution since the path AC as path cost 4 (less than 11)

Expand C to E

- Total path cost = $4 + 5 = 9$
- Path cost from A to D is 10 (greater than path cost, 9)
- Hence optimal path is ACE.



Uniform-Cost Search

- Completeness: Complete
- Optimality: Optimal in the sense that the node that it always expands is the node with the least path cost.
- Space Complexity: Uniform cost search is guided by path cost rather than depth, so its complexity cannot be characterized in terms of b and d . Instead, let c be the cost of optimal solution and every action to reach solution costs at least e , then in worst case space complexity = $O(b^{c/e})$
- Time Complexity: $O(b^{c/e})$

Summary of algorithms

Criterion	Breadth-First	Uniform-Cost	Depth-First	Depth-Limited	Iterative Deepening
Complete?	Yes	Yes	No	No	Yes
Time	$O(b^{d+1})$	$O(b^{\lceil C^*/\epsilon \rceil})$	$O(b^m)$	$O(b^l)$	$O(b^d)$
Space	$O(b^{d+1})$	$O(b^{\lceil C^*/\epsilon \rceil})$	$O(bm)$	$O(bl)$	$O(bd)$
Optimal?	Yes	Yes	No	No	Yes

even complete
 if step cost is not
 increasing with depth.

preferred
 uninformed
 search strategy

Informed Search Methods

2 Informed Search (Heuristic Search)

- Informed search have problem specific knowledge apart from problem definition.
- They use experimental algorithm which improves efficiency of search process.
- The idea is to develop a domain specific heuristic function $h(n)$ where $h(n)$ guesses the cost of getting to the goal from node n .

Heuristic Function = The heuristic function is a way to inform the search about the direction to a goal. It provides an informed way to guess which neighbor of a node will lead to a goal.

Best-First Search (BFS)

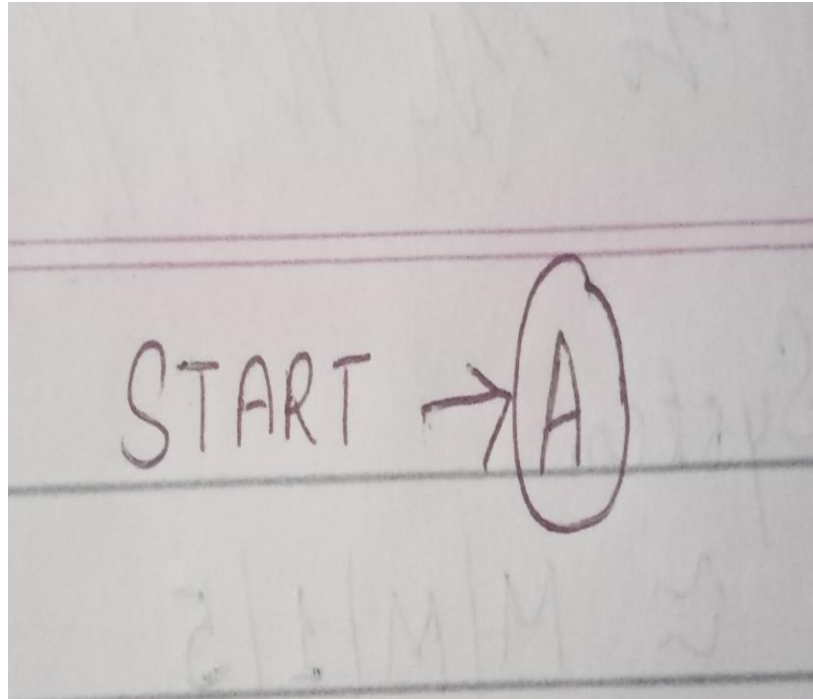
- Best-First search is a heuristic search algorithm.
- The name “best-first” refers to the method of exploring the node with the best “score” first.
- An evaluation function is used to assign a score to each candidate node. The evaluation function must represent some estimate of the cost of the path from state to the closest goal state.

Best-First Search (BFS)

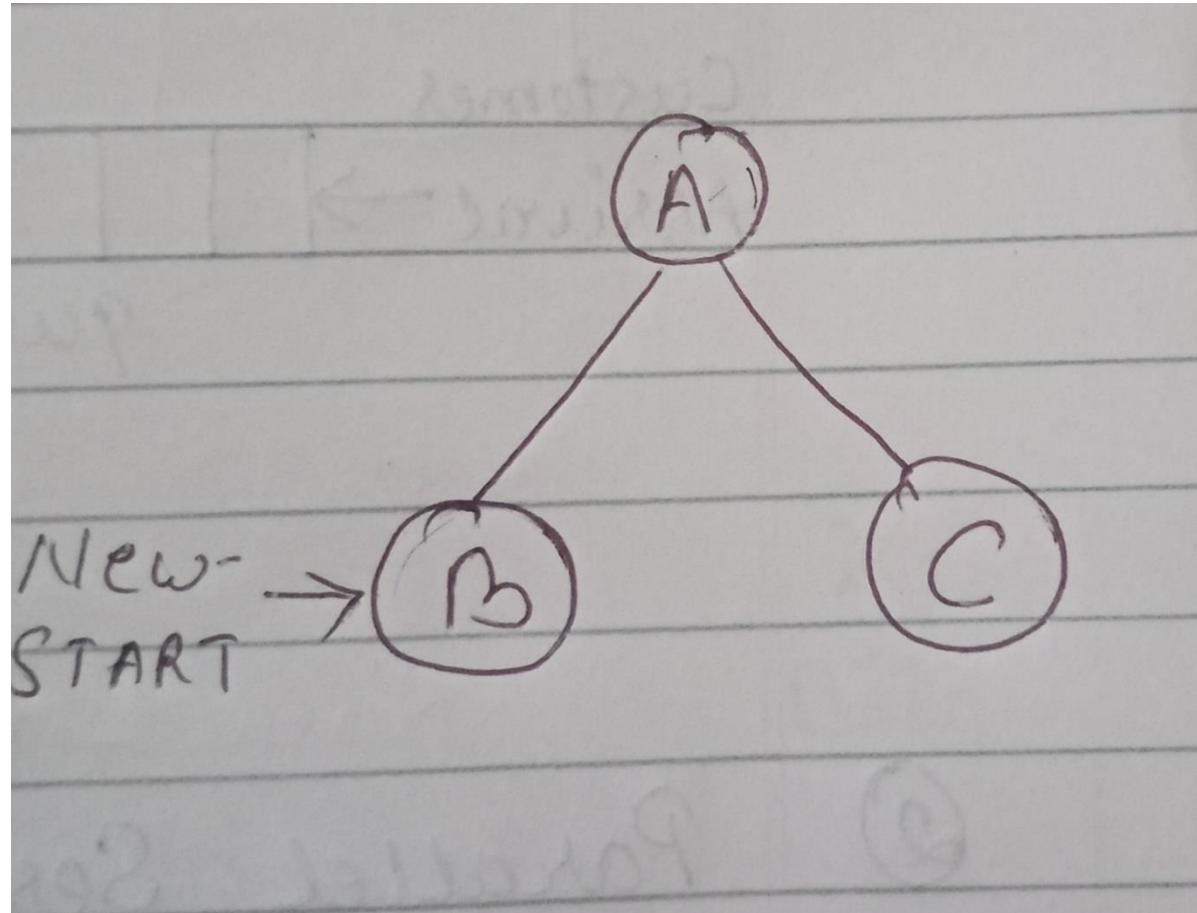
Algorithm

1. Put the initial node on a list START
2. If START = GOAL or START = EMPTY, then terminate search
3. Assign the next node as START and call this node-A
4. If A = GOAL, terminate the search with success
5. Else-if, node has successor and generate all of them. Find out how far they are from the GOAL node.
6. Sort all the children generated so far by remaining distance from the goal. Name the list as START-1. Replace START = START-1
7. Go to step-2

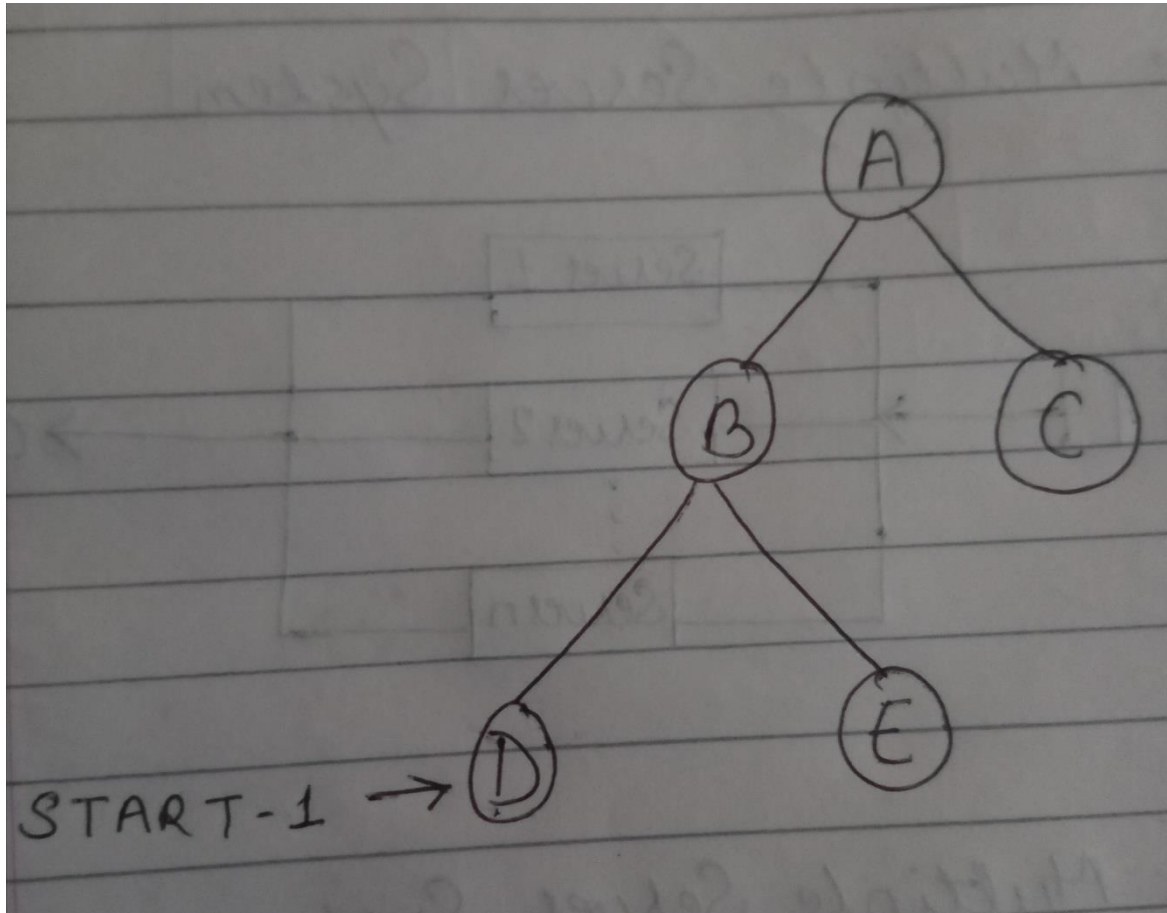
Best-First Search (BFS)



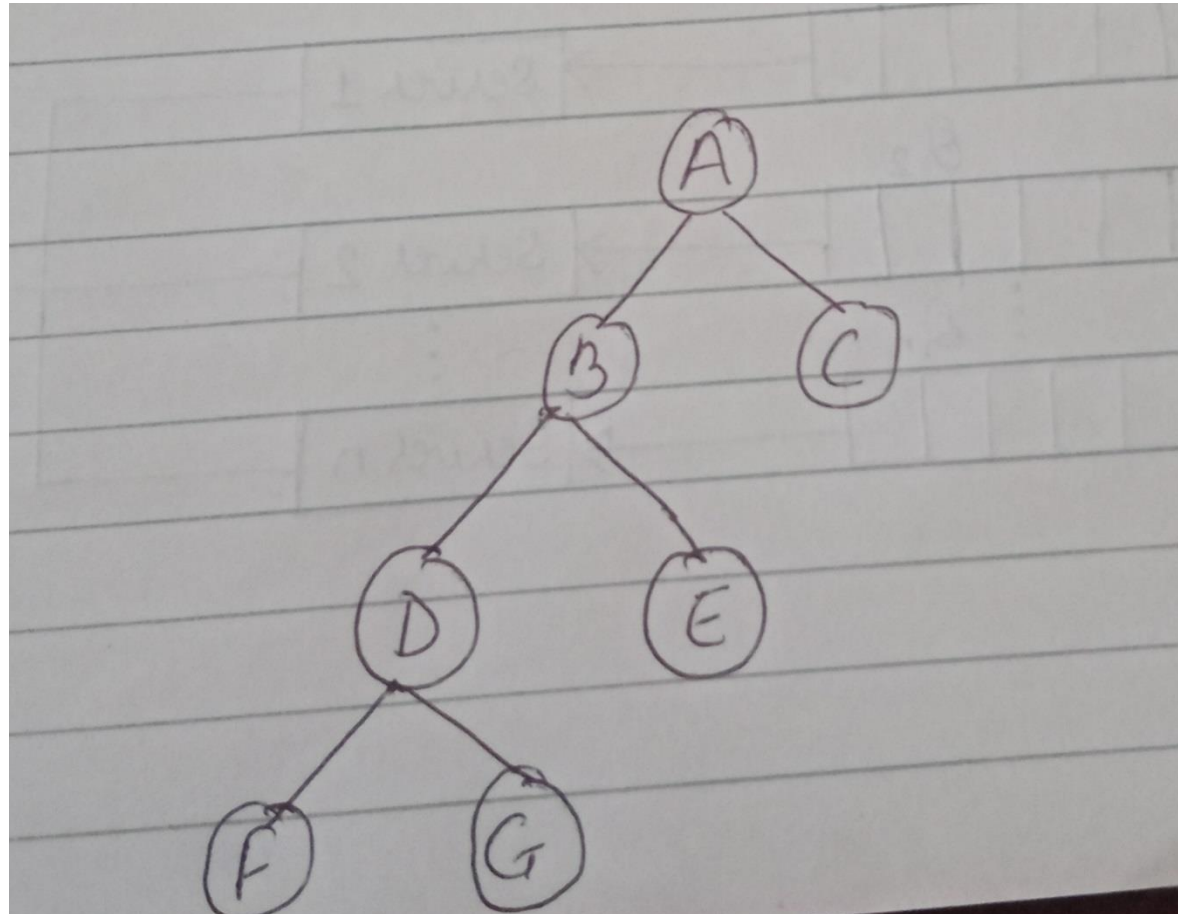
Best-First Search (BFS)



Best-First Search (BFS)



Best-First Search (BFS)



Types of BFS

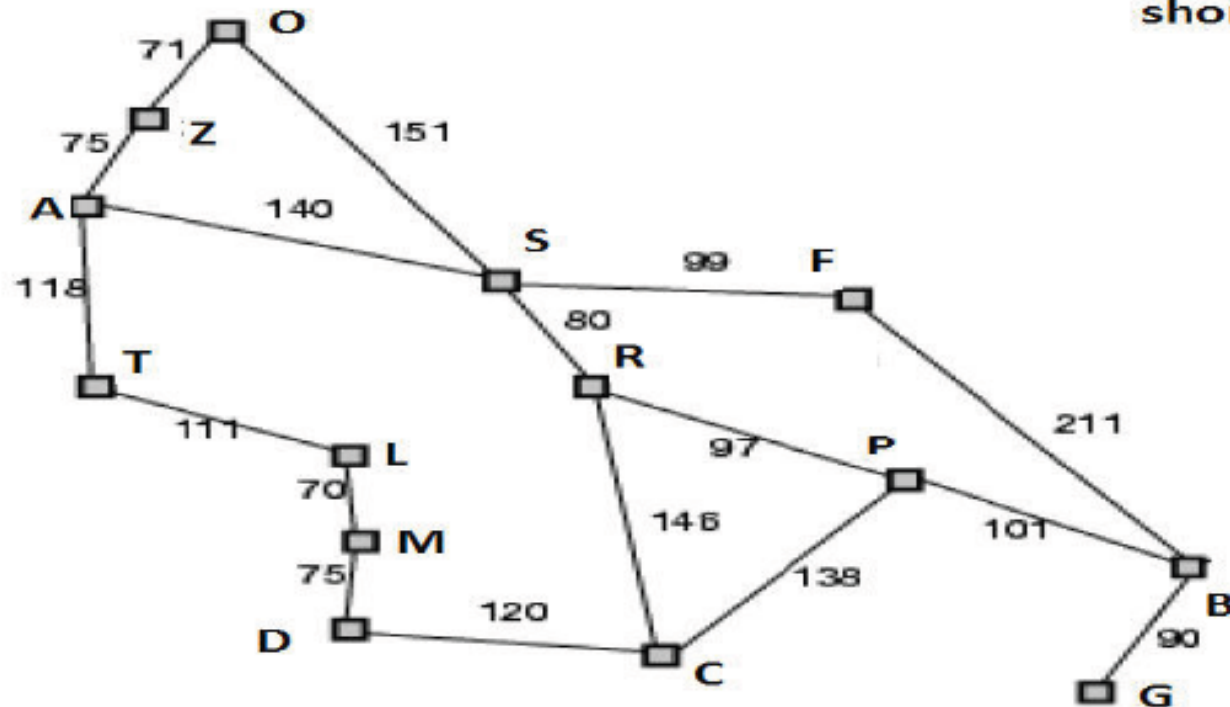
- Greedy Best First Search
- A* Search

Greedy Best First Search

- It tries to get as close as it can to the goal.
- It expands the node that appears to be closest to the goal.
- It evaluates the node by using heuristic function only.
- Evaluation function $f(n) = h(n)$
 - heuristic= (estimate of cost from n to goal)
 - $h(n) = 0$ for goal state

Greedy Best First Search

Given following graph of cities, starting at City “A”, problem is to reach to the “B”



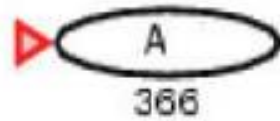
shortest line distance to B

A	366
B	0
C	160
D	242
F	176
G	77
L	244
M	241
O	380
P	101
R	193
S	253
T	329
Z	374

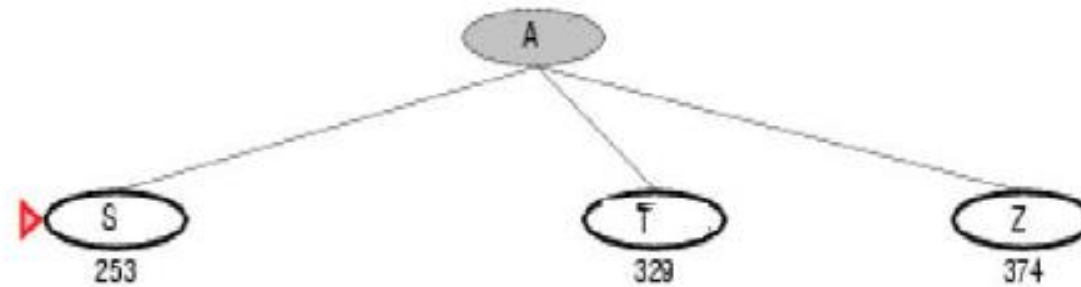
Greedy Best First Search

Solution using greedy best first can be as below:

Step 1: Initial State

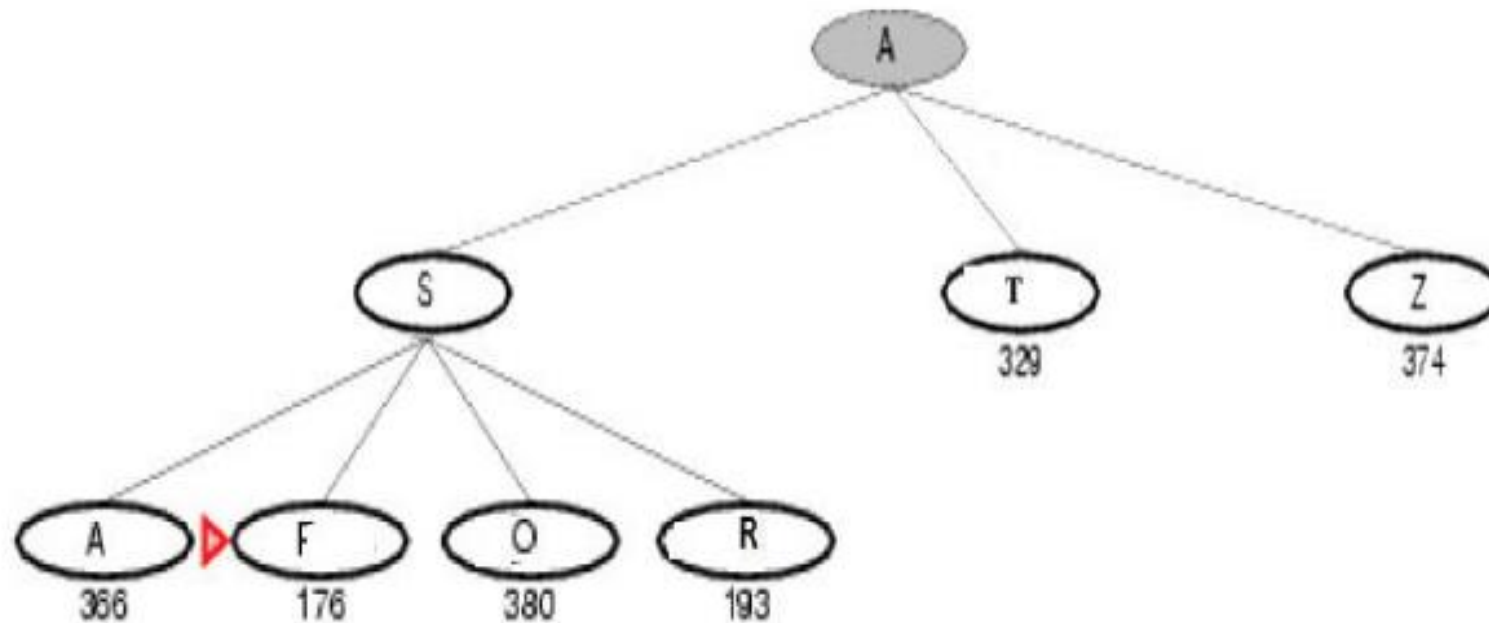


Step 2: After expanding A



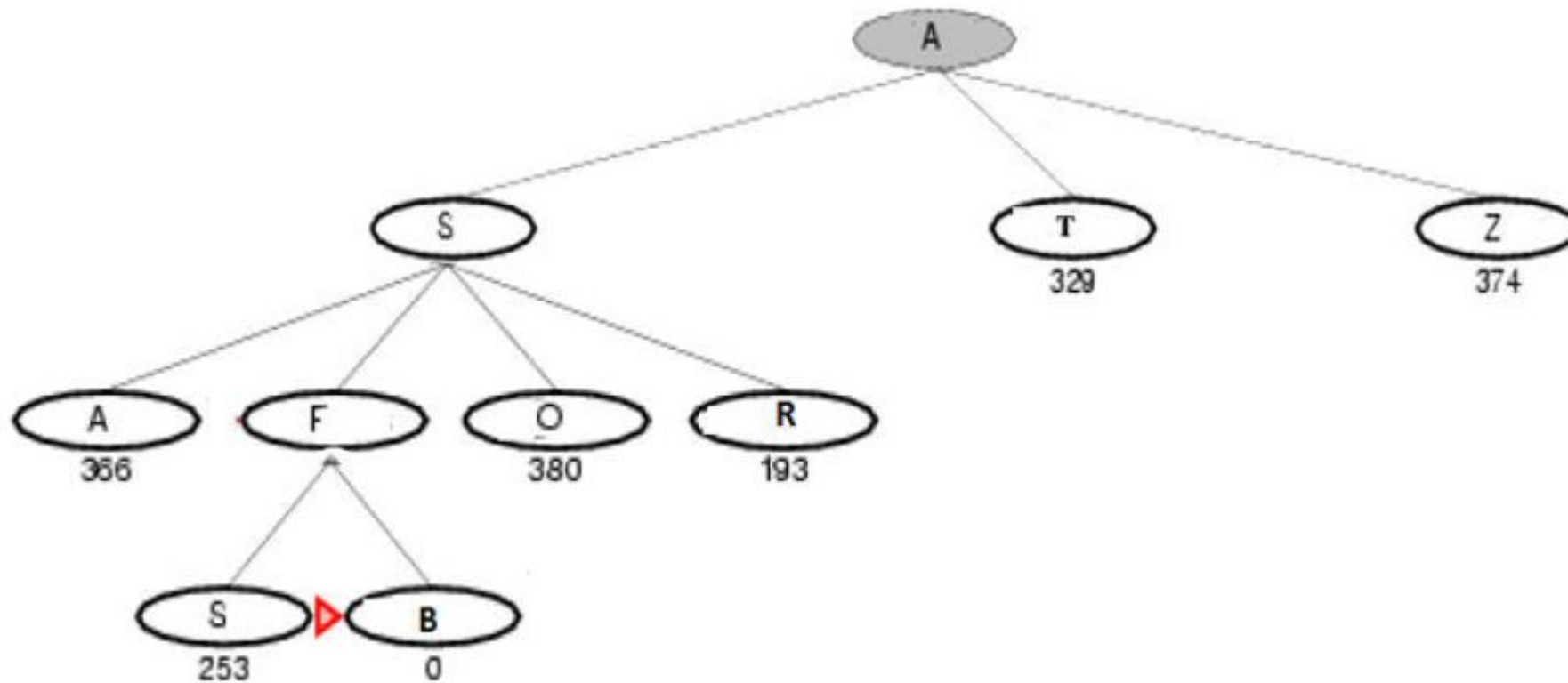
Greedy Best First Search

Step 3: After expanding S



Greedy Best First Search

Step 4: After expanding F



Greedy Best First Search

Properties:

- Completeness: No, can get stuck in loops.
- Time Complexity: $O(b^m)$, but a good heuristic can give dramatic improvement
- Space Complexity: $O(b^m)$, keeps all nodes in memory
- Optimality: No

Applications:

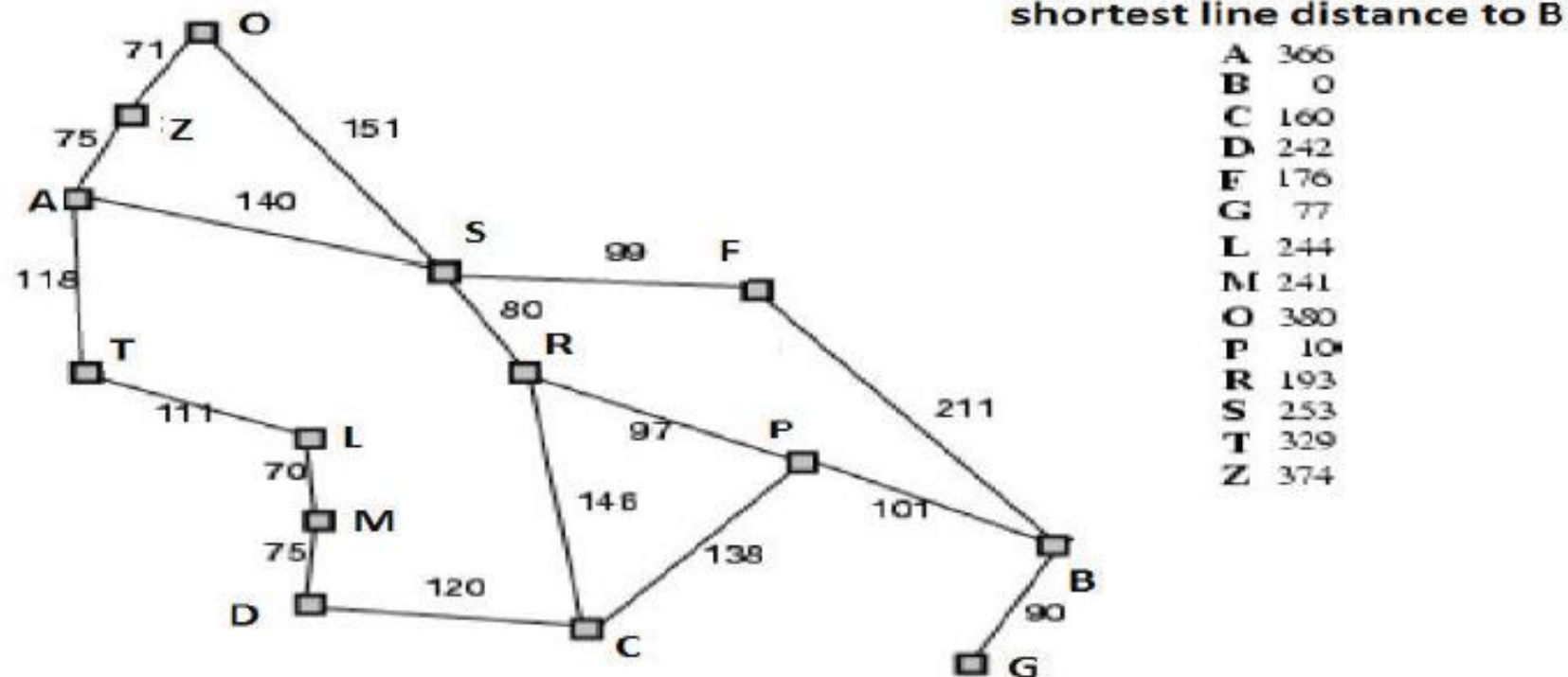
- This algorithm is used in Huffman encoding, minimum spanning tree, Dijkstra's algorithm etc.

A* Search

- It finds a minimal cost-path joining the start node and a goal node for node n.
- Evaluation function: $f(n) = g(n) + h(n)$ Where,
 $g(n)$ = cost so far to reach n from root
 $h(n)$ = estimated cost to goal from n
- Thus, $f(n)$ estimates always the lowest total cost of any solution path going through node n.
- Avoid expanding paths that are already expensive.

A* Search

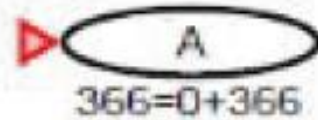
A* search example (Find path from A to B)



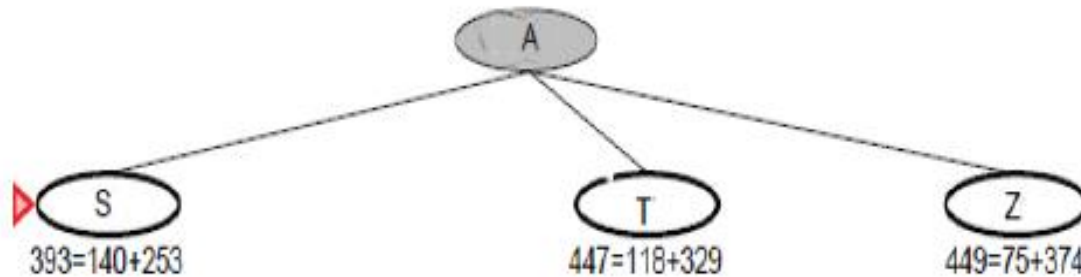
Here, evaluate nodes connected to source. Evaluate $f(n)=g(n)+h(n)$ for each node. Select node with lowest $f(n)$ value.

A* Search

Step 1:

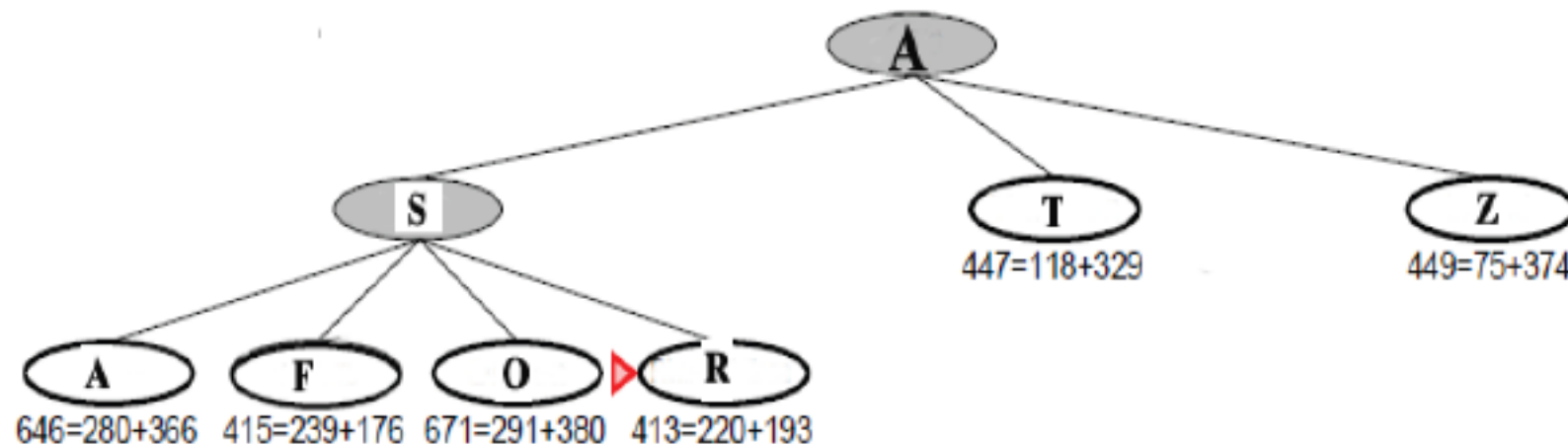


Step 2



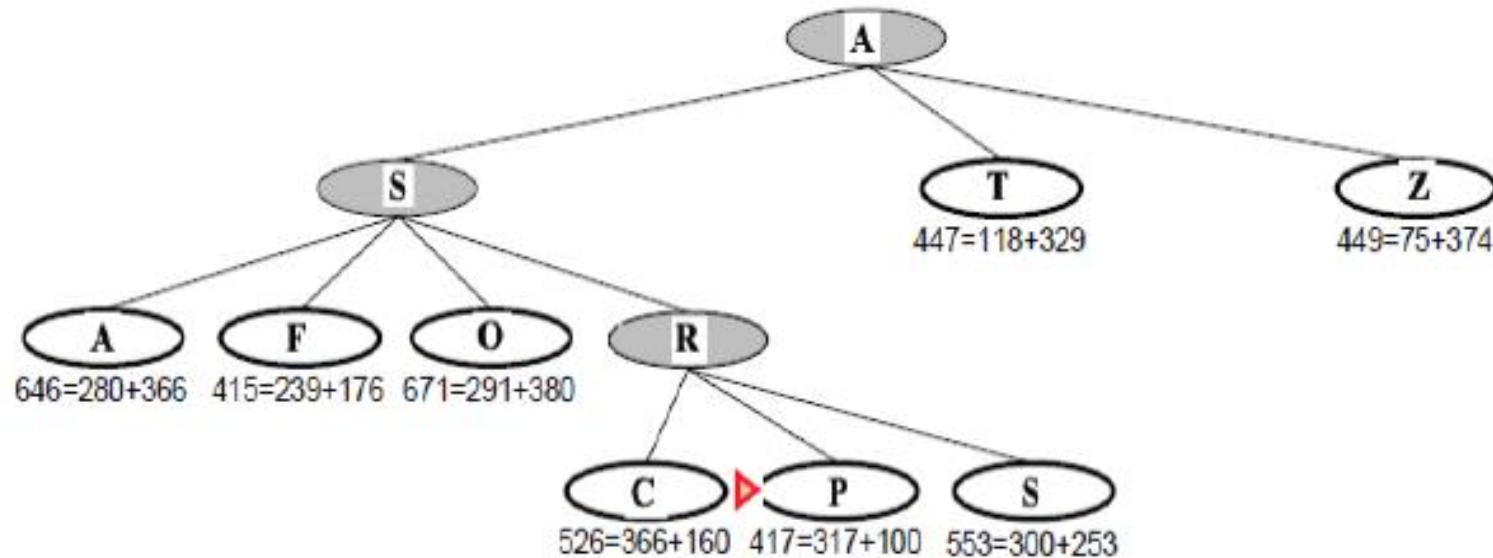
A* Search

Step 3:



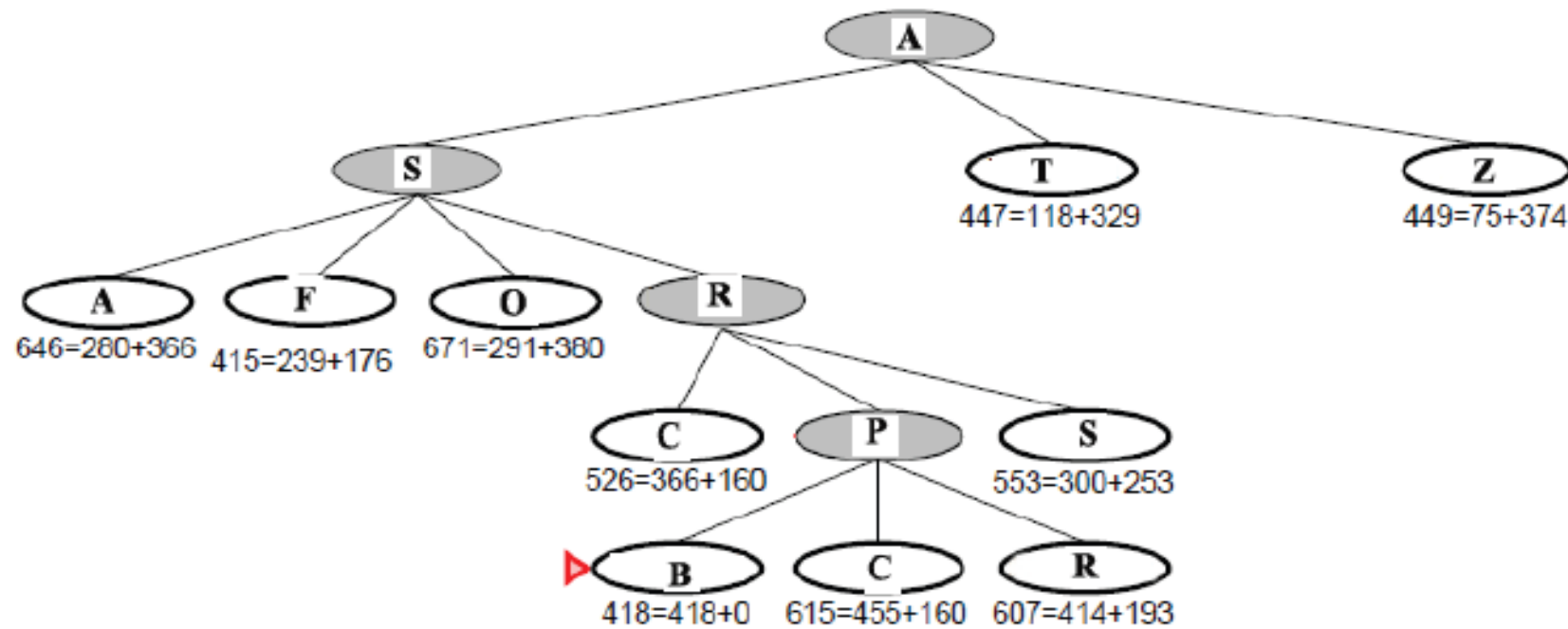
A* Search

Step 4:



A* Search

Step 5:



Properties of A* Search

- Complete ?

➤ Yes

- Optimal?

Yes

- Time Complexity

Exponential in [relative error in h * length of solution]

- Space

➤ Keeps all nodes in memory, hence **$O(b^d)$**

Adversarial Search

- Adversarial search used in game playing is the type of search where we examine the problems that arise when we try to plan ahead in the world where others are planning against us.
- A game can be formally defined as a kind of search problem as below:
 - Initial state: It includes the board position and identifies the player's to move.
 - Successor function: It gives a list of (move, state) pairs each indicating a legal move and resulting state.
 - Terminal test: This determines when the game is over. States where the game is ended are called terminal states.
 - Utility function: It gives numerical value of terminal states. E.g. win (+1), loose (-1) and draw (0).

Game playing

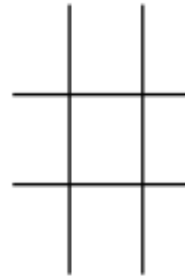
- Perfect play
 - The minimax algorithm
 - alpha-beta pruning

The minimax algorithm

- Perfect play for deterministic environments with perfect information
- **Basic idea:** choose move with highest minimax value
= best achievable payoff against best play
- **Algorithm:**
 1. Generate game tree completely
 2. Determine utility of each terminal state
 3. Propagate the utility values upward in the tree by applying MIN and MAX operators on the nodes in the current level
 4. At the root node use minimax decision to select the move with the max (of the min) utility value
- Steps 2 and 3 in the algorithm assume that the opponent will play perfectly.

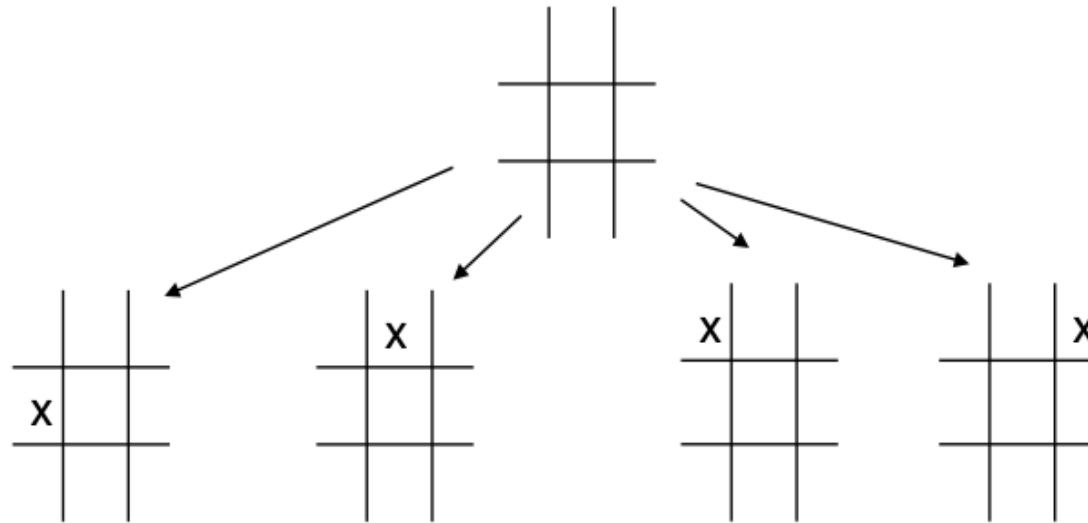
The Minimax Algorithm

Generate Game Tree



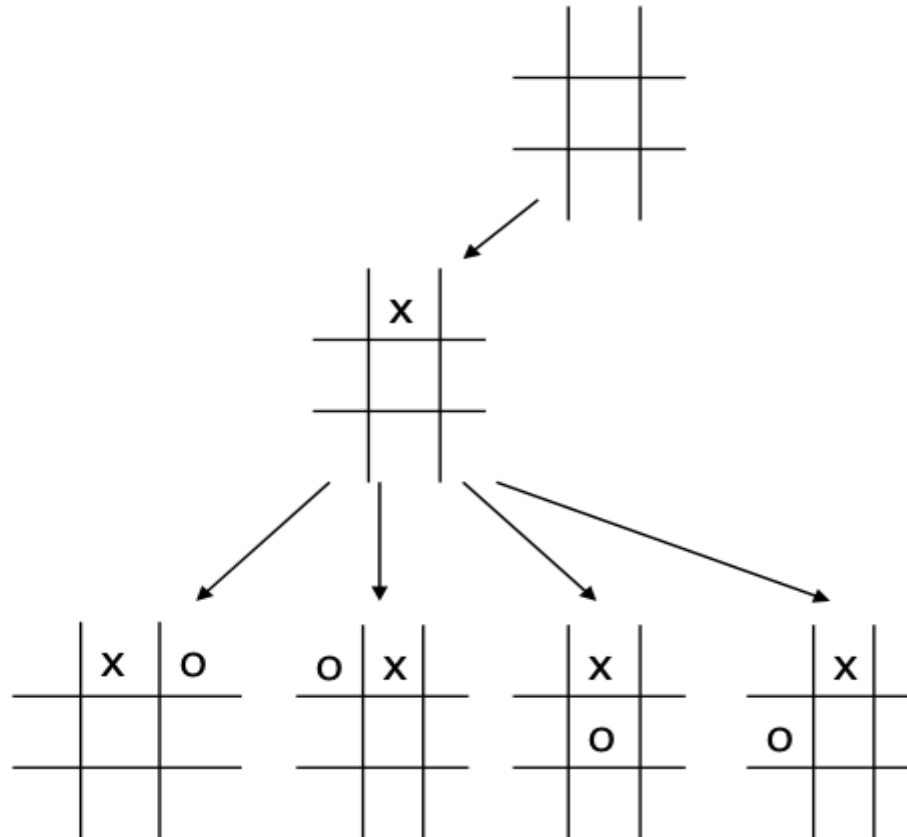
The Minimax Algorithm

Generate Game Tree



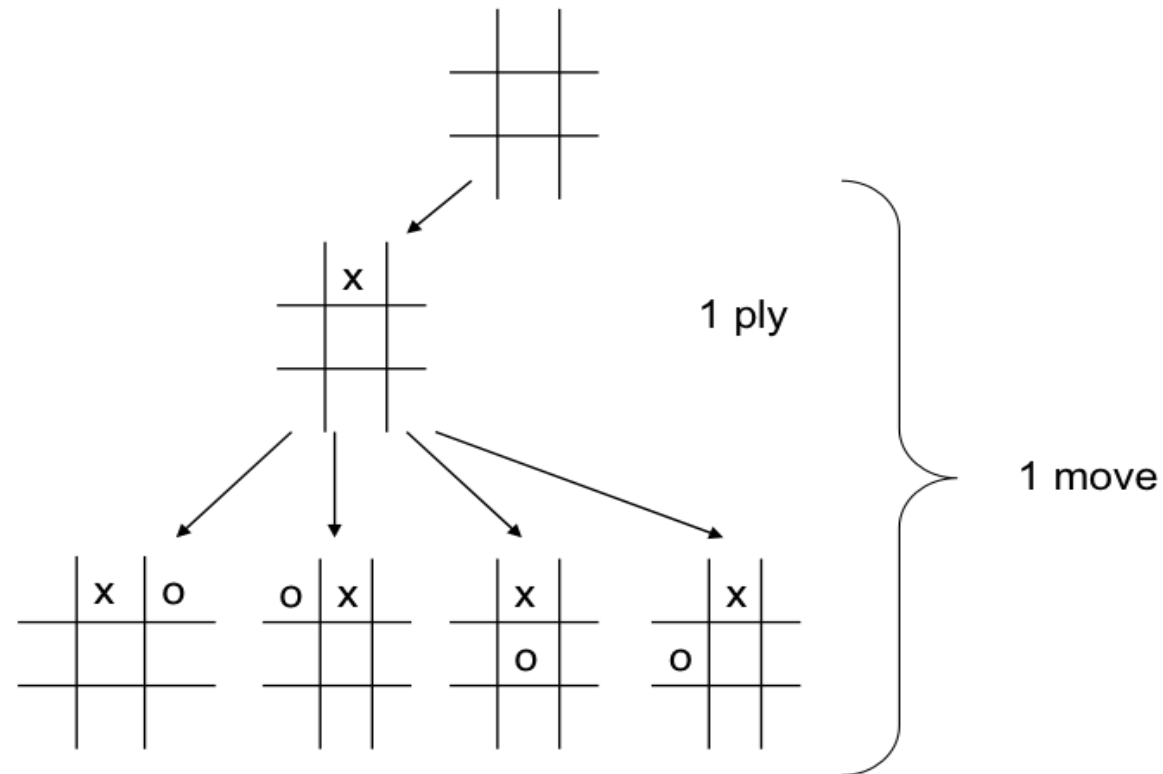
The Minimax Algorithm

Generate Game Tree

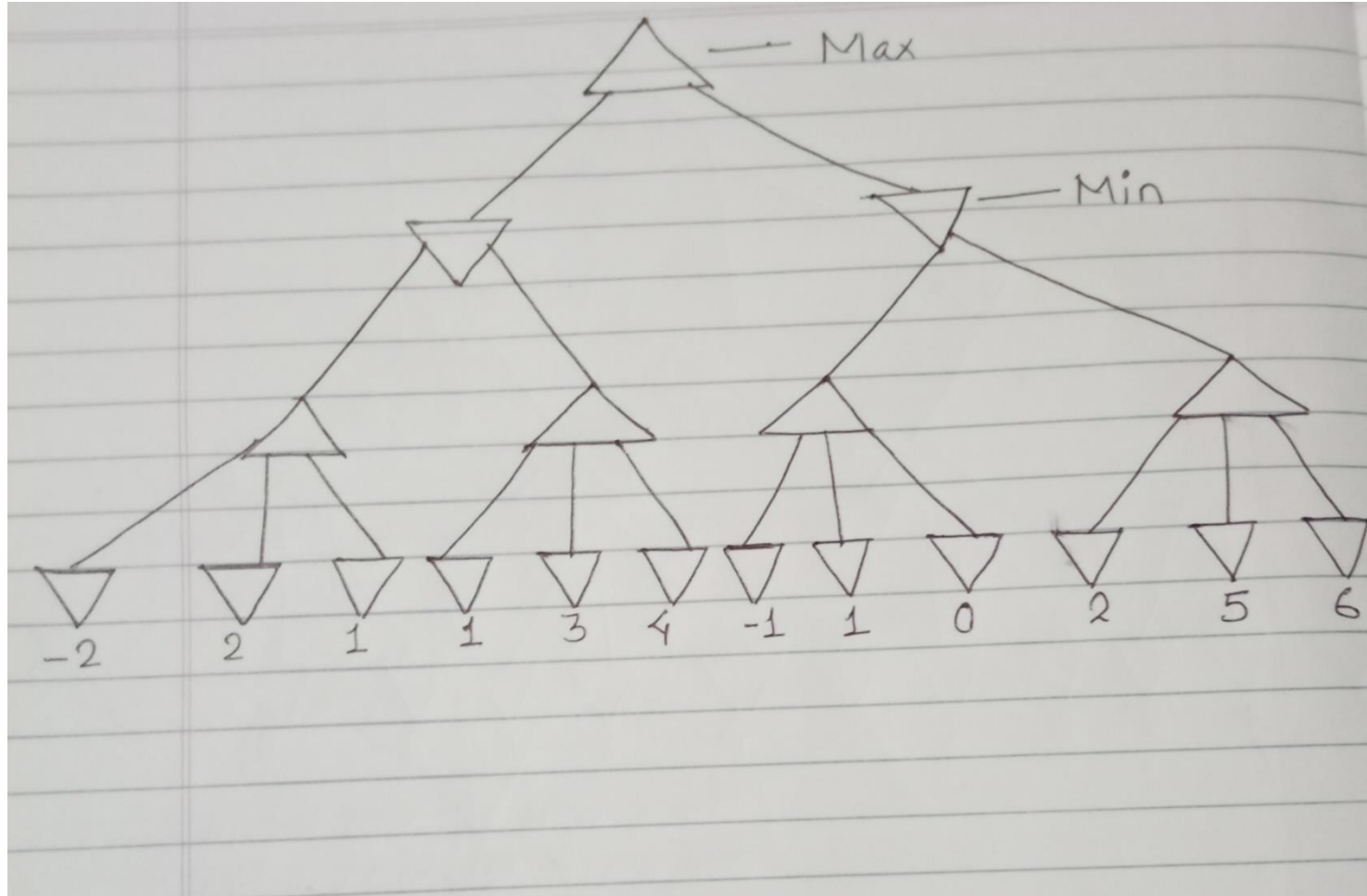


The Minimax Algorithm

Generate Game Tree



The Minimax Algorithm: Example 1



The Minimax Algorithm: Example 1

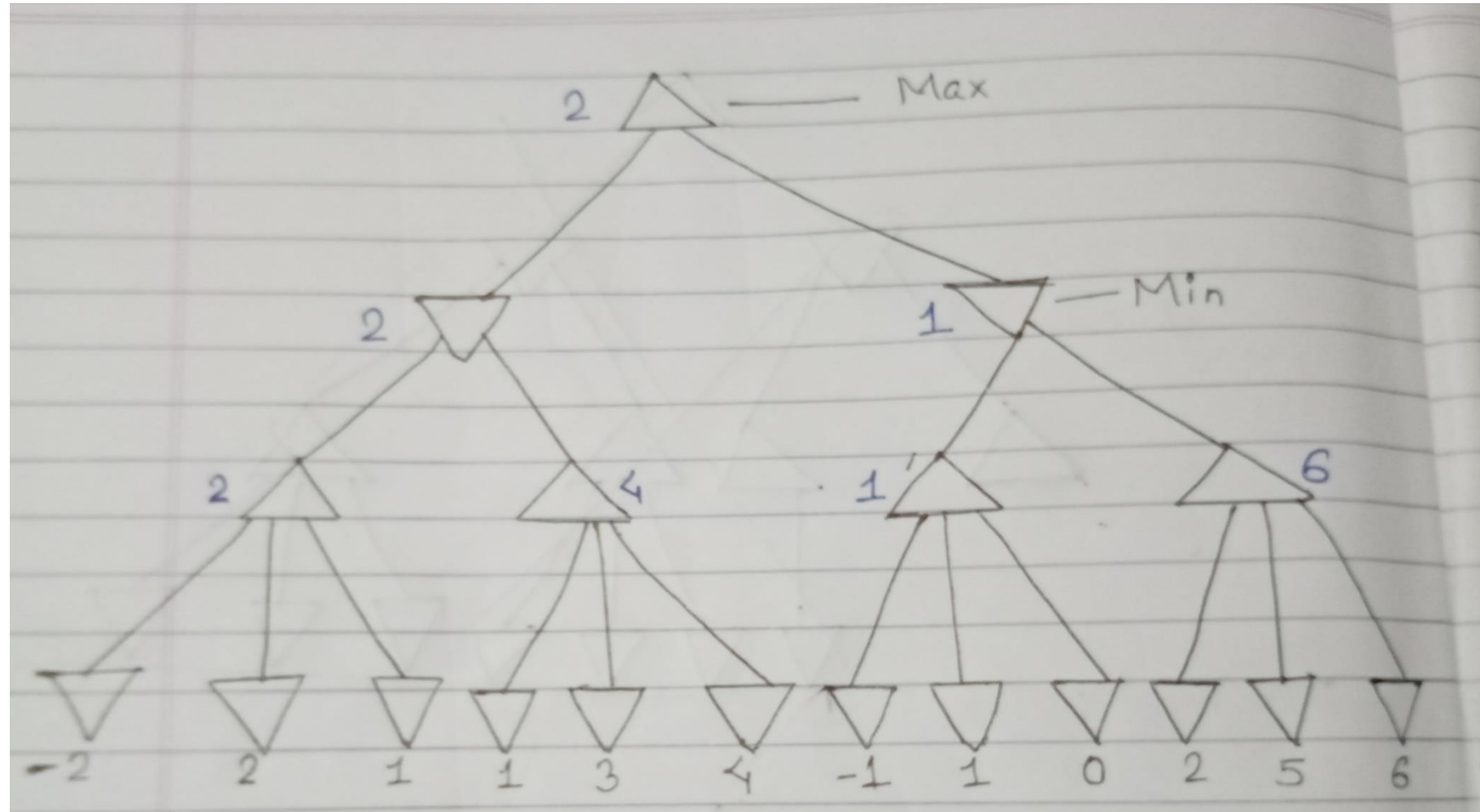
Worst case for max = $-\infty$
Worst case for min = $+\infty$

Step 1: $\text{Max}(-\infty, -2) \Rightarrow (-2, 2) \Rightarrow (2, 1) \Rightarrow 2$
 $\text{Max}(-\infty, 1) \Rightarrow (1, 3) \Rightarrow (3, 4) \Rightarrow 4$
 $\text{Max}(-\infty, -1) \Rightarrow (-1, 1) \Rightarrow (1, 0) \Rightarrow 1$
 $\text{Max}(-\infty, 2) \Rightarrow (2, 5) \Rightarrow (5, 6) \Rightarrow 6$

Step 2: $\text{Min}(+\infty, 2) \Rightarrow (2, 4) \Rightarrow 2$
 $\text{Min}(+\infty, 1) \Rightarrow (1, 6) \Rightarrow 1$

Step 3: $\text{Max}(-\infty, 2) \Rightarrow (2, 1) \Rightarrow 2$

The Minimax Algorithm: Example 1



Minimax Properties

- Optimal
 - Yes, against the perfect player.
- Time Complexity?
 - $O(b^m)$
- Space Complexity
 - $O(bm)$

α - β pruning: General Principle

The problem with minimax search is that the number of game states it has to examine is exponential in the number of moves. Unfortunately, we can not eliminate the exponent, but we can effectively cut it in half. The trick is that it is possible to compute the correct minimax decision without looking at every node in the game tree. That is we can borrow the idea of pruning in order to eliminate the large parts of the tree from consideration. The particular technique we will examine is called α - β pruning. When applied to a standard minimax tree, it returns the same moves as minimax would, but prunes away branches that cannot possibly influence the final decision.

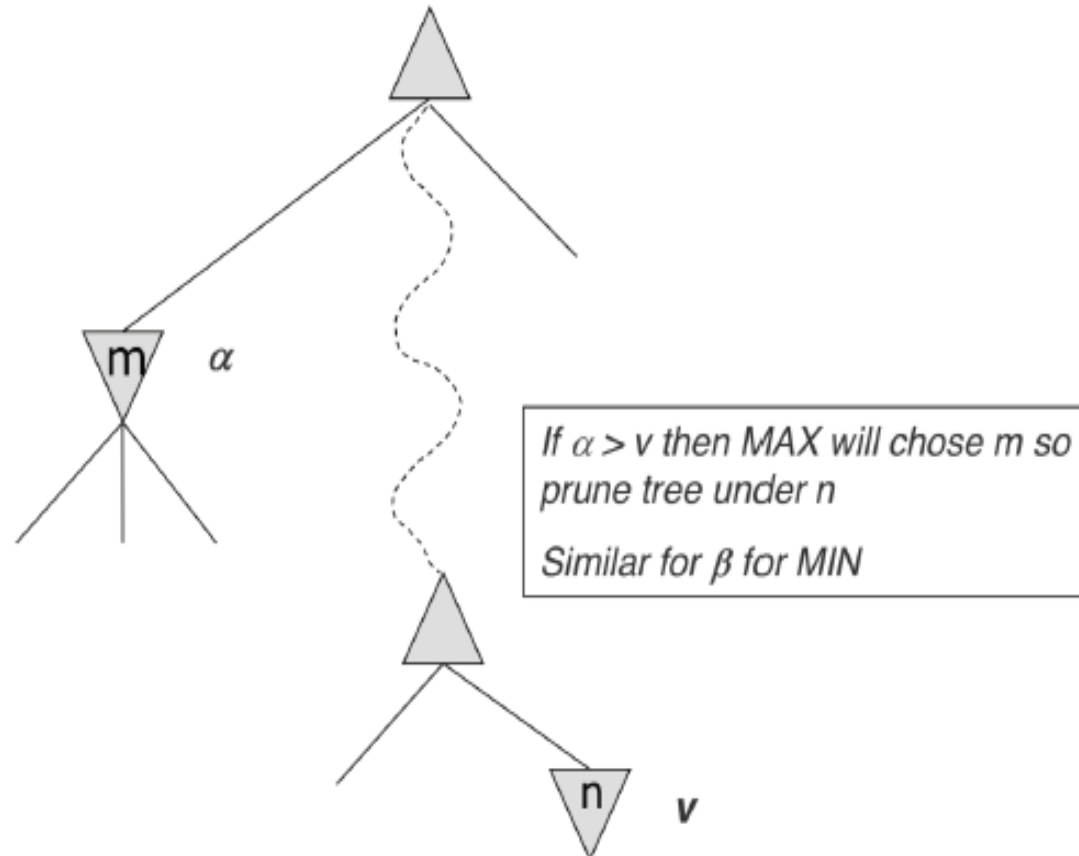
α - β pruning: general principle

Player

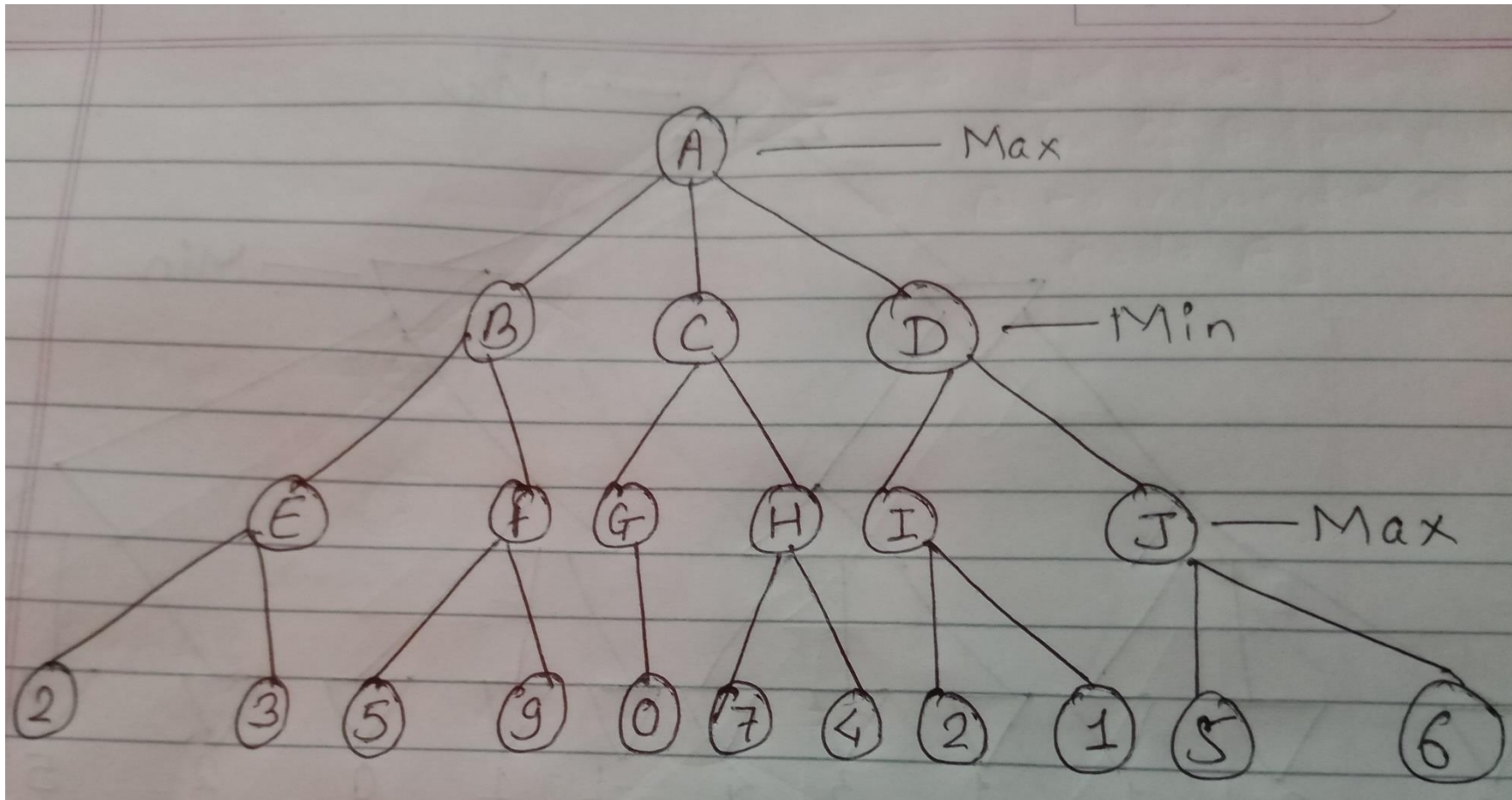
Opponent

Player

Opponent



α - β pruning: Example 1



α - β pruning: Example 1

→ Solⁿ

- For max, only value of α is changed.
 Worst case for max = $-\infty$
- For min, only value of β is changed.
 Worst case for min = $+\infty$

Step 1:

Max — (A)

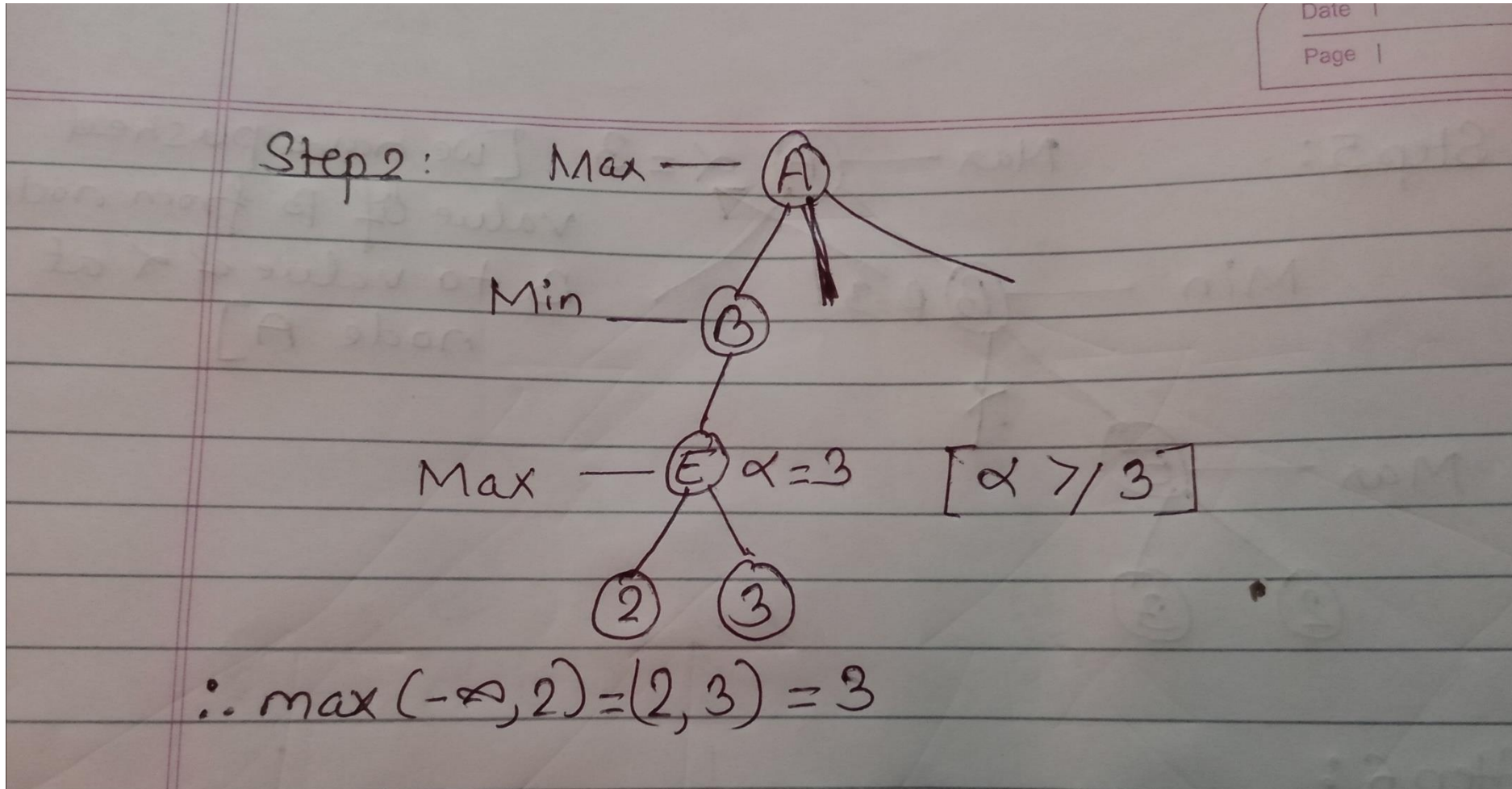
Min — (B)

Max — (C) $\alpha = 2$ $[\alpha \geq \beta, \text{i.e. min}^n 2 \text{ or max}]$

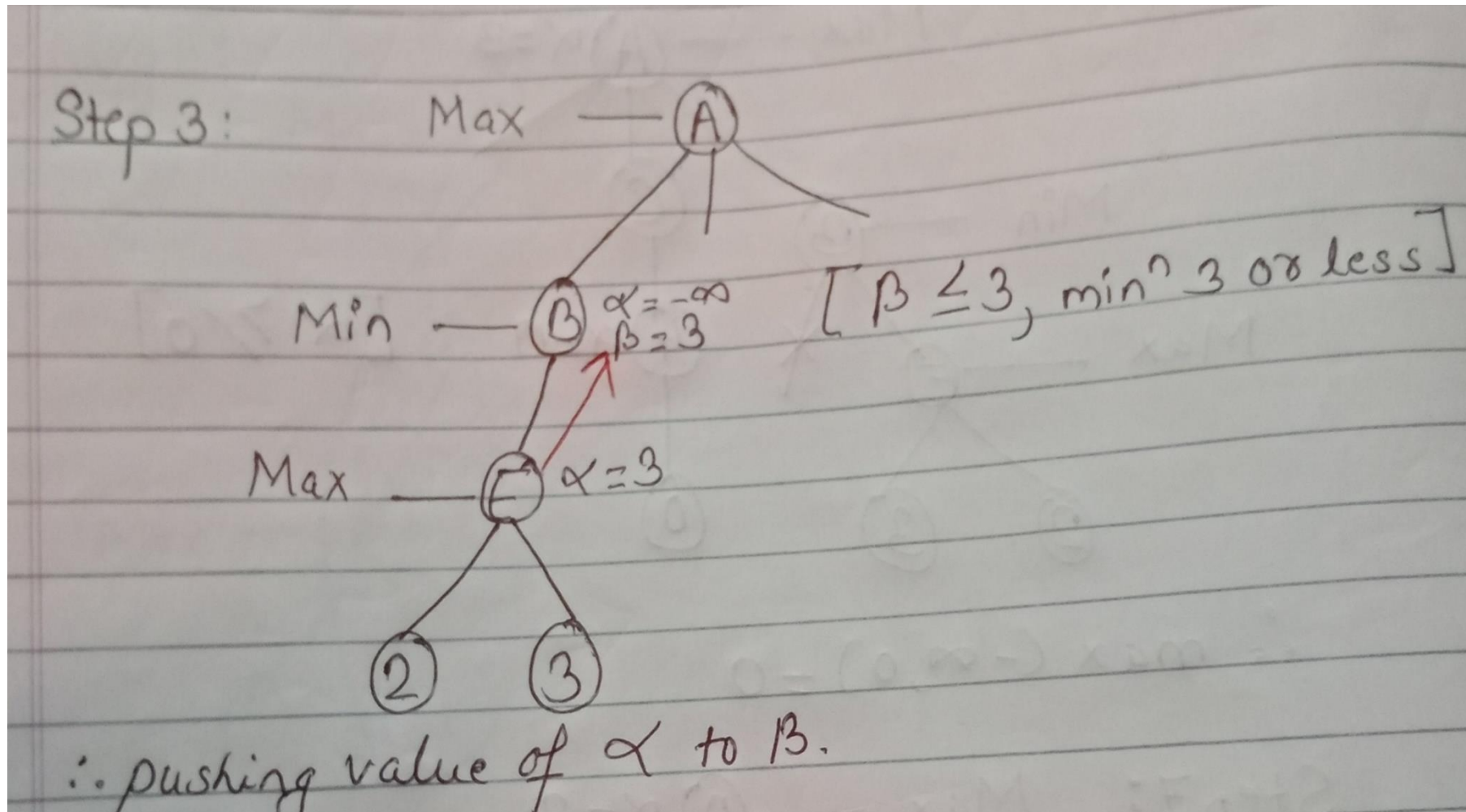
(2)

$\therefore \max(-\infty, 2) = 2$

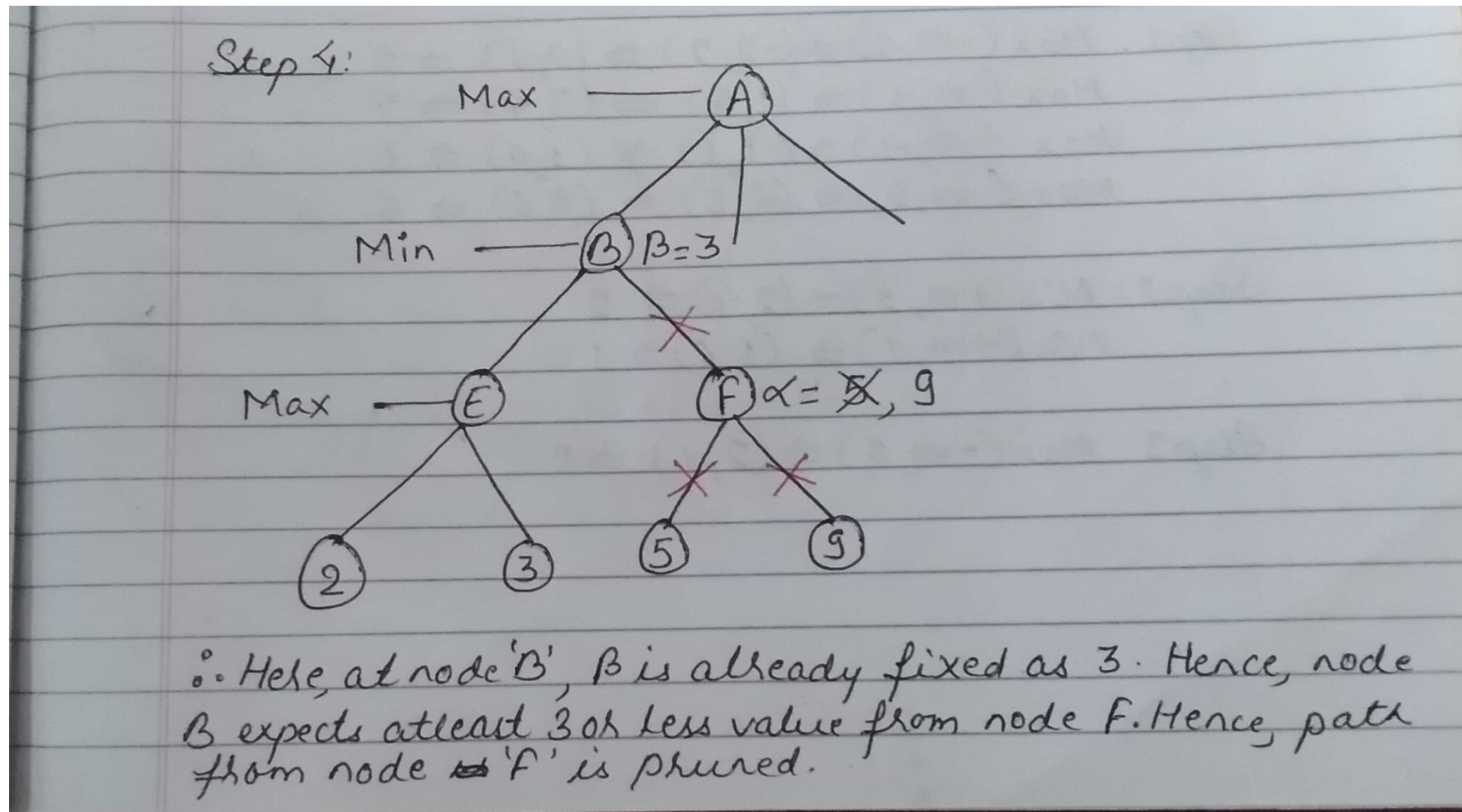
α - β pruning: Example 1



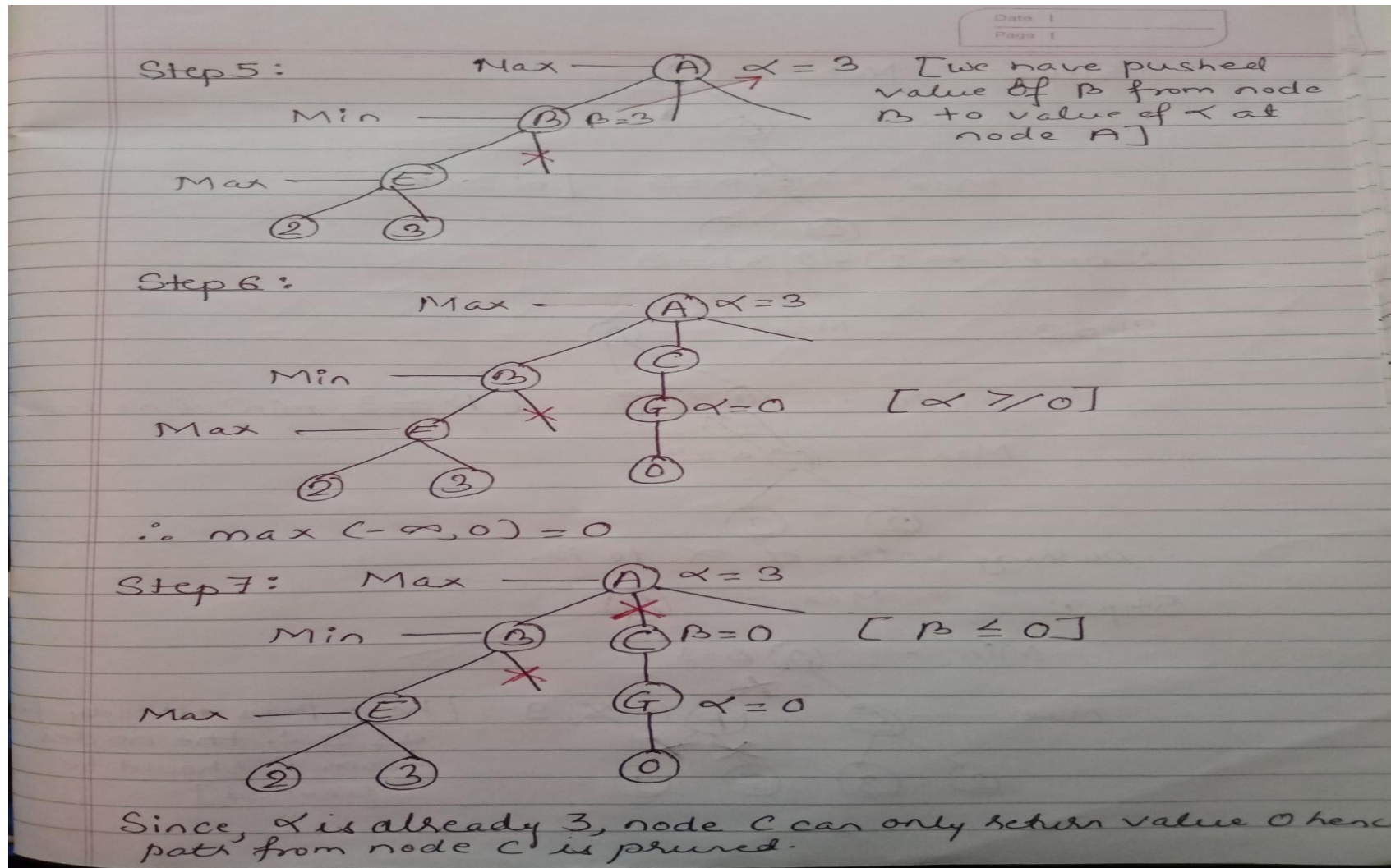
α - β pruning: Example 1



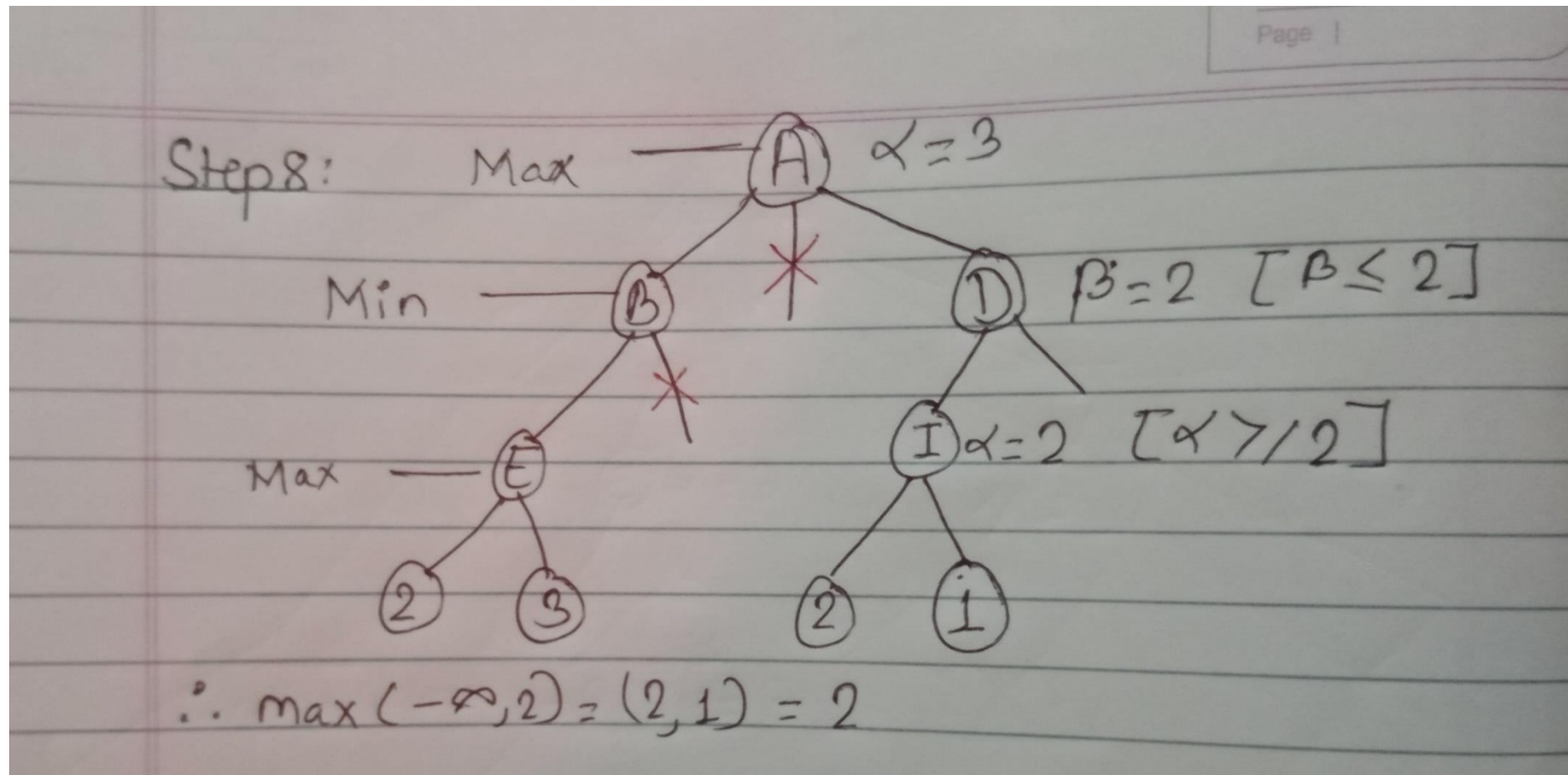
α - β pruning: Example 1



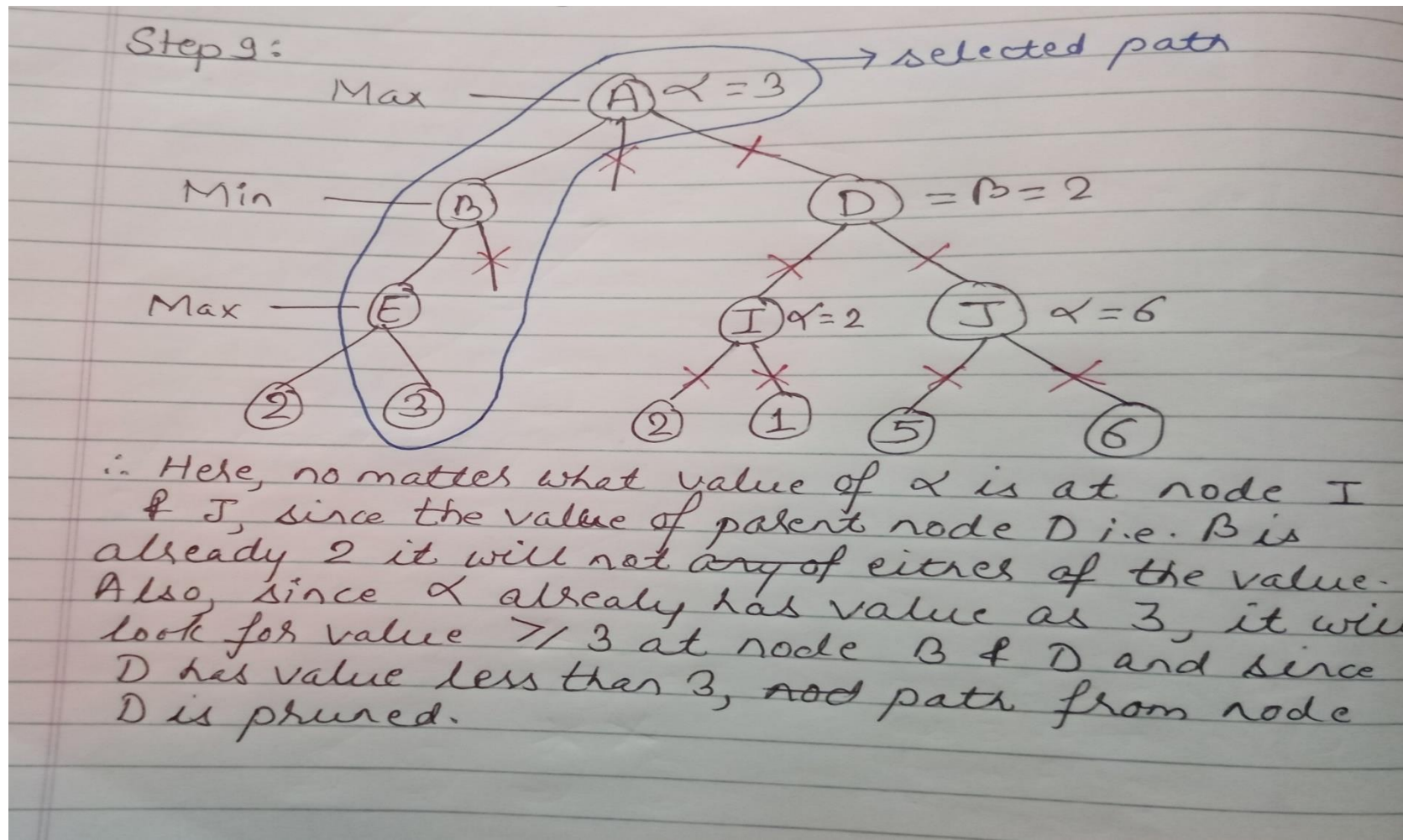
α - β pruning: Example 1



α - β pruning: Example 1



α - β pruning: Example 1



Properties of α - β pruning

- Pruning doesn't affect the final results.
- however good move ordering improves the effectiveness of pruning.
- With perfect ordering time complexity = $O(b^{m/2})$

References

- E. Rich and Knight, *Artificial Intelligence*, McGraw Hill, 1991.
- D. W. Patterson, *Artificial Intelligence and Expert Systems*, Prentice Hall, 2001.

Chapter 5

Knowledge, Reasoning and Planning

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MECE

Overview

- Knowledge Based Agents
- Propositional Logic
- First Order Logic
- Forward Chaining and Backward Chaining

Knowledge

- Knowledge is the information about a domain that can be used to solve problems in that domain.
- There are many other definitions such as:
- Knowledge is “information combined with experience, context, interpretation, and reflection. It is a high-value form of information that is ready to apply to decisions and actions.” (T. Davenport et al., 1998)
- Knowledge is “human expertise stored in person’s mind, gained through experience, and interaction with person’s environment.” (Sunasee and Sewery, 2002)
- Knowledge is information evaluated and organized by human mind so that it can be used purposefully, E.g. conclusions or explanations.” (Rousa, 2002)

Knowledge

Research literature classifies knowledge as follows:

Classification-based Knowledge	» Ability to classify information
Decision-oriented Knowledge	» Choosing the best option
Descriptive knowledge	» State of some world (heuristic)
Procedural knowledge	» How to do something
Reasoning knowledge	» What conclusion is valid in what situation?
Assimilative knowledge	» What its impact is?

Knowledge Based Agents

- Knowledge-based agents use a process of reasoning over an internal representation of knowledge to decide what actions to take.
- The central component of a knowledge-based agent is its knowledge base, or KB.
- A knowledge base is a set of sentences.
- Each sentence is expressed in a language called a knowledge representation language and represents some assertion about the world.
- A KB can initially contain some background information about the world, and a knowledge-based agent can add to the information in the KB through its observations of the world

Knowledge Based Agents

- There are two kinds of sentences:

Axiom: a sentence that is taken as being given and not derived from other sentences

Derived: a new sentence that is derived from others sentences.

- There must be a way to add new sentences to the knowledge base and a way to query what is known. The standard names for these operations are TELL and ASK, respectively.
- The process of deriving new sentences from old sentences is called inference.

Knowledge Representation

- Knowledge representation (KR) is the study of how knowledge about the world can be represented and what kinds of reasoning can be done with that knowledge.
- Knowledge Representation is the method used to encode knowledge in Intelligent Systems.
- The objective of knowledge representation is to express the knowledge about the world in a computer-tractable form.

Knowledge Representation using Logic

- Logic is defined as a formal language for expressing knowledge and ways of reasoning.
- Logic makes statements about the world which are true (or false).
- Compared to natural languages (expressive but context sensitive) and programming languages (good for concrete data structures but not expressive) logic combines the advantages of natural languages and formal languages.
- Logic is:
 - concise
 - unambiguous
 - context insensitive
 - expressive
 - effective for inferences

Knowledge Representation using Logic

- A logic is defined by the following:
 1. **Syntax** - describes the possible configurations that constitute sentences.
 2. **Semantics** - determines what facts in the world the sentences refer to i.e. the interpretation. Each sentence makes a claim about the world.
 3. **Proof theory** - set of rules for generating new sentences that are necessarily true given that the old sentences are true. The relationship between sentences is called **entailment**. The semantics link these sentences (representation) to facts of the world. The proof can be used to determine new facts which follow from the old.

Knowledge Representation using Logic

- 4. ***A set of sentences*** – A sentence is constructed from a set of primitives according to syntax rules.
- 5. ***A set of interpretations*** – An interpretation gives a semantic to primitives. It associates primitives with values.
- ***The valuation (meaning) function*** – Assigns a value (typically the truth value) to a given sentence under some interpretation. sentence $\times$ interpretation $\rightarrow \{\text{True}, \text{False}\}$.

Knowledge Representation using Logic

- **Types of logic:**

- Different types of logics are:
- Propositional logic
- First-order logic (First –order Predicate Logic).

Propositional Logic

- A proposition is a declarative sentence to which only one of the “Truth value” (i.e. TRUE or FALSE) can be assigned (but not both).
- Hence, the propositional logic is also called Boolean logic. When a proposition is true, we say that its truth value is T, otherwise its truth value is F.
- For example:
 - The square of 4 is 16 $\rightarrow$ T
 - The square of 5 is 7 $\rightarrow$ F

Atomic Sentences(Simple)

- The atomic sentences consist of a single proposition symbol.
- Each such symbol stands for a proposition that can be true or false.
- We use symbols that start with an uppercase letter and may contain other letters or subscripts, for example: P, Q,R,S, etc.
- For example:

P = Sun rises in West. (False sentence)

Complex Sentences (Molecular or Combined or Compound)

- The two or more statements connected together with some logical connectives such as AND ($\wedge$), OR ($\vee$), Implication ($\rightarrow$), etc.
- eg: $(A \wedge B)$, $(A \vee B)$, $(A \Rightarrow B)$, $(A \Leftrightarrow B)$, $(A \vee B) \wedge (A \vee \neg B)$
- A sentence (well formed formula) is defined as follows:
 - A symbol is a sentence
 - If S is a sentence, then $\neg S$ is a sentence
 - If S is a sentence, then (S) is a sentence
 - If S and T are sentences, then $(S \wedge \neg T)$, $(S \vee T)$, $(S \wedge T)$, and $(S \Leftrightarrow T)$ are sentences.

Formal Logic Connectives

- A set of connectives:

$\wedge$	AND
$\vee$	OR
$\neg$	NOT
$\Rightarrow$	implies
$\Leftrightarrow$	mutual implication

Formal Logic Connectives

- Negation (NOT):
 - Given a proposition p , negation operator ($\neg p$) is used to get negation of p denoted by $\neg p$ called “not p ”.
 - Example: Negation of the proposition “I love birds” is “I do not love birds” if the sentence I love birds is denoted by p then its negation is denoted by $\neg p$.
 - Truth Table:

p	$\neg p$
T	F
F	T

Formal Logic Connectives

- Conjunction (AND)

- Let p and q be propositions. The conjunction of p and q , denoted by $p \wedge q$, is the proposition “ p and q .”
- The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.
- Example: If we have propositions p = “Ram is intelligent” and q = “Ram is diligent” the conjunction of $p \wedge q$ is Ram is intelligent and diligent.
- This proposition is true only when Ram is intelligent and he is diligent also, false otherwise.

- Truth Table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Formal Logic Connectives

- Disjunction (OR)
 - Given two propositions p and q , the proposition “ p or q ” denoted by $p \vee q$ is the proposition that is false whenever both the propositions p and q are false, true otherwise.
 - Example: If we have propositions p = “Ram is intelligent” and q = “Ram is diligent” the disjunction of $p \vee q$ is Ram is intelligent or he is diligent.
 - This proposition is false only when Ram is not intelligent and not diligent, true otherwise.
 - Truth Table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Formal Logic Connectives

- Implication (Conditional Statements)
 - Given two propositions p and q , “if p , then q ”, the proposition implication $p \rightarrow q$ is the proposition that is false when p is true and q is false, true otherwise.
 - p is called “hypothesis” or “antecedent” or “premise” and q is called “conclusion” or “consequence”.
 - Example: p = “today is Sunday” q = “it is hot” then the implication can be “if today is Sunday then it is hot today” or “today is Sunday only if it is hot today”.

Formal Logic Connectives

- Implication (Conditional Statements)
- Variety of terminology is used to express $p \rightarrow q$.

“if p , then q ”

“if p , q ”

“ p is sufficient for q ”

“ q if p ”

“ q when p ”

“a necessary condition for p is q ”

“ q unless $\neg p$ ”

“ p implies q ”

“ p only if q ”

“a sufficient condition for q is p ”

“ q whenever p ”

“ q is necessary for p ”

“ q follows from p ”

Truth Table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Formal Logic Connectives

- Biconditional/Bi-implication

- Let p and q be propositions. The bi-conditional statement $p \leftrightarrow q$ is the proposition “ p if and only if q .”
- The bi-conditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.
- There are some other common ways to express $p \leftrightarrow q$: “ p is necessary and sufficient for q ” “if p then q , and conversely” “ p iff q .”
- Example: - Let p be the statement “You can take the flight,” and let q be the statement “You buy a ticket.” Then $p \leftrightarrow q$ is the statement? “You can take the flight if and only if you buy a ticket.”

Truth Table:

P	Q	$P \leftrightarrow Q$
false	false	true
false	true	false
true	false	false
true	true	true

Truth Tables For Logic Connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	false	true	false	false	true	true
false	true	true	false	true	true	false
true	false	false	false	true	false	false
true	true	false	true	true	true	true

Logic

Converse:	If $p \rightarrow q$ is an implication, then its converse is $q \rightarrow p$
Inverse:	If $p \rightarrow q$ is an implication, then its inverse is $\neg p \rightarrow \neg q$
Contrapositive:	If $p \rightarrow q$ is an implication, then its contrapositive is $\neg q \rightarrow \neg p$

Logical equivalence

- Two proposition p and q are logically equivalent and written as if both p and q have identical truth values:

$$\neg(p \wedge q) \equiv (\neg p \vee \neg q)$$

The following logical equivalences apply to any statements; the p's, q's and r's can stand for atomic statements or compound statements.

i. Double Negative Law

$$\neg(\neg p) \equiv p$$

ii. De Morgan's Laws

$$\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$$

$$\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$$

iii. Distributive Laws

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

Logic

- **Tautology:** Tautology is notation in formal logic which is always true (valid).
- **Contradiction (Unsatisfiable):** notation in formal logic which is always false.
- **Satisfiable:** If at least one sentence is true in the set.

Converting the sentences into propositional logic:

- Example 1: - “You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

➤ Solution,

a=You can access the Internet from campus

c= You are a computer science major

f =You are a freshman

$a \rightarrow (c \vee \neg f)$

- Example 2: - If it is not snowing and I have a time, then I will go to beach.

➤ Solution,

p= It is snowing

q= I have a time

r= I will go to beach

$(\neg p \wedge q) \rightarrow r$

- Example 3: Write implication, converse, inverse, negation, contra positive of the following integration. “The program is well structured only if it is readable”.

p = “the program is well structured”

q = “the program is readable”

Implication: ($p \rightarrow q$)

- If the program is well structured, then it is readable.

➤ Converse: ($q \rightarrow p$)

- If the program is readable, then it is well structured.

➤ Inverse: ($\neg p \rightarrow \neg q$)

- If the program is not well structured, then it is not readable.

➤ Contrapositive: ($\neg q \rightarrow \neg p$)

- If the program is not readable, then it is not well structured.

➤ Negation of p : ($\neg p$)

- The program is not well structured.

Logical Equivalences

- Two propositions p and q are logically equivalent if any of the truth values of each statement have the same truth.

$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$	commutativity of $\wedge$
$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$	commutativity of $\vee$
$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$	associativity of $\wedge$
$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$	associativity of $\vee$
$\neg(\neg\alpha) \equiv \alpha$	double-negation elimination
$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$	contraposition
$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$	implication elimination
$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$	biconditional elimination
$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$	De Morgan
$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$	De Morgan
$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$	distributivity of $\wedge$ over $\vee$
$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$	distributivity of $\vee$ over $\wedge$

Q. There are two restaurants next to each other.

One has a sign board as: “Good food is not cheap”.

The other has a sign board as “Cheap food is not good”.

Are both the sign board saying the same thing?

G= "Food is Good"

C= "Food is Cheap"

Sentence 1:- "Good food is not cheap" can be symbolically written as $G \rightarrow \neg C$

Sentence 2:- "Cheap food is not good" can be symbolically written as $C \rightarrow \neg G$

Now, The Truth Table is:

G	C	$\neg G$	$\neg C$	$G \rightarrow \neg C$	$C \rightarrow \neg G$
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

Since, $G \rightarrow \neg C$ and $C \rightarrow \neg G$ are logically equivalent. So both are saying same thing.

Inference rules in propositional logic

Rule of inference	Tautology	Name
$\frac{p \rightarrow q}{p} \therefore q$	$[p \wedge (p \rightarrow q)] \rightarrow q$	Modus ponens
$\frac{\neg q}{p \rightarrow q} \therefore \neg p$	$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$	Modus tollens
$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\frac{p \vee q \quad \neg p}{\therefore q}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism

Inference rules in propositional logic

$\frac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition
$\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification
$\frac{p}{q} \quad \frac{q}{\therefore p \wedge q}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\frac{p \vee q}{\neg p \vee r} \quad \frac{\neg p \vee r}{\therefore q \vee r}$	$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (p \vee r)$	Resolution

Proofs using Propositional Logic

Example 1: “If you send me an e-mail message then I will finish writing the program”, “If you do not send me an e-mail message then I will go to sleep early”, and “If I go to sleep early then I will wake up feeling refreshed”. Lead to the conclusion “If I do not finish writing the program then I will wake up feeling refreshed”.

Solution:

Let

p = “You send me an e-mail message”

q = “I will finish writing the program”

r = “I will go to sleep early”

s = “I will wake up feeling refreshed”

Hypothesis:

a. $p \rightarrow q$

b. $\neg p \rightarrow r$

c. $r \rightarrow s$

Conclusion: $\neg q \rightarrow s$

Steps	Operations	Reasons
1	$p \rightarrow q$	Given hypothesis
2	$\neg q \rightarrow \neg p$	Using contra positive on 1
3	$\neg p \rightarrow r$	Given hypothesis
4	$\neg q \rightarrow r$	Using hypothetical syllogism on 2 and 3
5	$r \rightarrow s$	Given hypothesis
6	$\neg q \rightarrow s$	Using hypothetical syllogism on 4 and 5

Hence the given hypotheses lead to the conclusion $\neg q \rightarrow s$

Example 2: Use rules of inference to show that the hypotheses “Randy works hard,” “If Randy works hard, then he is a dull boy,” and “If Randy is a dull boy, then he will not get the job” imply the conclusion “Randy will not get the job.”

SOLUTION

Let us assume:

p = “Randy works hard”

q = “Randy is a dull boy”

r = “Randy will get the job”

We can then rewrite the given statements (1), (2) and (3) using the above interpretations.

Step		Reason
1.	p	<i>Premise</i>
2.	$p \rightarrow q$	<i>Premise</i>
3.	$q \rightarrow \neg r$	<i>Premise</i>
4.	q	Modens ponens from (1) and (2)
5.	$\neg r$	Modens ponens from (3) and (4)

Limitation of Propositional Logic

- We cannot represent relations like ALL, some, or none with propositional logic.
- Example:
 - All the girls are intelligent.
 - Some apples are sweet.
- **Propositional logic has limited expressive power.**

Predicate Logic

- Predicate is a part of declarative sentences describing the properties of an object or relation among objects.
- For example: “is a student” is a predicate as ‘A is a student’ and ‘B is a student’.
- A predicate logic is a formal system that uses objects/ variables and quantifiers ($\forall$, $\exists$) to formulate propositions.

First Order Predicate Logic (FOPL)

- Whereas propositional logic assumes world contains facts, first-order logic (like natural language) assumes the world contains:
 - **Objects:** wheel, door, body, engine, seat, car, passenger, driver, people, houses, numbers, theories, Ronald McDonald, colors, baseball games, wars, centuries, ...
 - **Predicates:** Inside(car, passenger), Beside(driver, passenger), BrotherOf(person, person), BiggerThan(object, object), Inside(), part of(), HasColor(), OccurredAfter(), Owns(), ComesBetween(), ...
 - **Functions:** ColorOf(car), FatherOf(person), BestFriend(person), ThirdInningOf(), OneMoreThan(), EndOf(), ...
- Functions are relations with single value for each object



First Order Predicate Logic (FOPL)

Semantics/ Interpretation

there is a correspondence between

- functions, which return values
- predicates, which are true or false

Function: $\text{father_of}(\text{Mary}) = \text{Bill}$

Predicate: $\text{father_of}(\text{Mary}, \text{Bill})$

First Order Predicate Logic (FOPL)

- Atomic Sentences:

Atomic Sentences $\rightarrow$ Predicate (Term,.....)

Term $\rightarrow$ Function (Term,.....) | Constant | Variable

Example:

- Brother (Richard, John) which means Richard is the brother of John.
- Married(Father(Richard), Mother(John)) which means father of Richard and Mother of John are married.

First Order Predicate Logic (FOPL)

- Complex Sentences:

- Complex sentences are made from atomic sentences using connectives.
- E.g. $\neg S$, $S1 \wedge S2$, $S1 \vee S2$, etc
- Example: $\text{Colleague}(\text{Ram}, \text{Shyam}) \wedge \text{Colleague}(\text{Shyam}, \text{Ram})$ which means Shyam is the colleague of Ram and also Ram is colleague of Shyam.
- $\text{King}(\text{Richard}) \vee \text{King}(\text{John})$ which means either Richard or John is a king.

Quantification

- A quantifier is a symbol that permits one to declare the range or scope of variables in a logical expression.
- The process of binding propositional variable over a given domain is called quantification.
- Two common quantifier are the existential quantifier (“there exists or for some or at least one”) and universal quantifier (“for all or for each or for any or for every and or for arbitrary”).

Universal Quantifier ($\forall$: For All)

- It is denoted by $\forall$ and used for universal quantification.
- The universal quantification of $p(x)$ denoted by $\forall x$ $p(x)$ is proposition that is true for all values in universal set
- The universal quantifier is read as:
 - For all x , $p(x)$ holds
 - For each x , $p(x)$ holds
 - For every x , $p(x)$ holds



Existential Quantifier($\exists$: For Some)

- It is denoted by $\exists$ and used for existential quantification.
- The existential quantification of $p(x)$ denoted by $\exists x p(x)$ is proposition that is true for some values in universal set.
- The existential quantifier is read as:
 - There is an x , such that $p(x)$
 - There is at least one x such that $p(x)$
 - For some x , $p(x)$

Q. Assume that

P (x) denotes “x is an accountant.”

Q (x) denotes “x owns a maruti.”

Now, represent the following statement symbols.

a) All accountants own maruti.

Meaning: For all x, if x is an accountant, then x owns maruti

$$\underbrace{\forall x}_{\forall x} \underbrace{\text{if } x \text{ is an accountant}}_{p(x)} \underbrace{\text{then } x \text{ owns maruti}}_{q(x)} \\ \Rightarrow \forall x (p(x) \rightarrow q(x))$$

b) Some accountants own maruti

Meaning: For some x, x is an accountant and x owns maruti

$$\underbrace{\text{For some } x}_{\exists x} \underbrace{\text{ } x \text{ is an accountant}}_{p(x)} \underbrace{\text{and } x \text{ owns maruti}}_{q(x)} \\ \Rightarrow \exists x (p(x) \wedge q(x))$$



C) All owners of maruti are accountants.

Meaning: For all x, if x is a owner of maruti, then x is an accountant.

$$\Rightarrow \forall x (q(x) \rightarrow p(x))$$

d) Someone who owns a Maruti, is an accountant.

Meaning: For some x, who owns a maruti and x is an accountant.

$$\Rightarrow \exists x (q(x) \wedge p(x))$$

Nested Quantifiers

- We often express more complex sentences using multiple quantifiers.

- Examples:

1. Brothers are siblings.

$$\forall x \forall y \text{ Brother}(x,y) \rightarrow \text{Sibling}(x,y)$$

- To say siblinghood is a symmetric relationship we can write:

$$\forall x,y \text{ Sibling}(x,y) \Leftrightarrow \text{Sibling}(y,x)$$

2. Everybody loves Somebody, which means that for every person, there is someone that person loves.

$$\Rightarrow \forall x \exists y \text{ Loves}(x, y)$$

Nested Quantifiers

3. There is someone who is loved by everyone.

➤ $\exists y \forall x \text{ Loves}(x, y)$

- The order of quantification is therefore important.

$\forall x (\exists y \text{ Loved}(x, y))$ says that everyone has a particular property, namely, the property that somebody loves them.

$\exists x (\forall y \text{ Loved}(x, y))$ says that someone in the world has a particular property, namely the property of being loved by everybody.

Connection between $\exists$ and $\forall$

- The two quantifiers are connected with each other through negation.
- “Everyone likes ice cream” means that there is no one who does not like ice cream.
- Therefore,

$\forall x \text{ Likes } (x, \text{IceCream})$ is equivalent to $\neg \exists x \neg \text{Likes } (x, \text{IceCream})$

- Because $\forall$ is really a conjunction over the universe of objects and $\exists$ is a disjunction, they do obey de Morgan's rules.

$$\forall x \neg P \equiv \neg \exists x P$$

$$\neg \forall x P \equiv \exists x \neg P$$

$$\forall x P \equiv \neg \exists x \neg P$$

$$\exists x P \equiv \neg \forall x \neg P$$

Properties of Quantifiers

- From above examples we can derive following properties of quantifiers:

- $\forall x \forall y$ is same as $\forall y \forall x$
- $\exists x \exists y$ is same as $\exists y \exists x$
- $\exists x \forall y$ is not same as $\forall y \exists x$ because :

$$\exists x \forall y \text{ Loves}(x, y)$$

"There is a person who loves everyone in the world"

$$\forall y \exists x \text{ Loves}(x, y)$$

"Everyone in the world is loved by at least one person"

Translating English into FOPL

1. Every gardener likes the sun.
2. You can fool some of the people all of the time.
3. You can fool all of the people some of the time.
4. All purple mushrooms are poisonous.
5. No purple mushroom is poisonous.

Translating English to FOL

- Every gardener likes the sun.
 $\forall x \text{ gardener}(x) \Rightarrow \text{likes}(x, \text{Sun})$
- You can fool some of the people all of the time.
 $\exists x \forall t (\text{person}(x) \wedge \text{time}(t)) \Rightarrow \text{can-fool}(x, t)$

Translating English to FOL

- You can fool all of the people some of the time.

$$\forall x \exists t (\text{person}(x) \wedge \text{time}(t)) \Rightarrow \text{can-fool}(x, t)$$

$$\exists t \forall x (\text{person}(x) \wedge \text{time}(t)) \Rightarrow \text{can-fool}(x, t)$$

- All purple mushrooms are poisonous.

$$\forall x (\text{mushroom}(x) \wedge \text{purple}(x)) \Rightarrow \text{poisonous}(x)$$

Translating English to FOL...

- No purple mushroom is poisonous.

$$\neg(\exists x) \text{ purple}(x) \wedge \text{mushroom}(x) \wedge \text{poisonous}(x)$$

or, equivalently,

$$(\forall x) (\text{mushroom}(x) \wedge \text{purple}(x)) \Rightarrow \neg \text{poisonous}(x)$$

Q. Convert into FOPL

- a. All men are people.**
- b. Marcus was Pompeian.**
- c. All Pompeian were Roman.**
- d. Ram tries to assassinate Hari.**
- e. All Romans were either loyal to caser or hated him.**
- f. Socrates is a man. All men are mortal; therefore Socrates is mortal.**
- g. Some student in this class has studied mathematics.**

a. All men are people.

$\Rightarrow \forall x \text{ MAN}(x) \rightarrow \text{PEOPLE}(x)$

b. Marcus was Pompeian.

$\Rightarrow \text{POMPEIAN}(\text{Marcus})$

c. All Pompeian were Roman.

$\Rightarrow \forall x \text{ POMPEIAN}(x) \rightarrow \text{ROMAN}(x)$

d. Ram tries to assassinate Hari.

$\Rightarrow \text{ASSASSINATE}(\text{Ram}, \text{Hari})$

e. **All Romans were either loyal to caser or hated him.**

$$\Rightarrow \forall x \text{ ROMAN}(x) \rightarrow \text{LOYAL}(x, \text{caser}) \vee \text{HATES}(x, \text{caser})$$

f. **Socrates is a man. All men are mortal; therefore Socrates is mortal.**

$$\Rightarrow \text{MAN}(\text{Socrates}), \forall x \text{ MAN}(x) \rightarrow \text{MORTAL}(x), \text{MORTAL}(\text{Socrates})$$

g. **Some student in this class has studied mathematics.**

$\Rightarrow$ Let

- $S(x) = \text{"x is a student in this class"}$
- $M(x) = \text{"x has studied mathematics"}$

Hence, required expression is: $\exists x [S(x) \wedge M(x)]$

Rules of Inference for Quantified Statements

a. Universal Instantiation

$$\frac{\forall x p(x)}{\therefore p(d)}$$

Where d is in the domain of discourse D .

For example: If all balls in a box are red then any randomly drawn ball is also red .

b. Universal Generalization

$$\frac{p(d)}{\therefore \forall x p(x)}$$

Where, d is the domain of discourse D .

For example: If all the ball in a box are taken one by one in randomly manner and if all are red then we conclude that all balls in the box are red.

c. Existential Instantiations

$$\frac{\exists x p(x)}{\therefore p(d)}$$

For some d in the domain of discourse.

For example: If some balls in a box are red then resulting ball after experiment will also be red.

d. Existential Generalization

$$\frac{p(d)}{\therefore \exists x p(x)}$$

For some d in the domain of discourse.

For example: If we take only one random experiment for the ball drawn and apply the result of ball to the domain.

Q. Given Expression: All men are mortal. Einstein is a man. Prove that “Einstein is mortal” using FOPL.

Solution: Let

$M(x) = \text{“}x \text{ is a man”}$

$N(x) = \text{“}x \text{ is mortal”}$

Hypothesis: $\forall x [M(x) \rightarrow N(x)], M(\text{Einstein})$

Conclusion: $N(\text{Einstein})$

Steps	Operations	Reasons
1.	$\forall x [M(x) \rightarrow N(x)]$	Given Hypothesis
2.	$M(\text{Einstein}) \rightarrow N(\text{Einstein})$	Using universal instantiation on 1
3.	$M(\text{Einstein})$	Given Hypothesis
4.	$N(\text{Einstein})$	Using modus pollens on 2 and 3

Hence the given hypotheses lead to the conclusion “Einstein is mortal”.



Q. Given Expression: “Lions are dangerous animals”, and “There are lions”. Prove that “There are dangerous animals” using FOPL.

Solution: Let

$D(x)$ = “x is a dangerous animal”

$L(x)$ = “x is a lion”

Hypothesis: $\forall x [L(x) \rightarrow D(x)], \exists x L(x)$

Conclusion: $\exists x D(x)$

Steps	Operations	Reasons
1.	$\forall x [L(x) \rightarrow D(x)]$	Given Hypothesis
2.	$L(a) \rightarrow D(a)$	Using universal instantiation on 1
3.	$\exists x L(x)$	Given Hypothesis
4.	$L(a)$	Using existential instantiation on 3
5.	$D(a)$	Using modus <u>pollenson</u> 2 and 4
6.	$\exists x D(x)$	Using existential generalization on 5

Hence the given hypotheses lead to the conclusion “There are dangerous animals”



Activate \n
Go to PC set

Rule Based Deduction System

- Rule-based systems (RBS) provide automatic problem-solving tools for capturing the human expertise and decision making.
- RBS are means for codifying the problem solving of human experts.
- Experts typically express most of their problem-solving techniques in terms of antecedent-consequent rules.
- The main properties of rule-based systems are:
 - They incorporate practical human knowledge in if-then rules.
 - Their skill increases proportionally to the enlargement of their knowledge.
 - They can solve a wide range of potentially complex problems by selecting relevant rules and then combining the results.
 - They adaptively determine the best sequence of rules to examine.

Forward Chaining

- Forward chaining is one of the two main methods of reasoning when using inference rules.
- Described logically as repeated application of modus ponens.
- Forward chaining is a popular implementation strategy for expert systems, business and production rule systems.
- Forward chaining starts with the available data and uses inference rules to extract more data until a goal is reached.

Forward Chaining

- An inference engine using forward chaining searches the inference rules until it finds one where the antecedent (If clause) is known to be true. When such a rule is found, the engine can conclude, or infer, the consequent (Then clause), resulting in the addition of new information to its data.
- Inference engines will iterate through this process until a goal is reached.
- Because the data determines which rules are selected and used, this method is called data-driven.

Forward Chaining: Example

Example

- The rule base contains the following four rules:
 - If X croaks and eats flies - Then X is a frog.
 - If X chirps and sings - Then X is a canary.
 - If X is a frog - Then X is green.
 - If X is a canary - Then X is yellow.

Goal: conclude the color of a pet named Fritz, given that he croaks and eats flies.

Forward Chaining: Example

- **Inference Mechanism:**
- This rule base would be searched and the first rule would be selected, because its antecedent (If Fritz croaks and eats flies) matches our data. Now the consequents (Then X is a frog) is added to the data.
- The rule base is again searched and this time the third rule is selected, because its antecedent (If Fritz is a frog) matches our data that was just confirmed. Now the new consequent (Then Fritz is green) is added to our data.
- Nothing more can be inferred from this information, but we have now accomplished our goal of determining the color of Fritz.

Backward Chaining

- Backward chaining (or backward reasoning) is an inference method that can be described as working backward from the goal(s).
- It is used in automated theorem provers, proof assistants and other artificial intelligence applications.
- Rules are based on the modus ponens inference rule.
- It is one of the two most commonly used methods of reasoning with inference rules and logical implications.

Backward Chaining

- Backward chaining starts with a list of goals and works backwards from the consequent to the antecedent to see if there is data available that will support any of these consequents.
- An inference engine using backward chaining would search the inference rules until it finds one which has a consequent (Then clause) that matches a desired goal.
- If the antecedent (If clause) of that rule is not known to be true, then it is added to the list of goals.
- Because the list of goals determines which rules are selected and used, this method is called goal-driven, in contrast to data-driven forward-chaining inference. The backward chaining approach is often employed by expert systems.

Backward Chaining: Example

- **Example**
- The rule base contains the following four rules:
 - If X croaks and eats flies - Then X is a frog.
 - If X chirps and sings - Then X is a canary.
 - If X is a frog - Then X is green.
 - If X is a canary - Then X is yellow.

Goal: conclude the color of a pet named Fritz, given that he croaks and eats flies.

Backward Chaining: Example

- **Inference Mechanism:**
- This rule base would be searched and the third and fourth rules would be selected, because their consequents (Then Fritz is green, Then Fritz is yellow) match the goal (to determine Fritz's color).
- It is not yet known that Fritz is a frog, so both the antecedents (If Fritz is a frog, If Fritz is a canary) are added to the goal list.
- The rule base is again searched and this time the first two rules are selected, because their consequents (Then X is a frog, Then X is a canary) match the new goals that were just added to the list.
- The antecedent (If Fritz croaks and eats flies) is known to be true and therefore it can be concluded that Fritz is a frog, and not a canary.
- The goal of determining Fritz's color is now achieved (Fritz is green if he is a frog, and yellow if he is a canary, but he is a frog since he croaks and eats flies; therefore, Fritz is green).

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- E. Rich and Knight, *Artificial Intelligence*, McGraw Hill, 1991.
- D. W. Patterson, *Artificial Intelligence and Expert Systems*, Prentice Hall, 2001.

Chapter 6

Machine Learning

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MECE

Overview

- Forms and Learning
- Supervised Learning
- The Theory of Everything
- Linear Regression and Classification
- Developing Machine Learning Systems

Concepts of Learning

Machine Learning

- Machine learning is a branch of AI that uses algorithm to allow computer to evolve behaviors based on data collected from databases or gathered through sensors.
- Machine learning focuses on prediction based on known properties learned from the training data.
- The performance is usually evaluated learned with respect to reproduce known knowledge.

Concepts of Learning

Learning

- Learning is the process of acquiring new or modifying existing knowledge, behaviors, skills, values, or preferences.
- Learning is the improvement of performance with experience over time.
- Learning element is the portion of a learning AI system that decides how to modify the performance element and implements those modifications.
- We all learn new knowledge through different methods, depending on the type of material to be learned, the amount of relevant knowledge we already possess, and the environment in which the learning takes place.

Learning

- Learning involves 3 factors:

Changes: Learning changes the learner. For machine learning the problem is determining the nature of these changes and how to best represent them.

Generalization: Learning leads to generalization. Performance must improve not only on the same task but on similar tasks.

Improvement: Learning leads to improvements. Machine learning must address the possibility that changes may degrade performance and find ways to prevent it.

What is Machine Learning?

- The subfield of computer science that “gives computers the ability to learn without being explicitly programmed”.

(Arthur Samuel, 1959)

- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E .”

(Tom Mitchell, 1997)

- Using data for answering questions.

Learning Methods

- There are two different kinds of information processing which must be considered in a machine learning system.
 - *Inductive learning* is concerned with determining general patterns, organizational schemes, rules, and laws from raw data, experience or examples.
 - *Deductive learning* is concerned with determination of specific facts using general rules or the determination of new general rules from old general rules. $A=B$, $B=C$, then $A=C$.
 - Deductive learning is accomplished through a sequence of deductive inference steps using known facts.
 - From the known facts, new facts or relationships are logically derived.
 - Deductive learning usually requires more inference than the other methods.

Features of Machine Learning

- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data-driven technology.
- Machine learning is much similar to data mining as it also deals with the huge amount of the data.

Why Machine learning is important?

- Rapid increment in the production of data.
- Solving complex problems, which are difficult for a human.
- Decision making in various sector including finance.
- Finding hidden patterns and extracting useful information from data.

Learning may be classified into three categories:

Supervised

Unsupervised

Reinforcement

Types of Learning

Supervised

Learn through examples of which we know the desired output (what we want to predict).

The training data is labeled.

Unsupervised

Is this a cat or a dog?

Are these emails spam or not?

Reinforcement

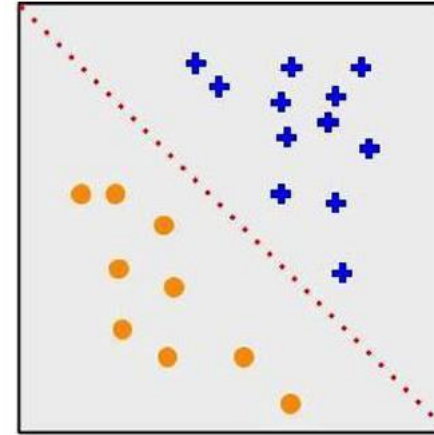
Predict the market value of houses, given the square meters, number of rooms, neighborhood, etc.

Types of Learning

- Supervised Machine Learning

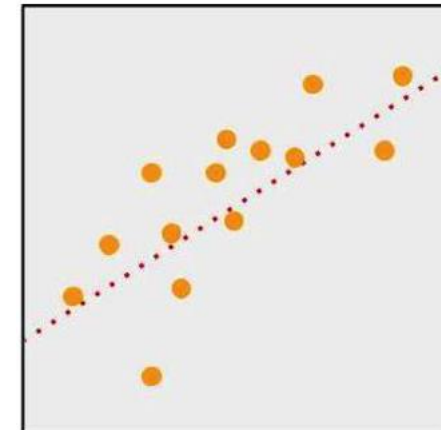
1. Classification

- Output is a discrete variable(yes/no)
- Example: cat/dog, spam filtering, etc



2. Regression

- Output is continuous (numerical)
- Example: price, temperature, housing price, etc



Types of Learning

Supervised

There is no *desired* output. Learn something about the data.
Latent relationships.

Unsupervised

The training data is unlabeled.

I have photos and want to put them in 20 group

Reinforcement

I want to find anomalies in the credit card usage patterns of my customers.

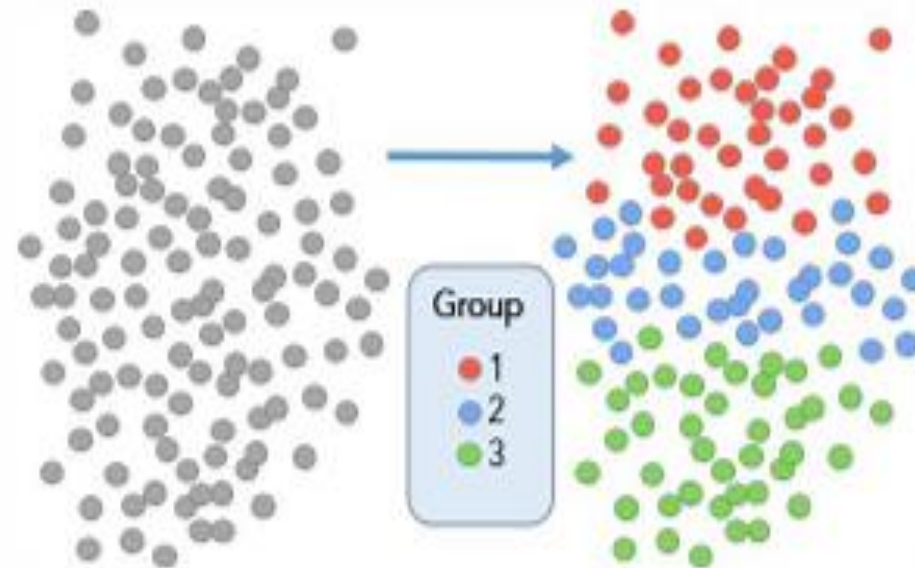
Types of Learning

Supervised

Useful for learning structure in the data (clustering), hidden correlations, reduce dimensionality, etc.

Unsupervised

Reinforcement



Types of Learning

Supervised

Unsupervised

Reinforcement

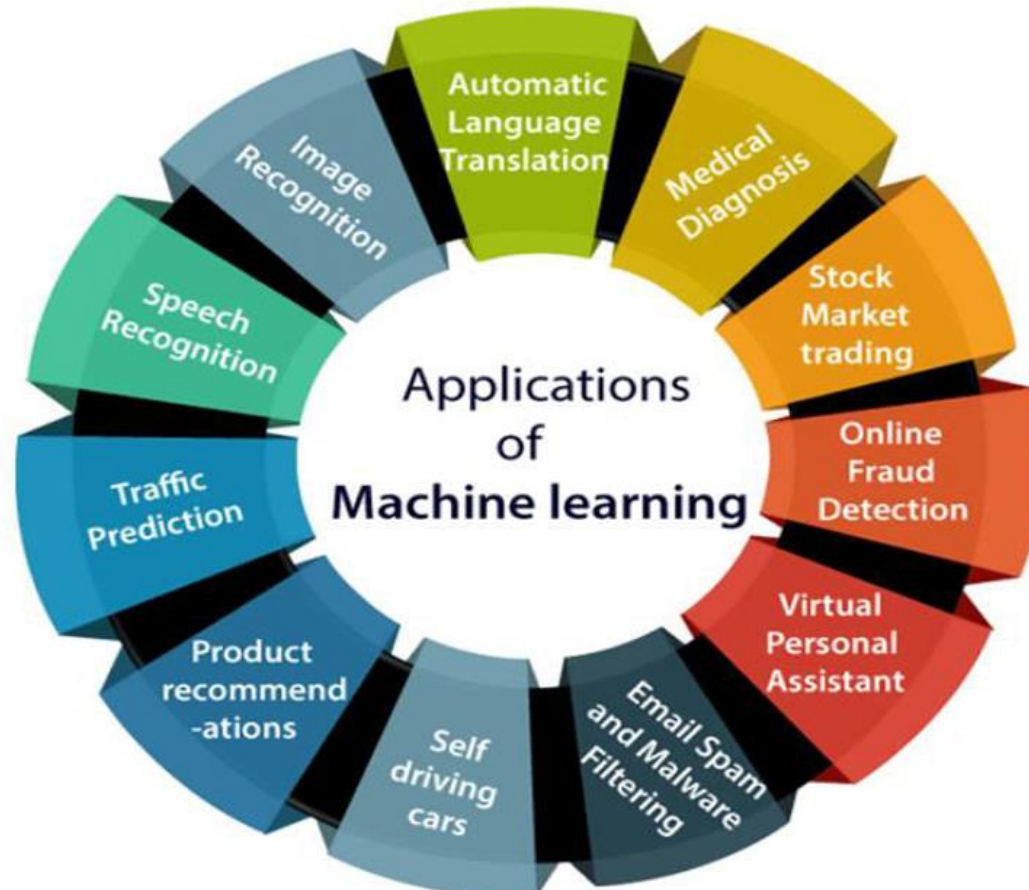
An agent interacts with an environment and watches the result of the interaction

Environment gives feedback via a positive or negative reward signal.

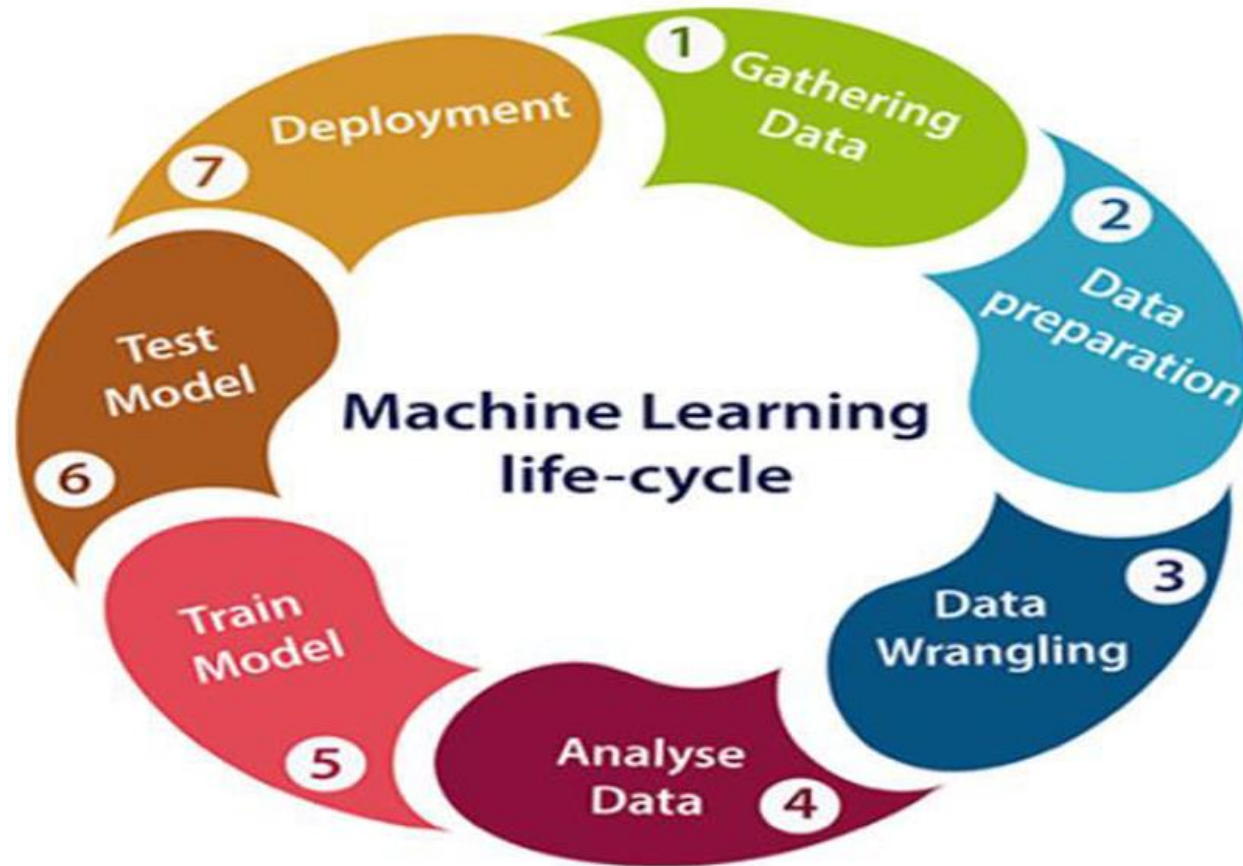
Types of Learning

Criteria	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labeled data	The machine is trained on unlabeled data without any guidance	An agent interacts with its environment by performing actions & learning from errors or rewards
Type of problems	Regression & classification	Association & clustering	Reward-based
Type of data	Labeled data	Unlabeled data	No predefined data
Training	External supervision	No supervision	No supervision
Approach	Maps the labeled inputs to the known outputs	Understands patterns & discovers the output	Follows the trial-and-error method

Applications of Machine Learning



Machine learning life cycle



Machine learning life cycle

1. Gathering Data

- To solve a problem, we create a machine learning system called "model", and this model is created by providing "training". But to train a model, we need data, hence, life cycle starts by collecting data.
- The quantity and quality of the collected data will determine the efficiency of the output.
- The more will be the data, the more accurate will be the prediction. • This step includes the below tasks:
 - ✓ Identify various data sources
 - ✓ Collect data
 - ✓ Integrate the data obtained from different sources

Machine learning life cycle

2. Data Preparation

- The process of data preparation can be further divided into two processes:

Data exploration:

- It is used to understand the nature of data that we have to work to understand the characteristics, format, and quality of data.
- In this, we find Correlations, general trends, and outliers.

Data Pre-processing:

- Now the next step is preprocessing of data for its analysis.

Machine learning life cycle

3. Data Wrangling

- It is the process of cleaning and converting raw data into a useable format.
- It is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format to make it more suitable for analysis in the next step.
- In real-world applications, collected data may have various issues, including:
 - ✓ Missing Values
 - ✓ Duplicate data
 - ✓ Invalid data
 - ✓ Noise
- So, we use various filtering techniques to clean the data.

Machine learning life cycle

4. Data Analysis

- Now the cleaned and prepared data is passed on to the analysis step.
This step involves:
 - ✓ Selection of analytical techniques
 - ✓ Building models
 - ✓ Review the result
- It starts with the determination of the type of the problems, where we select the machine learning techniques such as Classification, Regression, Cluster analysis, Association, etc. then build the model using prepared data, and evaluate the model.

Machine learning life cycle

5. Train Model

- In this step we train our model to improve its performance for better outcome of the problem.
- We use datasets to train the model using various machine learning algorithms.

Machine learning life cycle

6. Test Model

- Once our machine learning model has been trained on a given dataset, then we test the model.
- In this step, we check for the accuracy of our model by providing a test dataset to it.
- Testing the model determines the accuracy percentage of the model as per the requirement of project or problem.

Machine learning life cycle

7. Deployment

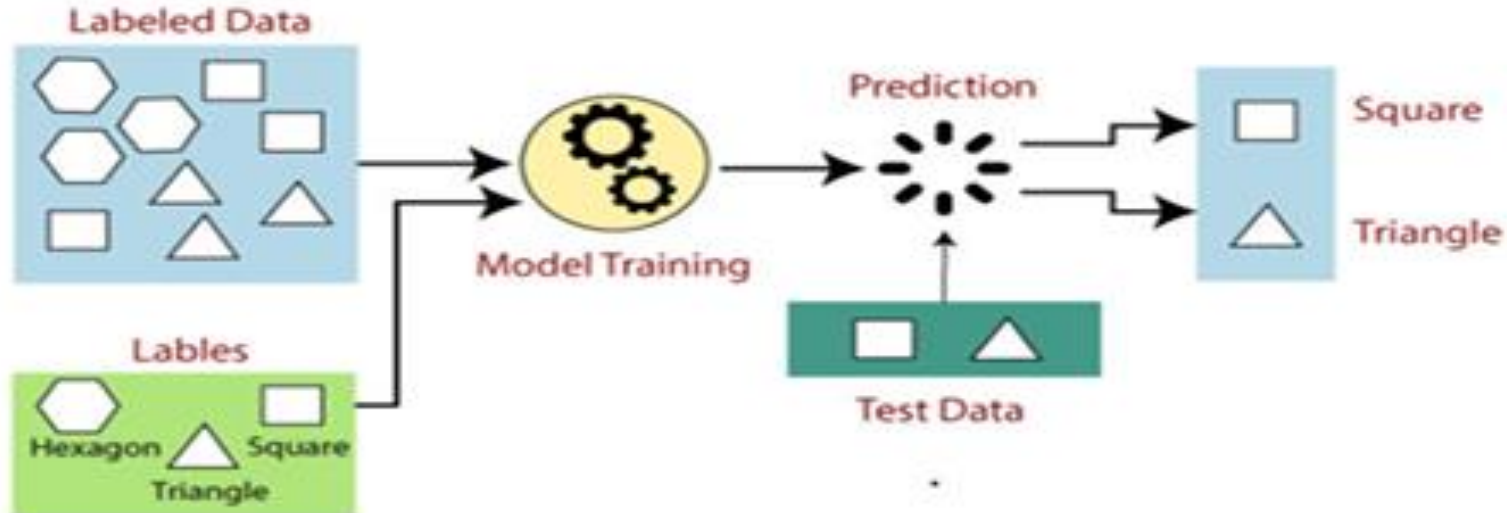
- The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system.
- If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system.
- But before deploying the project, we will check whether it is improving its performance using available data or not.

Supervised Machine Learning

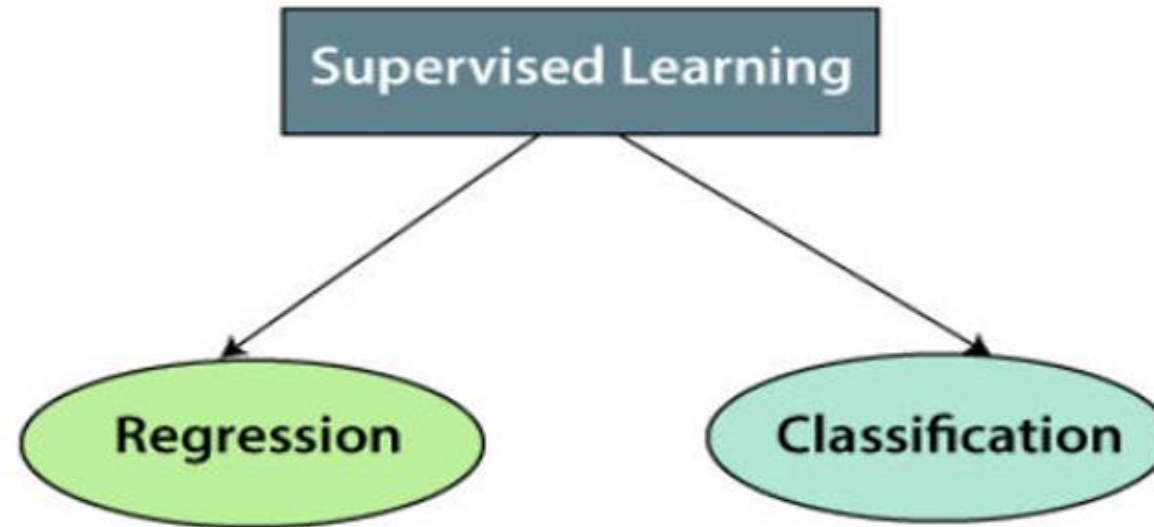
- Supervised learning is the types of machine learning in which machines are trained using well "labelled" training data.
- The labelled data means some input data is already tagged with the correct output.
- On basis of that data, machines predict the output.
- In supervised learning, the training data provided to the machines work as the supervisor that teaches the machines to predict the output correctly.
- The aim of a supervised learning algorithm is to find a mapping function to map the input variable(x) with the output variable(y).
- Example: Risk Assessment, Image classification, Fraud Detection, spam filtering, etc.

How Supervised Learning Work?

- After the training process is completed, the model is tested on the basis of test data (a subset of the training set), and then it predicts the output.



Types of supervised Machine learning Algorithms



Regression

- It is used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc.
- Below are some popular Regression algorithms which come under supervised learning:

Linear
Regression

Regression
Trees

Non-Linear
Regression

Bayesian
Linear
Regression

Polynomial
Regression

Classification

- Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc.
- Below mentioned are some of the types of Classification:

Random
Forest

Decision Trees

Logistic
Regression

Support
vector
Machines

Advantages of Supervised Learning

- With the help of supervised learning, the model can predict the output on the basis of prior experiences.
- Supervised learning is a simple process for you to understand. In the case of unsupervised learning, we don't easily understand what is happening inside the machine, how it is learning, etc.
- Supervised learning can be very helpful in classification problems
- Supervised learning model helps us to solve various real-world problems such as fraud detection, spam filtering, etc.

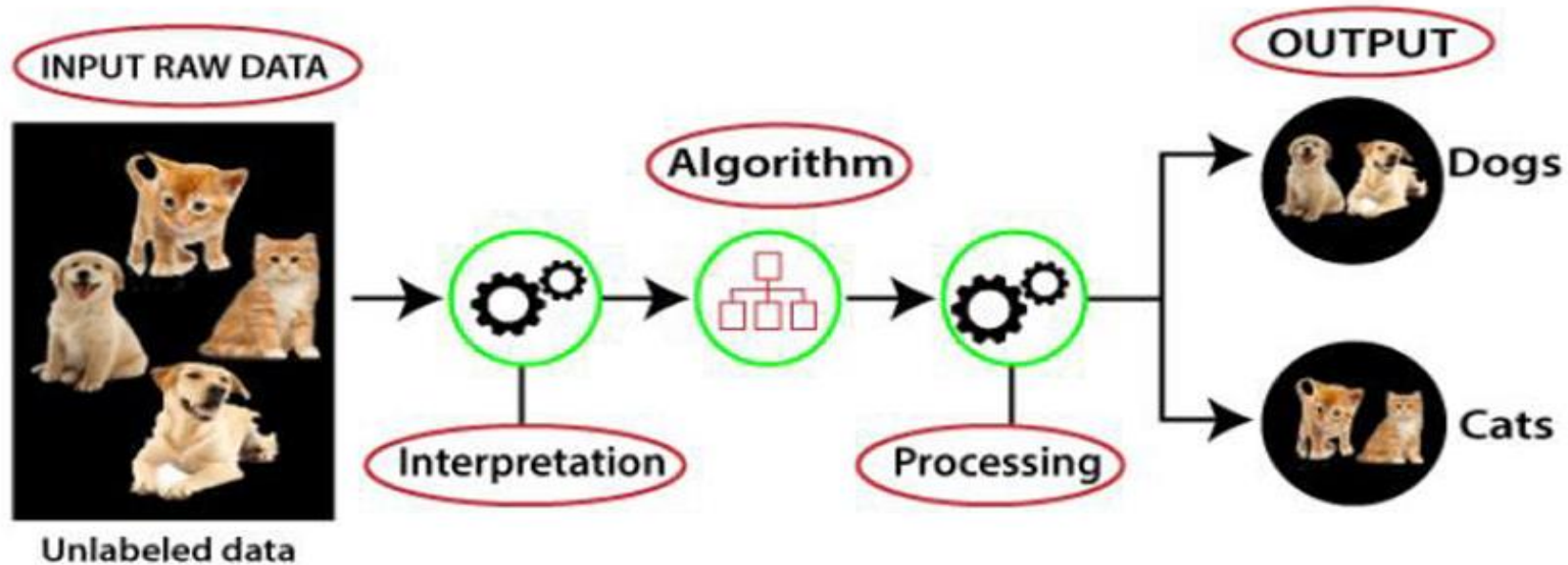
Unsupervised Learning

- Unsupervised learning is a type of machine learning in which models are trained using unlabeled dataset and are allowed to act on that data without any supervision.
- Here models are not supervised using training dataset. Instead, models itself find the hidden patterns and insights from the given data.

Why Unsupervised Learning?

- Unsupervised learning is helpful for finding useful insights from the data.
- Unsupervised learning is much similar as a human learns to think by their own experiences, which makes it closer to the real AI.
- Unsupervised learning works on unlabeled and uncategorized data which make unsupervised learning more important.
- In real-world, we do not always have input data with the corresponding output so to solve such cases, we need unsupervised learning.

Working of Unsupervised Learning



Types of Unsupervised Learning Algorithm:

Clustering

- Clustering is a method of grouping the objects into clusters such that objects with most similarities remains into a group
- Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.

Association

- It is used for finding the relationships between variables in the large database.
- Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item.

Advantages of Unsupervised Learning

- Labeling of data demands a lot of manual work and expenses. Unsupervised learning solves the problem by learning the data and classifying it without any labels.
- Unsupervised learning is preferable as it is easy to get unlabeled data in comparison to labeled data.
- It is very helpful in finding patterns in data, which are not possible to find using normal methods.

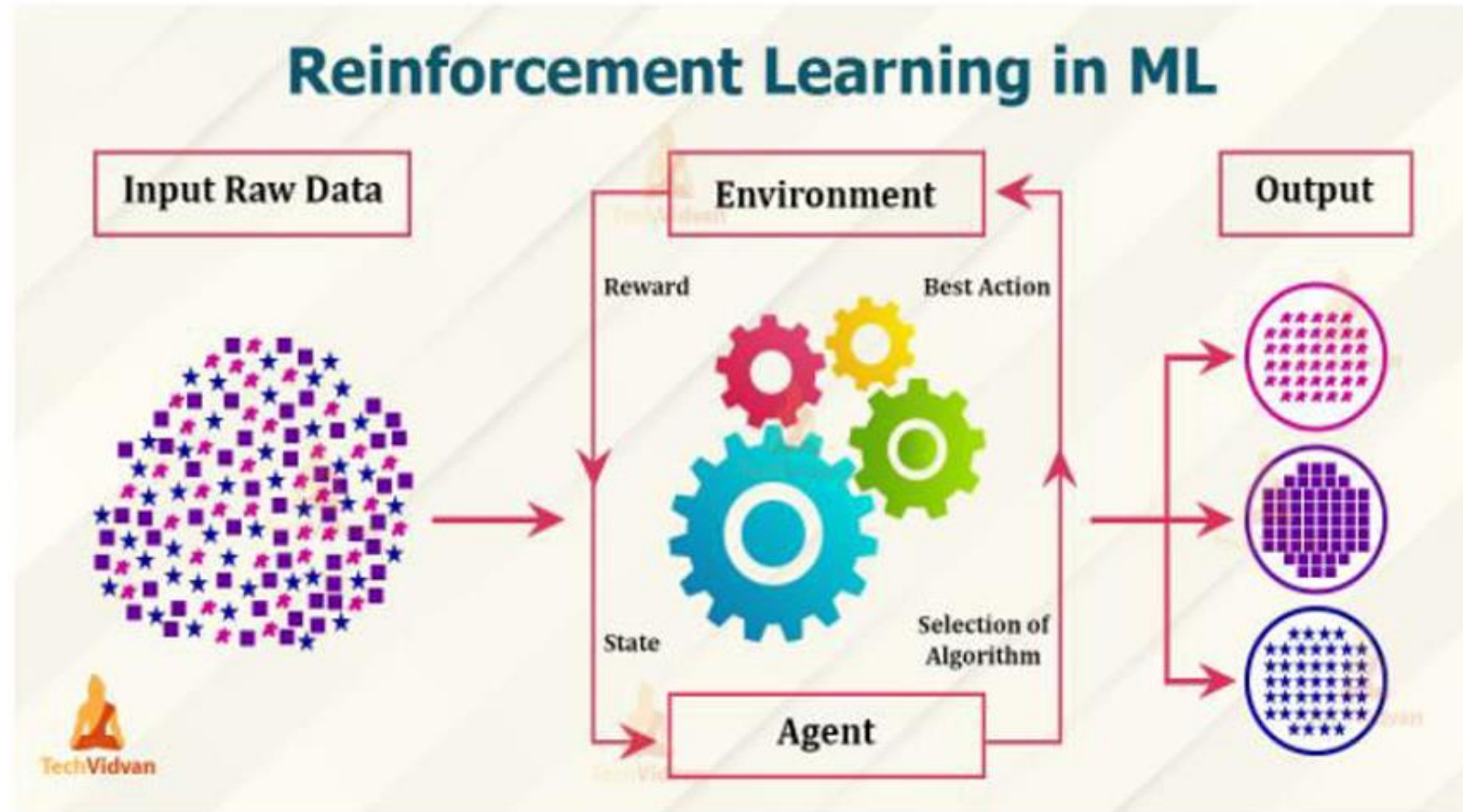
Disadvantages of Unsupervised Learning

- Unsupervised learning is intrinsically more difficult than supervised learning as it does not have corresponding output.
- The result of the unsupervised learning algorithm might be less accurate as input data is not labeled, and algorithms do not know the exact output in advance.
- The more the features, the more the complexity increases.
- It is also a time-consuming process. The learning phase of the algorithm might take a lot of time, as it analyses and calculates all possibilities.

Reinforcement Learning

- Reinforcement Learning is a feedback-based Machine learning technique.
- Reinforcement Learning solves a specific type of problem where decision making is sequential, and the goal is long-term, such as game-playing, robotics, etc.
- The agent interacts with the environment and explores it by itself. The primary goal of an agent in reinforcement learning is to improve the performance by getting the maximum positive rewards.
- The agent learns with the process of hit and trial, and based on the experience, it learns to perform the task in a better way.

How Does it Work?



Key Features of Reinforcement Learning

- In Reinforcement Learning, the agent is not instructed about the environment and what actions need to be taken.
- It is based on the hit and trial process.
- The agent takes the next action and changes states according to the feedback of the previous action.
- The agent may get a delayed reward.
- The environment is stochastic, and the agent needs to explore it to reach to get the maximum positive rewards.

Types of Reinforcement Learning

Positive Reinforcement

- Here, we try to add a reward for every good result in order to increase the likelihood of a good result.
- Too much positive reinforcement may lead to an overload of states that can reduce the consequences.

Negative Reinforcement

- It is opposite to the positive reinforcement as it increases the tendency that the specific behavior will occur again by avoiding the negative condition.
- It can be more effective than the positive reinforcement depending on situation and behavior, but it provides reinforcement only to meet minimum behavior.

Advantages of Reinforcement Learning

- It can solve higher-order and complex problems. Also, the solutions obtained will be very accurate.
- This model will undergo a rigorous training process that can take time. This can help to correct any errors.
- Since the model learns constantly, a mistake made earlier would be unlikely to occur in the future.

Disadvantages of Reinforcement Learning

- The usage of reinforcement learning models for solving simpler problems won't be correct. The reason being, the models generally tackle complex problems.
- We need lots of data to feed the model for computation. Reinforcement Learning models require a lot of training data to develop accurate results.
- This consumes time and lots of computational power.

Application of Reinforcement Learning

- Reinforcement Learning models of companies can analyze customer preferences and help in the better advertisement of the products.
- We have Deep learning algorithms like AlphaGo, AlphaZero that are gaming algorithms for games like chess, shogi and go.
- RL is now a big help in recommendation systems like news, music apps, and web-series apps like Netflix, etc. These apps work as per customer preferences.
- Reinforcement learning helps to understand chemical reactions. We can try to have cleaner reactions that yield better products.

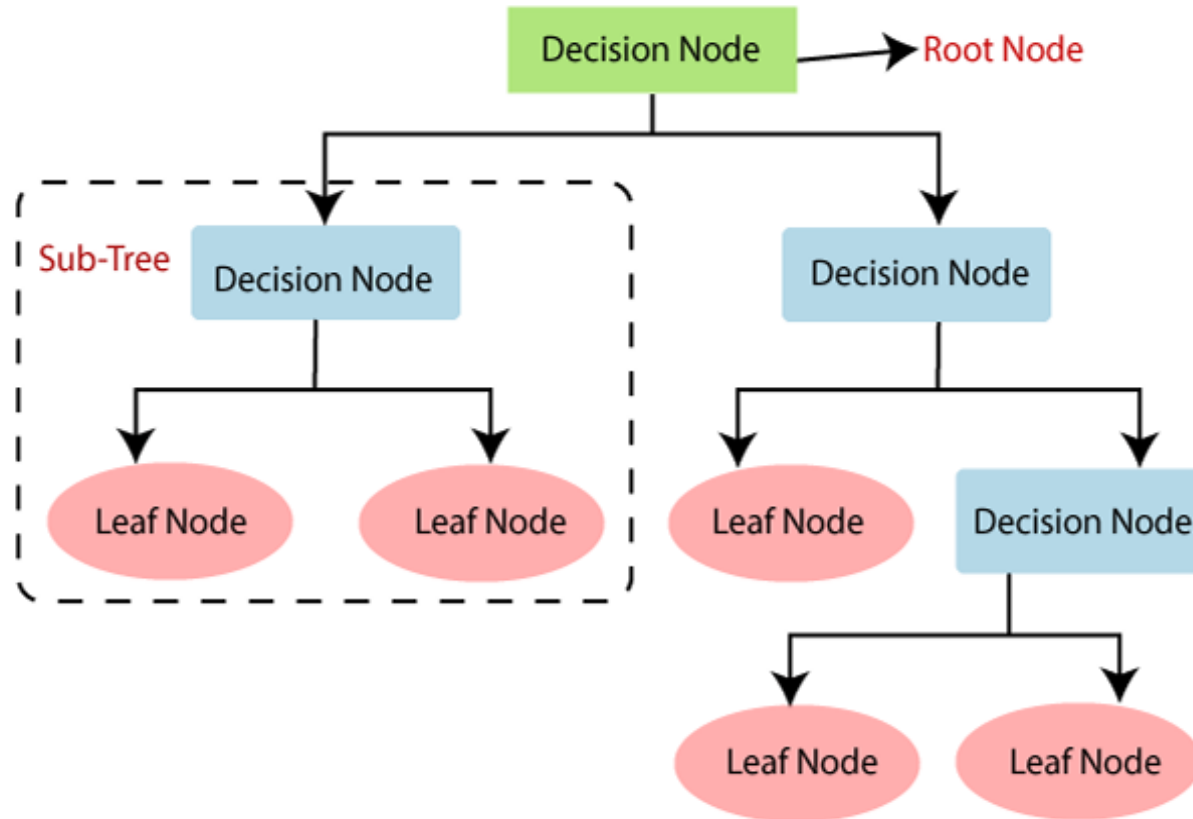
Reinforcement Learning Vs Supervised Learning

Reinforcement Learning	Supervised Learning
RL works by interacting with the environment.	Supervised learning works on the existing dataset.
The RL algorithm works like the human brain works when making some decisions.	Supervised Learning works as when a human learns things in the supervision of a guide.
There is no labeled dataset is present	The labeled dataset is present.
No previous training is provided to the learning agent.	Training is provided to the algorithm so that it can predict the output.
RL helps to take decisions sequentially.	In Supervised learning, decisions are made when input is given.

Decision Tree Classification Algorithm

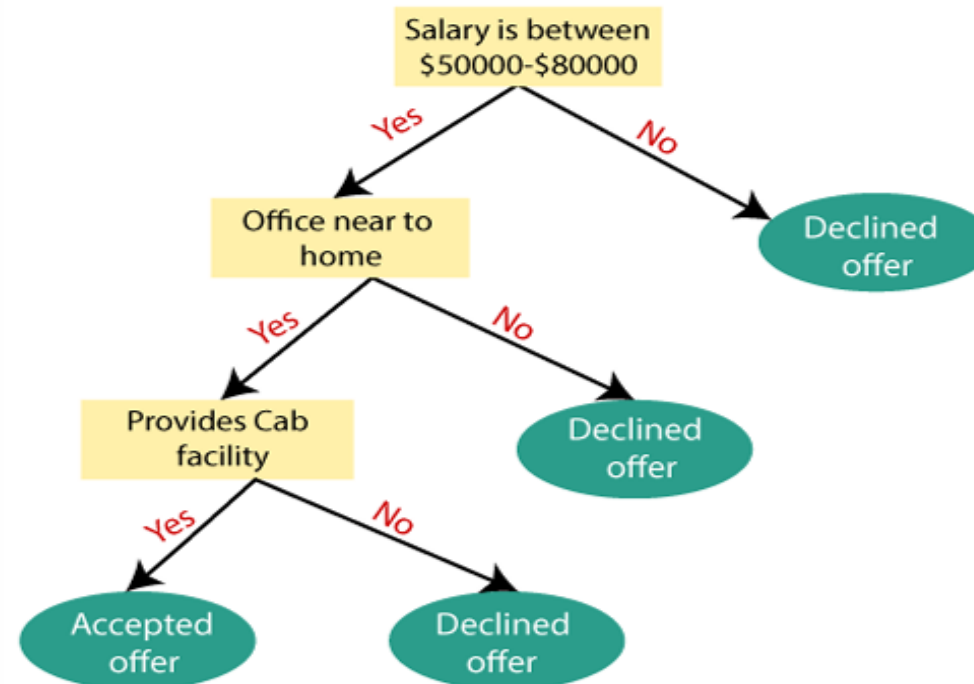
- Decision Tree is a Supervised learning technique that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems.
- It is a tree-structured classifier. In a Decision tree, there are two nodes, which are the Decision Node and Leaf Node.
- Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.
- The decisions or the test are performed on the basis of features of the given dataset.

Decision Tree



Example

- Suppose there is a candidate who has a job offer and wants to decide whether he should accept the offer or Not. So, to solve this problem, the decision tree starts with the root node (Salary attribute by ASM). The root node splits further into the next decision node (distance from the office) and one leaf node based on the corresponding labels.



Naive Bayes Classifier

- Naïve Bayes algorithm is a supervised learning algorithm, which is based on Bayes theorem and used for solving classification problems.
- It is mainly used in text classification that includes a high-dimensional training dataset.
- Naïve Bayes Classifier is one of the simple and most effective Classification algorithms which helps in building the fast machine learning models that can make quick predictions.
- It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.
- It is called Naïve because it assumes that the occurrence of a certain feature is independent of the occurrence of other features.

Naive Bayes Classifier

- The Naive Bayes classifier works on the principle of conditional probability, as given by the Bayes theorem.
- The Bayes theorem gives us the conditional probability of event A, given that event B has occurred.
 - According to Bayes theorem:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

where:

$P(A|B)$ = Conditional Probability of A given B

$P(B|A)$ = Conditional Probability of A given B

$P(A)$ = Probability of event A

$P(B)$ = Probability of event A

Naive Bayes Classifier

- At a certain University, 4% men are over 6 feet tall and 1% women are over 6 feet tall. The total student population is divided in the ratio 3:2 in favor of women. If a student is selected at random from among all those over 6 feet tall, what is the probability that the student is a woman?
 - $P(M) = 2/5 = 0.4$
 - $P(W) = 3/5 = 0.6$
 - $P(T/M) = 4\% = 0.04$
 - $P(T/W) = 1\% = 0.01$

Example

- We require $P(W|T)$:
- By Baye's Theorem:

$$P(W|T) = \frac{P(T|W)P(W)}{P(T|W)P(W) + P(T|M)P(M)}$$
$$= (0.01 * 0.6) / ((0.01 * 0.6) + (0.04 * 0.4))$$
$$= 3/11$$

Hence, the probability that the selected student being a woman is 3/11.

The Theory of Everything

- The "Theory of Everything" in AI, while not a formalized concept like in physics, is often used to describe the overarching goal of creating a unifying framework or model that explains and integrates all aspects of artificial intelligence.
- It encapsulates the idea of achieving a **General Artificial Intelligence (AGI)**—an AI system capable of understanding, learning, and performing any intellectual task a human can do.

1. What is the "Theory of Everything" in AI?

- In essence, it refers to:
- A comprehensive framework that unites the diverse subfields and methods in AI (e.g., machine learning, symbolic reasoning, natural language processing).
- The goal of creating a single, all-encompassing AI system capable of generalizing knowledge and solving tasks across domains without needing task-specific adjustments.
- It draws inspiration from physics, where the "Theory of Everything" seeks to unify quantum mechanics and general relativity into one coherent model.

2. Current Landscape in AI

- AI is currently divided into various approaches, each excelling in specific areas but lacking a unifying framework:
- Machine Learning (ML): Focuses on statistical models that learn patterns from data.
- Neural Networks (Deep Learning): Excels in image recognition, natural language processing, and more.
- Symbolic AI: Uses explicit rules and logic for reasoning but struggles with flexibility.
- Reinforcement Learning (RL): Excels in sequential decision-making tasks, such as games or robotics.
- Despite advancements, most current systems are "Narrow AI," meaning they specialize in one domain or task.

3. Components of a Unified Theory

For a "Theory of Everything" in AI, the following elements are key:

1. Representation Learning: A universal way to encode all forms of knowledge—visual, textual, numerical—into a format machines can process and generalize from.
2. Learning Mechanisms: A framework that combines supervised, unsupervised, and reinforcement learning seamlessly.
3. Reasoning and Problem-Solving: Integrating symbolic reasoning with statistical learning for a hybrid approach.
4. Memory and Transfer Learning: Retaining and applying learned knowledge across tasks and domains.
5. Consciousness and Self-Reflection: Building models that can understand their own limitations and improve autonomously.
6. Ethics and Alignment: Ensuring that AGI systems align with human values and operate safely.

4. Prominent Theories and Approaches

- Unified Theories of Cognition (e.g., Soar, ACT-R): Early attempts to model all human cognitive processes computationally.
- Deep Learning for Generalization: Current advancements suggest large-scale models like GPT, which can perform diverse tasks with minimal fine-tuning, as a step toward unification.
- Emergent Behavior in AI: Large-scale systems sometimes display unexpected abilities, hinting at potential paths toward AGI.

5. Challenges to a Unified Theory

- Scalability: Developing models that can handle vast and diverse datasets.
- Interpretability: Understanding the inner workings of large AI systems.
- Integration: Combining symbolic reasoning, probabilistic methods, and neural networks in one coherent system.
- Safety and Alignment: Preventing unintended consequences and ensuring ethical use.

6. Why Does It Matter?

"Theory of Everything" in AI would:

- Enable the creation of highly capable AGI systems.
- Provide insights into human intelligence and cognition.
- Revolutionize fields like science, medicine, and education by solving previously unsolvable problems.

While the journey to such a theory is long and complex, ongoing advances in machine learning, neuroscience-inspired models, and interdisciplinary research are paving the way.

References

- E. Rich and Knight, *Artificial Intelligence*, McGraw Hill, 1991.
- D. W. Patterson, *Artificial Intelligence and Expert Systems*, Prentice Hall, 2001.

Chapter 7

Deep Learning and Other AI Applications

Er. Sharmila Bista

MECE

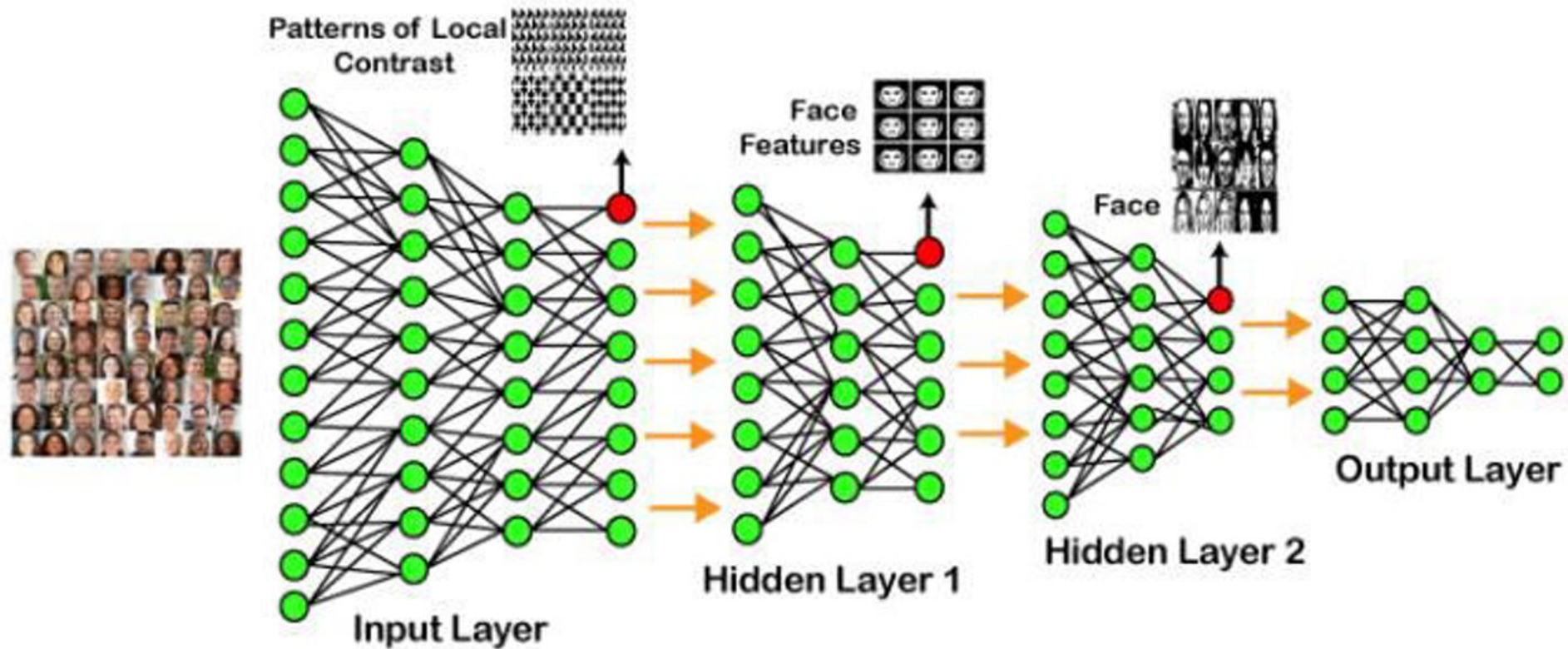
Overview

- Deep Learning Fundamentals
- Neural Networks and Nature Language Processing
- Deep Learning for NLP
- Computer Vision

Deep Learning

- Deep learning is based on the branch of machine learning, which is a subset of artificial intelligence.
- Basically, it is a machine learning class that makes use of numerous nonlinear processing units so as to perform feature extraction as well as transformation.
- The output from each preceding layer is taken as input by each one of the successive layers.
- Deep learning is implemented by the help of deep networks, which are nothing but neural networks with multiple hidden layers.

Example of Deep Learning



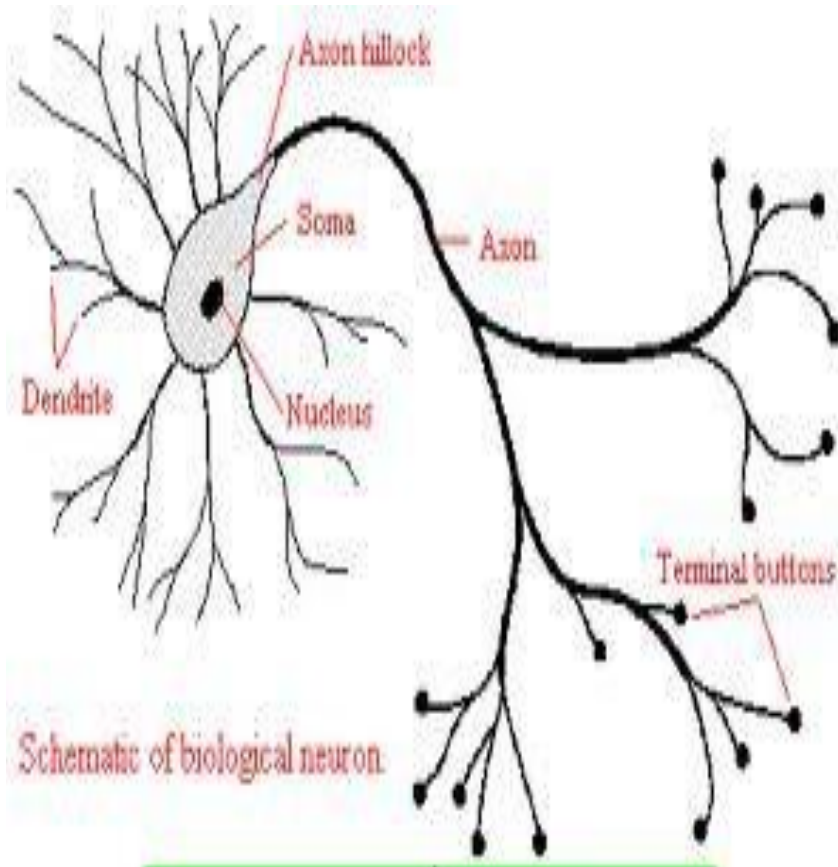
Neural Networks

- A neuron is a cell in brain whose principal function is the collection, Processing, and dissemination of electrical signals. Brains Information processing capacity comes from networks of such neurons.
- Due to this reason some earliest AI work aimed to create such artificial networks/ connectionism / Parallel distributed processing/ Neural computing.
- An Artificial Neural Network (ANN) is an information processing paradigm that is inspired by the way biological nervous systems, such as the brain, process information.
- It is composed of a large number of highly interconnected processing elements (neurons) working to solve specific problems.
- ANNs, like people, learn by example.
- An ANN is configured for a specific application, such as pattern recognition or data classification, through a learning process.
- Artificial neurons are crude approximations of the neurons found in real brains. They may be physical devices, or purely mathematical constructs.

Neural Networks

- A neural network is a system of hardware and/or software patterned after the operation of neurons in the human brain.
- A neural network usually involves a large number of processors operating in parallel and arranged in tiers.
- The first tier receives the raw input information -- analogous to optic nerves in human visual processing.
- Each successive tier receives the output from the tier preceding it, rather than from the raw input – in the same way neurons further from the optic nerve receive signals from those closer to it.
- The last tier produces the output of the system.

Basic Components of Biological Neurons



- The majority of neurons encode their activation or outputs as a series of brief electrical pulses (i.e. spikes or action potentials).
- The neuron's cell body (soma) processes the incoming activations and converts them into output activations.
- The neuron's nucleus contains the genetic material (DNA).
- Dendrites are fibers which emanate from the cell body and provide the receptive zone that receive activation from other neurons.
- Axons are fibers acting as transmission lines that send action potentials to other neurons.
- The junctions that allow signal transmission between the axons and the dendrites are called synapses.
- The process of transmission is by diffusion of chemicals called neurotransmitters across the synaptic cleft.

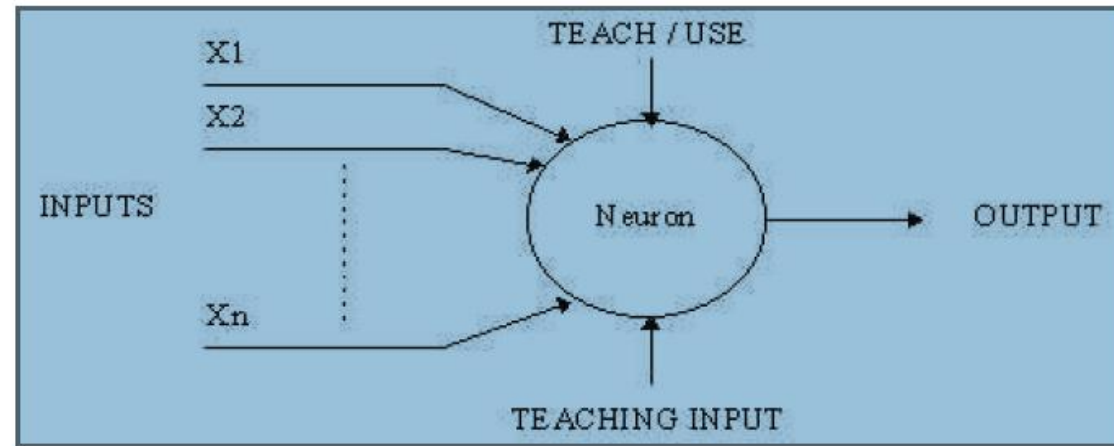
Basic Components of Biological Neurons

Biological NN	Artificial NN
Soma	Neuron
Dendrite	Input
Axon	Output
Synapse	weight

Units of Neural Network

- **Nodes (units)**
 - Nodes represent a cell of neural network.
- **Links**
 - Links are directed arrows that show propagation of information from one node to another node.
- **Activation**
 - Activations are inputs to or outputs from a unit.
- **Weight**
 - Each link has weight associated with it which determines strength and sign of the connection.
- **Activation function:**
 - A function which is used to derive output activation from the input activations to a given node is called activation function.
- **Bias:**
 - Bias is a feature of a statistical technique or of its results whereby the expected value of the results differs from the true underlying quantitative parameter being estimated.

A Simple Neuron



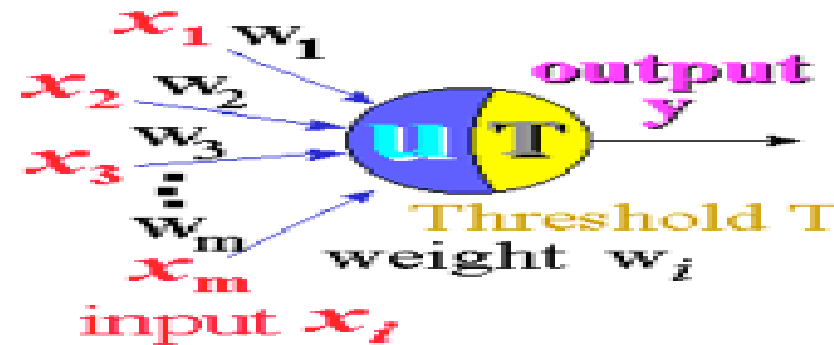
An artificial neuron is a device with many inputs and one output. The neuron has two modes of operation; the training mode and the using mode.

In the training mode, the neuron can be trained to fire (or not), for particular input patterns.

In the using mode, when a taught input pattern is detected at the input, its associated output becomes the current output.

Artificial Neuron

- An artificial neuron is designed to mimic the first-order characteristics of a biological neuron.
- A set of inputs, x_i .
- Each input is multiplied by the corresponding weight, w_i .
- All of the weighted inputs are summed to determine the activation level.
- An activation function is applied to determine the neuron output.



Mathematical Structure

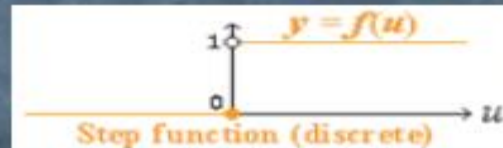
- A neuron is a simple sum and compare device.

$$u = x_1w_1 + x_2w_2 + \dots x_nw_n - T$$

$y = f(u)$

threshold

One possible function for y is a step function:

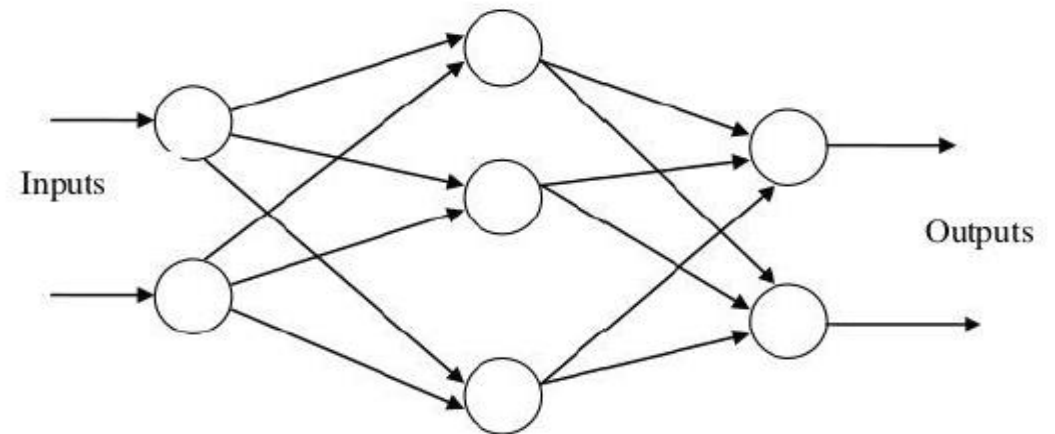


If the weighted sum of the inputs is greater than the threshold, then the neuron “fires”

Types of Neural Network

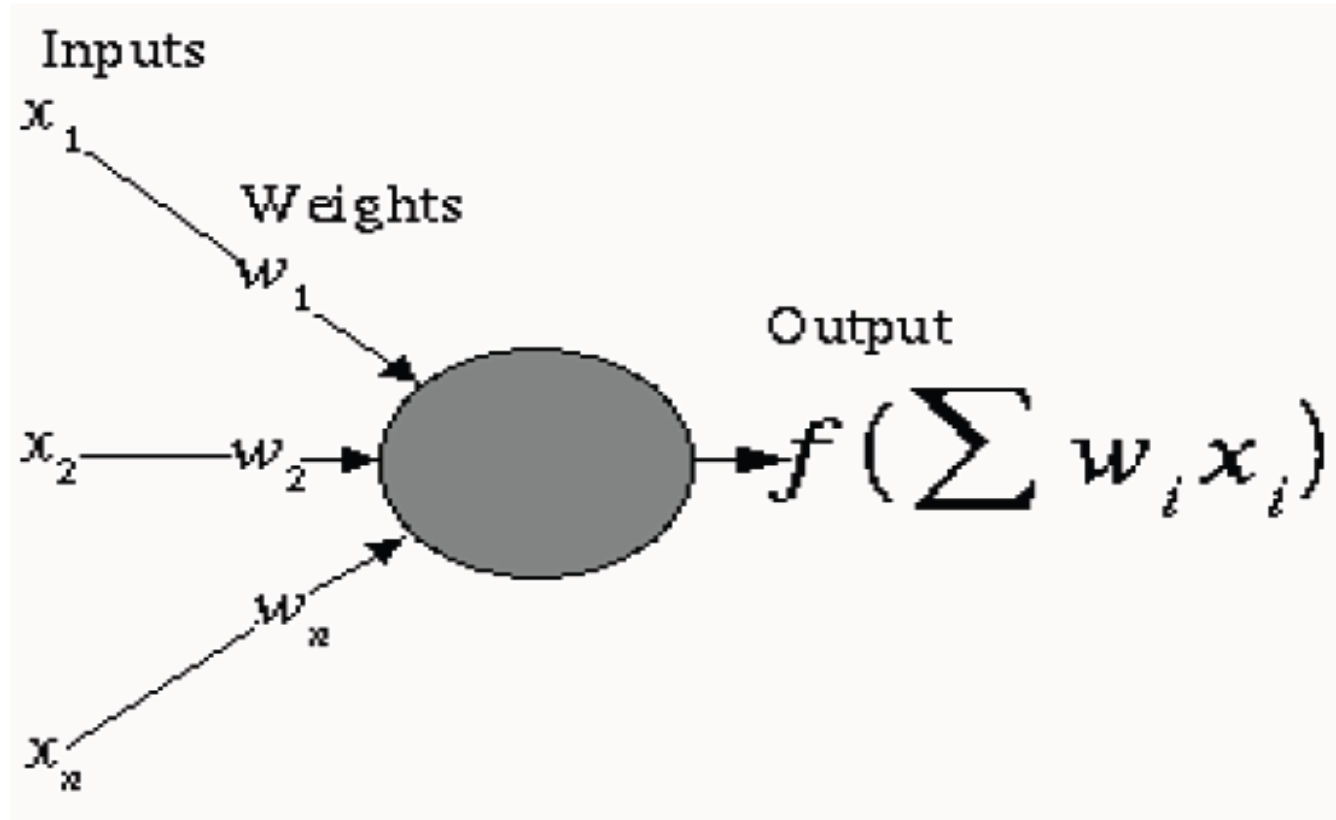
Feed-forward ANN

- Feed-forward ANNs allow signals to travel one way only, from input to output.
- There is no feedback (loops) i.e. the output of any layer does not affect that same layer.
- Feed forward ANNs tend to be straight forward networks that associate inputs with outputs.
- They are extensively used in pattern recognition.
- This type of organization is also referred to as bottom-up or top-down.



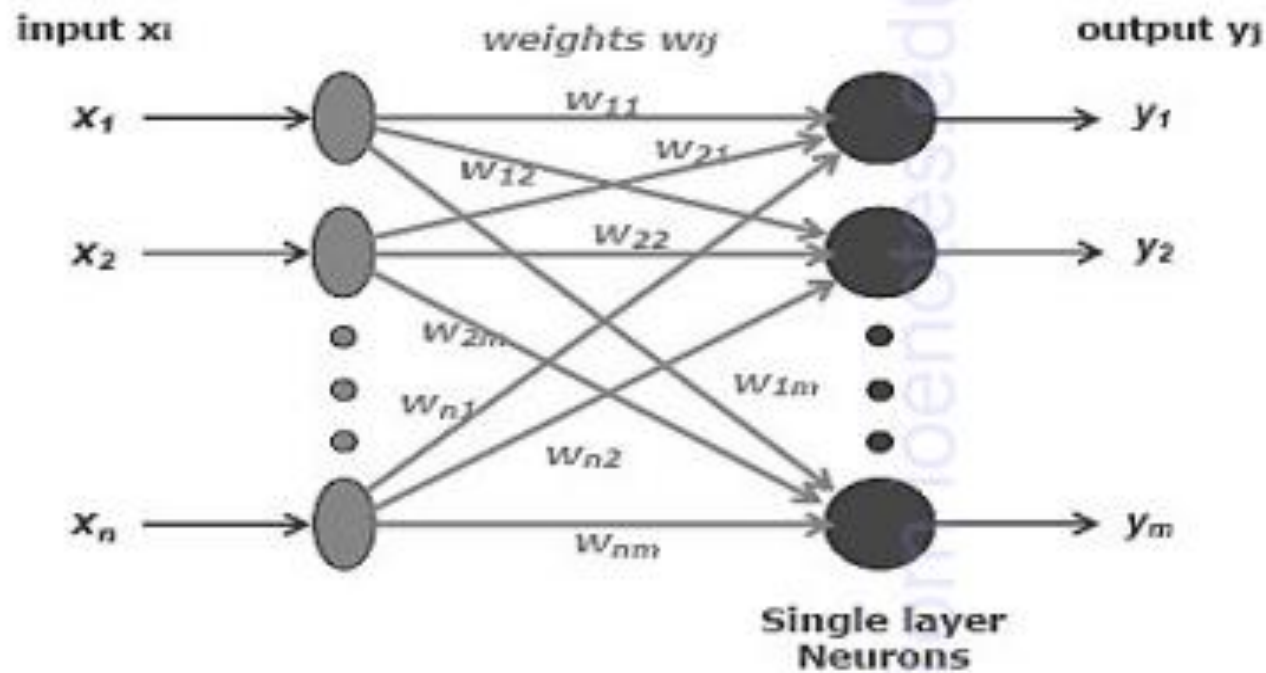
Types of Neural Network

- 1. Feed-forward ANN



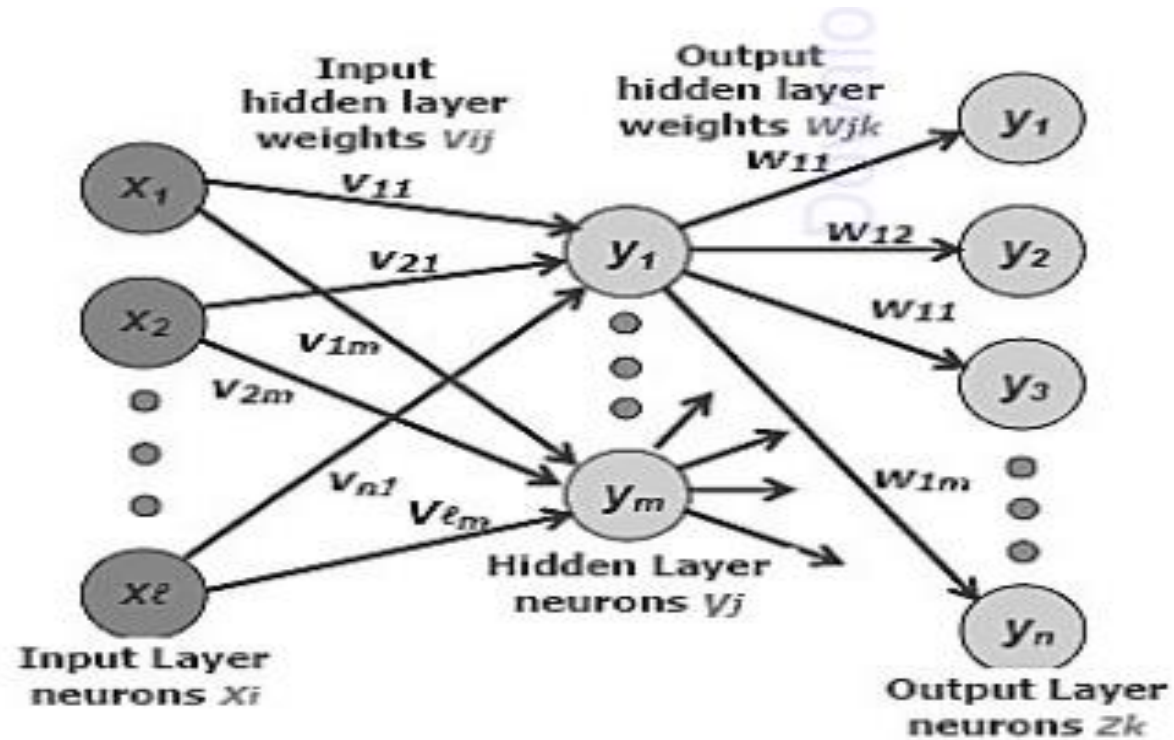
Types of Neural Network

a. Single Layer Feed Forward ANN



Types of Neural Network

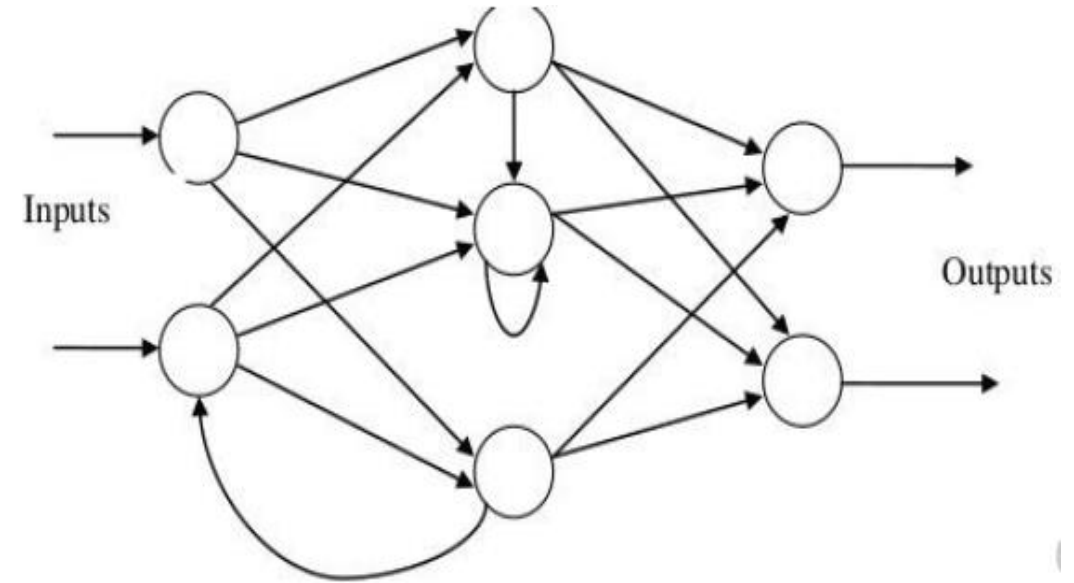
b. Multi Layer Feed Forward ANN



Types of Neural Network

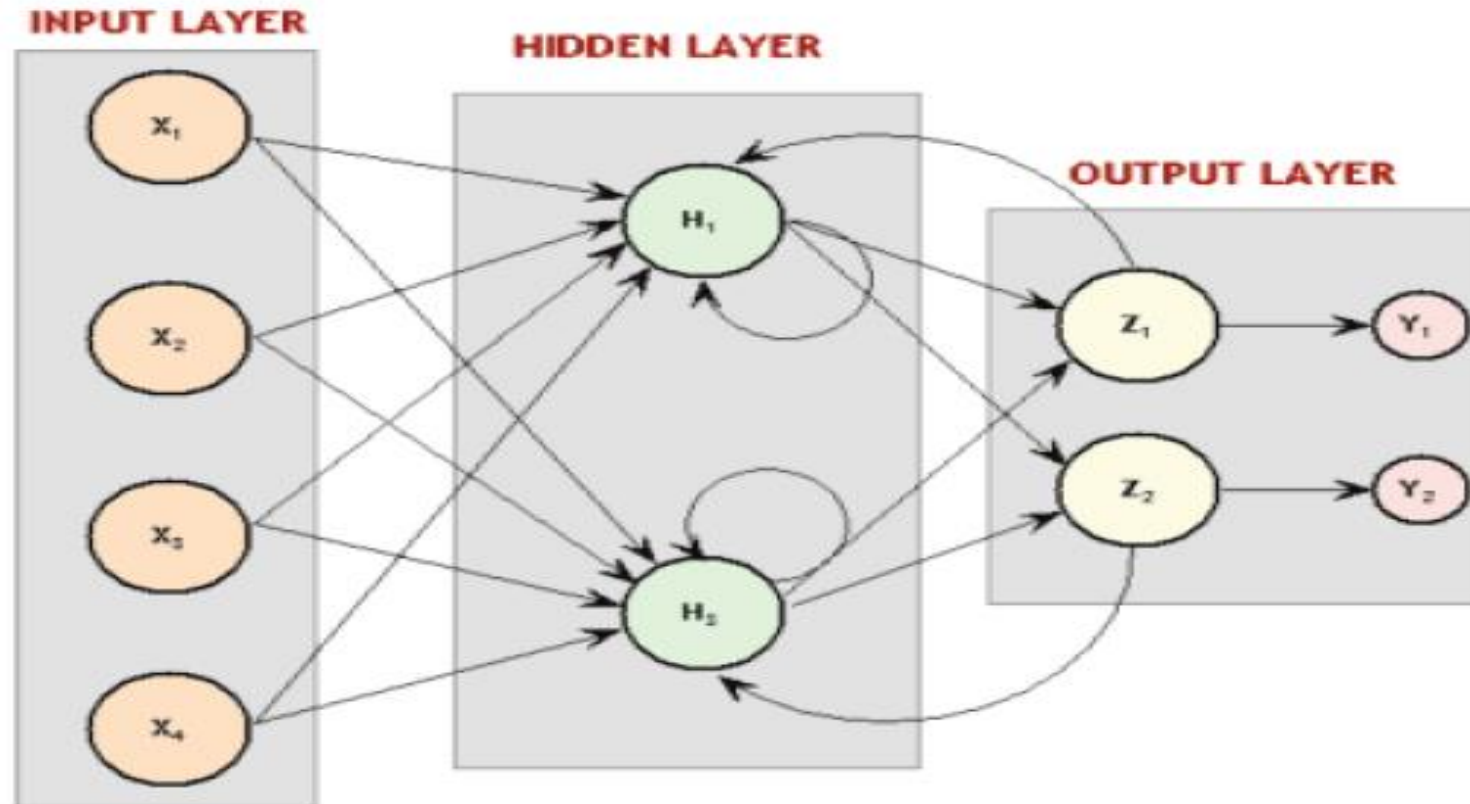
2. Feedback (Recurrent) ANN

- Feedback networks can have signals traveling in both directions by introducing loops in the network.
- Feedback networks are very powerful and can get extremely complicated.



Types of Neural Network

2. Feedback (Recurrent) ANN



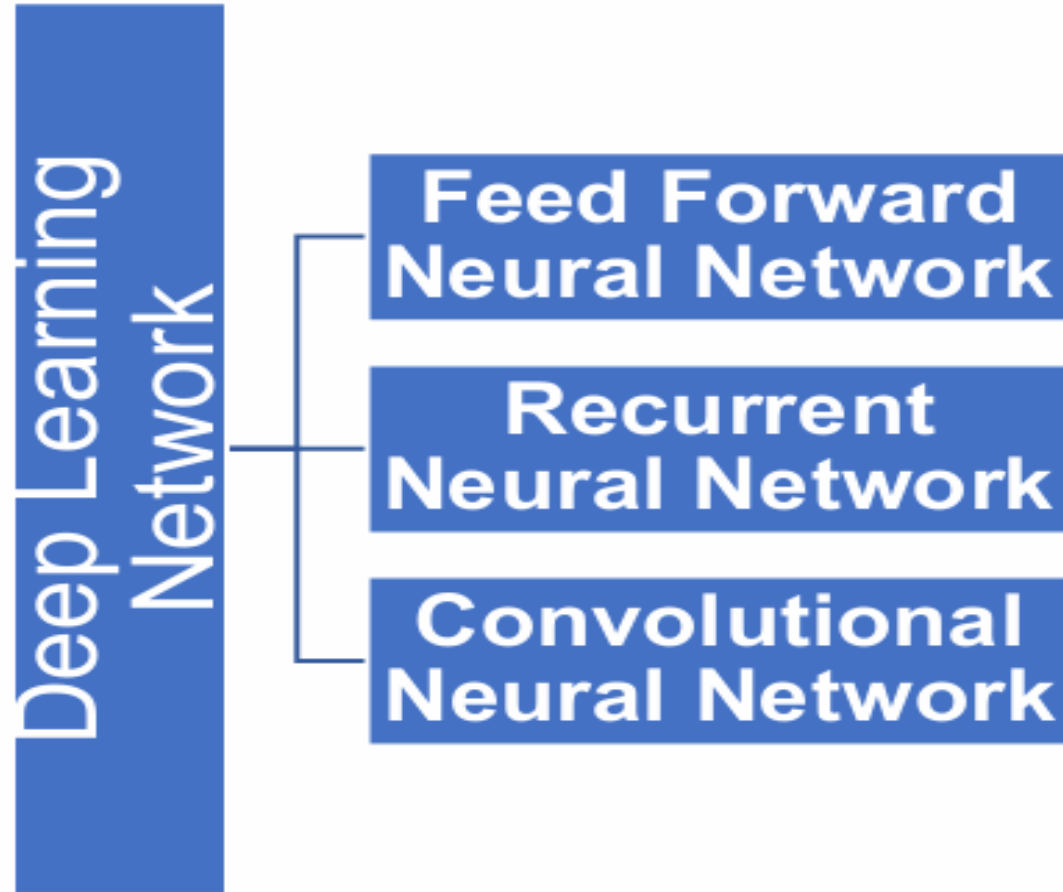
Deep Neural Network

- If you have one or a few hidden layers, then you have a shallow neural network. If you have many hidden layers, then you have a deep neural network.
- They are highly proficient on model and process non-linear associations.
- Neural networks are widely used in supervised learning and reinforcement learning problems. These networks are based on a set of layers connected to each other.
- In deep learning, the number of hidden layers, mostly non-linear, can be large; say about 1000 layers.
- DL models produce much better results than normal ML networks.

Deep Nets and Shallow Nets

- There is no clear threshold of depth that divides shallow learning from deep learning.
- The prediction accuracy of a neural net depends on its weights and biases.
- The process of improving the accuracy of neural network is called training.
- The point of training is to make the cost of training as small as possible across millions of training examples. To do this, the network tweaks the weights and biases until the prediction matches the correct output.

Types of Deep Learning Networks

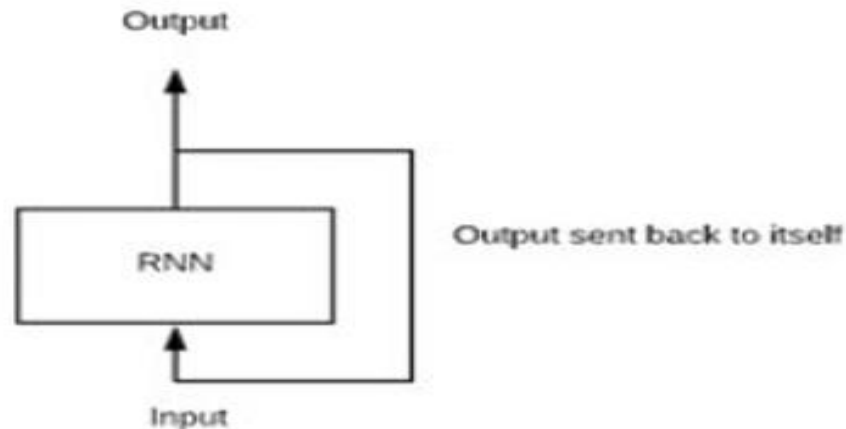


Feed Forward Neural Network

- This is the simplest type of neural network.
- With this type of architecture, information flows in only one direction, forward.
- It means, the information's flow starts at the input layer, goes to the "hidden" layers, and end at the output layer.
- The network does not have a loop. Information stops at the output layers.
- Some of the application can be data compression, pattern recognition, hand-written character recognition.

Recurrent Neural Network

- RNN is a multi-layered neural network that can store information in context nodes, allowing it to learn data sequences and output a number or another sequence.
- In simple words it an Artificial neural networks whose connections between neurons include loops.



Applications of RNN

- Help securities traders to generate analytic reports
- Detect abnormalities in the content of financial statement
- Detect fraudulent credit-card transaction
- Provide a caption for images
- Power chatbots
- Rhythm Learning
- Music Composition

Convolutional Neural Network

- Convolutional Neural Networks are a special kind of neural network mainly used for image classification, clustering of images and object recognition.
- CNNs are extensively used in computer vision; have been applied also in acoustic modelling for automatic speech recognition.
- The idea behind convolutional neural networks is the idea of a “moving filter” which passes through the image.
- Facebook as facial recognition software uses these nets.
- To achieve the best accuracy, deep convolutional neural networks are preferred more than any other neural network.

Applications of CNN

- Identify Faces, Street Signs, Tumors
- Image Recognition
- Video Analysis
- NLP

Deep Learning Applications

1. Vision

- Deep learning have been applied to a wide range of vision tasks, from self-driving cars to grading cucumbers.
- Driving is among the most demanding of vision tasks: not only must the algorithm detect, localize, track, and recognize pigeons, paper bags, and pedestrians, but it has to do it in real time with near-perfect accuracy.
- It has been also used to automatic image caption generation.

Deep Learning Applications

2. Natural Language Processing

- Deep learning has also had a huge impact on natural language processing (NLP) applications such as machine translation and speech recognition.
- As of 2020, machine translation systems are approaching human performance for language pairs such as French and English for which very large paired data sets are available, and they are usable for other language pairs covering the majority of Earth's population.
- There is even some evidence that networks trained on multiple languages do in fact learn an internal meaning representation: for example, after learning to translate Portuguese to English and English to Spanish, it is possible to translate Portuguese directly into Spanish.

Deep Reinforcement Learning

- Deep reinforcement learning (deep RL) is a subfield of machine learning that combines reinforcement learning (RL) and deep learning.
- RL considers the problem of a computational agent learning to make decisions by trial and error.
- Deep RL algorithms are able to take in very large inputs (e.g. every pixel rendered to the screen in a video game) and decide what actions to perform to optimize an objective (eg. maximizing the game score).
- The possibilities of deep reinforcement learning came to the attention of many during the well-publicized defeat of a Go grandmaster by DeepMind's AlphaGo.
- In addition to playing Go, deep reinforcement learning has achieved human-level prowess in other games such as chess, poker, Atari games and several other competitive video games

Natural Language Processing

- Natural Language Processing (NLP) refers to AI method of communicating with an intelligent systems using a natural language such as English.
- The field of NLP involves making computers to perform useful tasks with the natural languages humans use.
- The input and output of an NLP system can be:
 - Speech
 - Written Text

Advantages of NLP

- NLP helps users to ask questions about any subject and get a direct response within seconds.
- NLP offers exact answers to the question means it does not offer unnecessary and unwanted information.
- NLP helps computers to communicate with humans in their languages.
- It is very time efficient.
- Most of the companies use NLP to improve the efficiency of documentation processes, accuracy of documentation, and identify the information from large databases

Disadvantages of NLP

- NLP may not show context.
- NLP is unpredictable .
- NLP may require more keystrokes.
- NLP is unable to adapt to the new domain, and it has a limited function that's why NLP is built for a single and specific task only

Components of NLP

1. Natural Language Understanding (NLU)

- Natural Language Understanding (NLU) helps the machine to understand and analyze human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.
- NLU involves the following tasks
 - It is used to map the given input into useful representation.
 - It is used to analyze different aspects of the language.

Components of NLP

2. Natural Language Generation (NLG)

- Natural Language Generation (NLG) acts as a translator that converts the computerized data into natural language representation.
- It mainly involves Text planning, Sentence planning, and Text Realization.
 - Text planning – It includes retrieving the relevant content from knowledge base.
 - Sentence planning – It includes choosing required words, forming meaningful phrases, setting tone of the sentence.
 - Text Realization – It is mapping sentence plan into sentence structure.

Applications of NLP

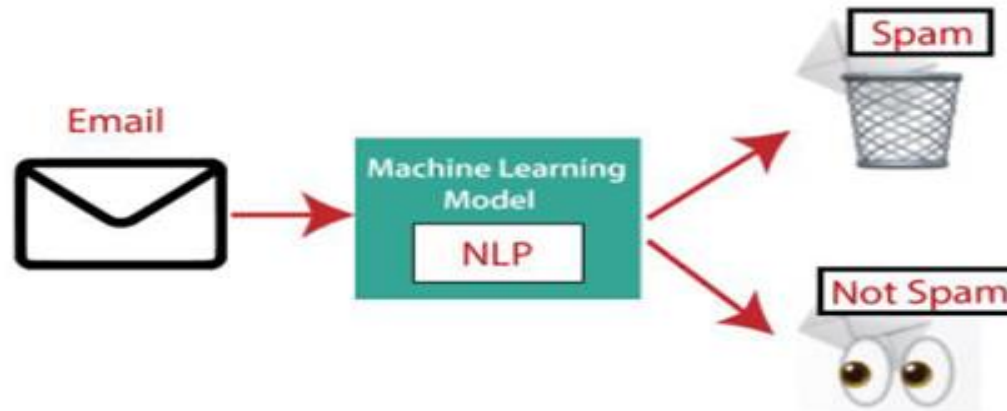
1. Question Answering

- Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.



Applications of NLP

2. Spam Detection Spam detection is used to detect unwanted e-mails getting to a user's inbox.



Applications of NLP

3. Sentiment Analysis

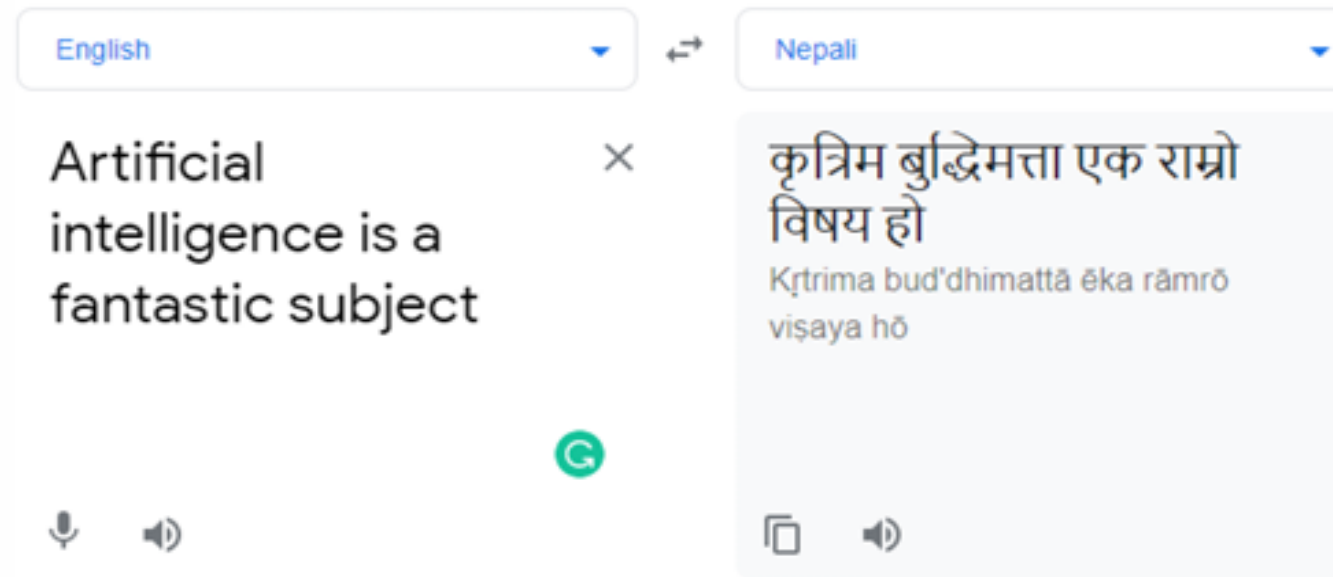
- Sentiment Analysis is also known as opinion mining. It is used on the web to analyze the attitude, behavior, and emotional state of the sender.
- This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or neutral), identify the mood of the context (happy, sad, angry, etc.)



Applications of NLP

4. Machine Translation

- Machine translation is used to translate text or speech from one natural language to another natural language.



Applications of NLP

5. Spelling Correction

- Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction(example).

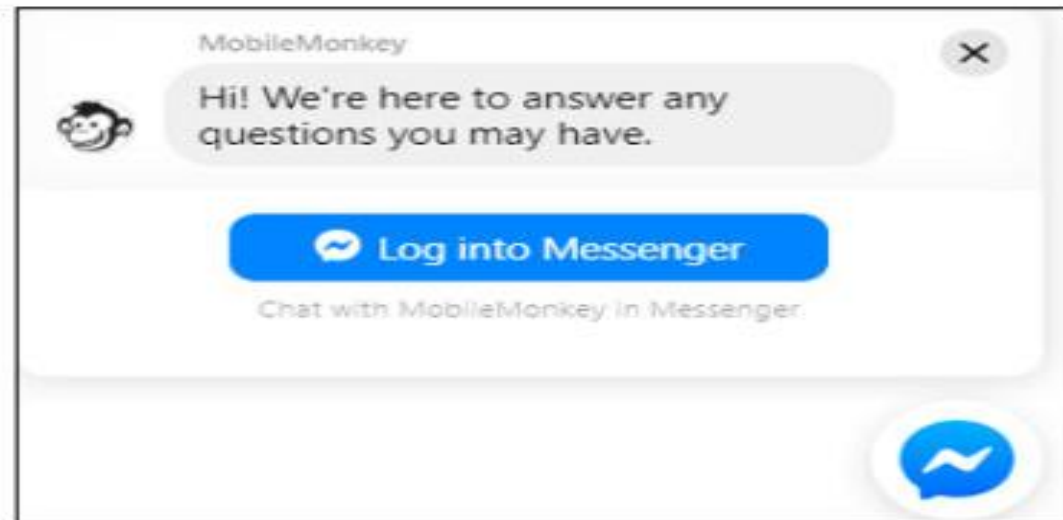
6. Speech Recognition

- Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on

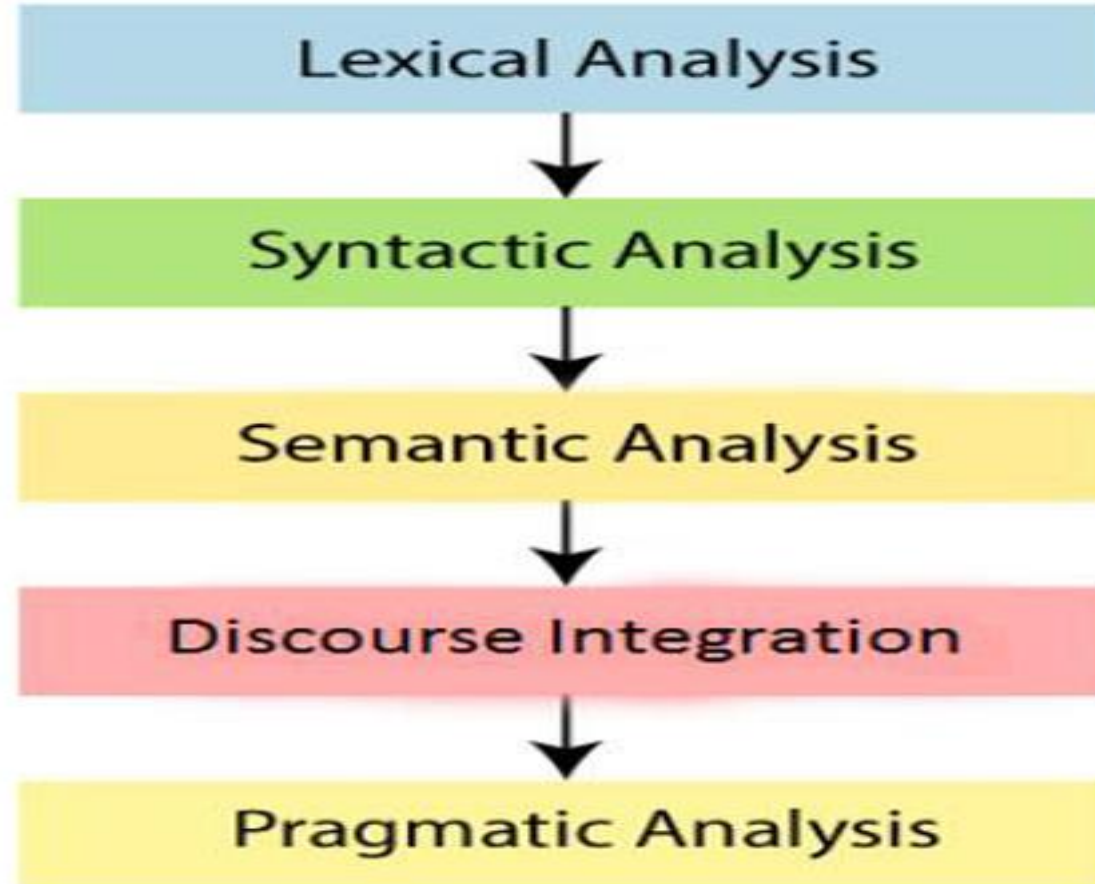
Applications of NLP

7. Chatbot

- Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.



Phases of NLP



Phases of NLP

- Lexical Analysis
 - It involves identifying and analyzing the structure of words
 - Lexicon of a language means the collection of words and phrases in a language.
 - Lexical analysis is dividing the whole chunk of txt into paragraphs, sentences, and words.
- Syntactic Analysis (Parsing)
 - It involves analysis of words in the sentence for grammar and arranging words in a manner that shows the relationship among the words.
 - The sentence such as “The school goes to boy” is rejected by English syntactic analyzer.

Phases of NLP

- Semantic Analysis

- Semantic analysis is concerned with the meaning representation.
- It draws the exact meaning or the dictionary meaning from the text.
- The semantic analyzer disregards sentence such as “hot ice-cream”.

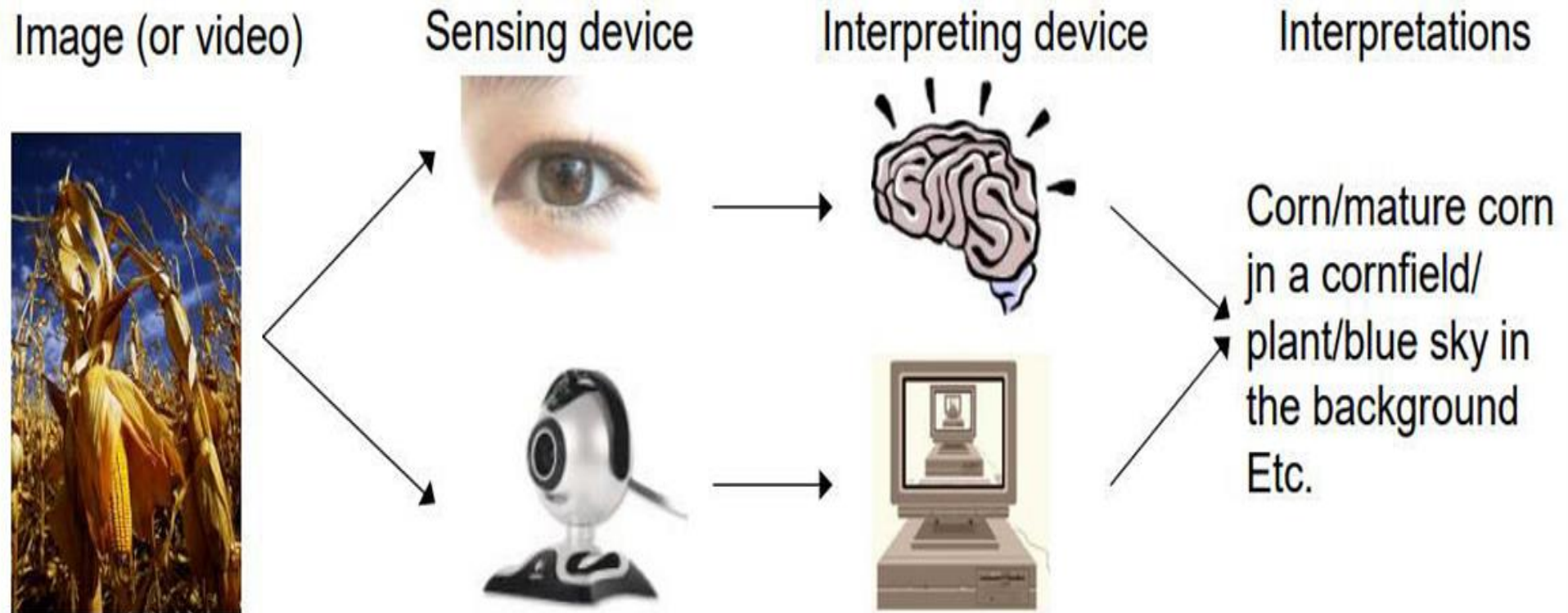
- Discourse Integration

- The meaning of any sentence depends upon the meaning of the sentence just before it.
- In addition, it also brings about the meaning of immediately succeeding sentence

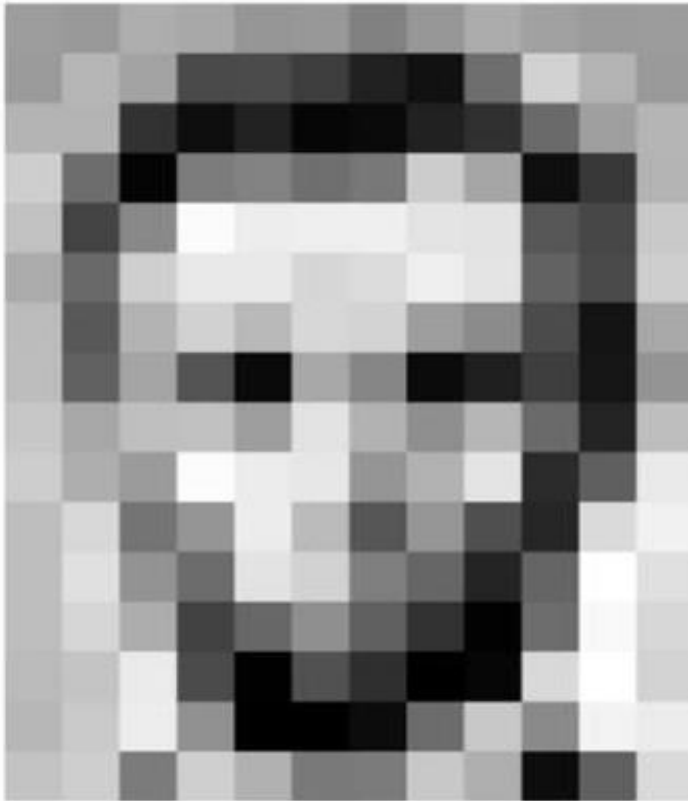
Phases of NLP

- Pragmatic Analysis
 - During this, what was said is re-interpreted on what it actually meant.
 - It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.
 - For Example: "Open the door" is interpreted as a request instead of an order.

What is (Computer) vision?

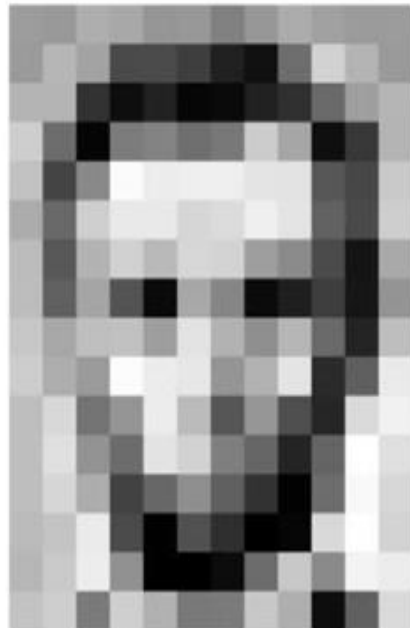


Images are numbers



157	153	174	168	150	152	129	153	172	161	165	166
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	169	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	157	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	95	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	265	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

Tasks in Computer Vision



Input Image



167	163	174	168	160	162	129	151	172	161	155	156
165	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	6	124	131	111	120	204	166	16	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	195	215	211	168	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	198	227	178	143	182	106	36	190
205	174	195	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	3	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

Pixel Representation



classification

Lincoln

Washington

Jefferson

Obama

$$\begin{bmatrix} 0.8 \\ 0.1 \\ 0.05 \\ 0.05 \end{bmatrix}$$

High level feature detection

Let's identify key features in each image category



Nose,
Eyes,
Mouth

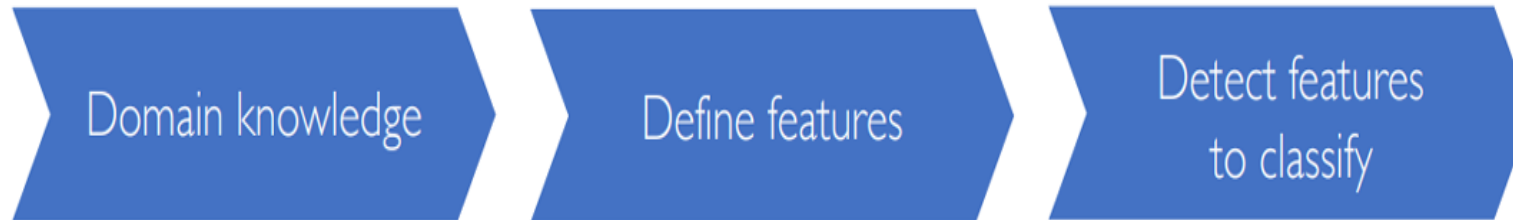


Wheels,
License Plate,
Headlights



Door,
Windows,
Steps

Manual feature extraction



Problems?

Some problems in feature extraction

Viewpoint variation



Scale variation



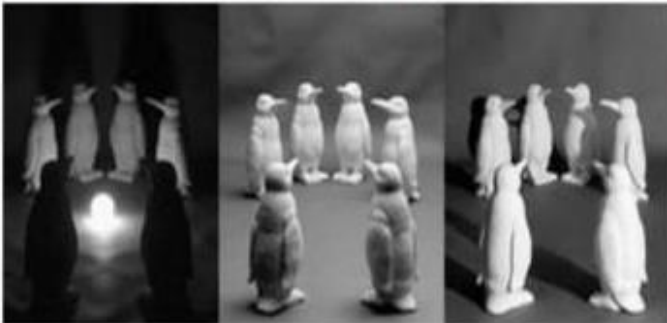
Deformation



Occlusion



Illumination conditions



Background clutter

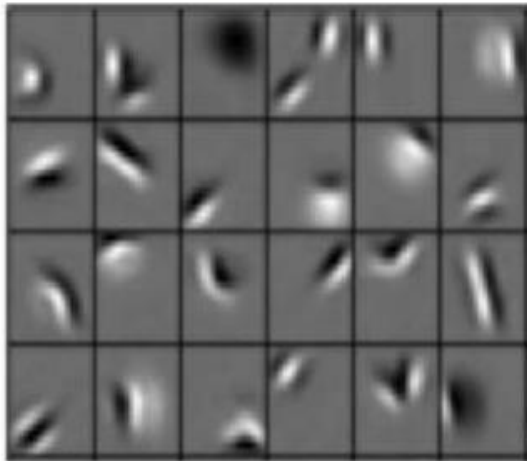


Intra-class variation



Learning Feature Representations

Low level features



Edges, dark spots

Mid level features



Eyes, ears, nose

High level features



Facial structure

Visual Recognition Techniques

Classification



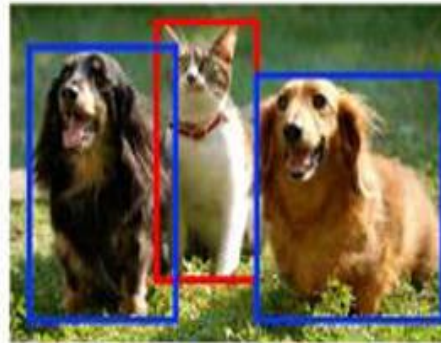
CAT

Classification
+ Localization



CAT

Object Detection



CAT, DOG

Instance Segmentation



CAT, DOG

Terminologies

- Classification: Labelling the image based on what it consists of is classification. Here the class is 'Cat'.
- Localization: The next question comes where the object is situated in the image. When we identify the location of the object in the frame and create a bounding box around it, It is known as localization.
- Object Detection: In object detection, we use a bounding box that is either square or rectangular in shape, but it does not tell anything about the shape of the objects.
- Instance segmentation creates a pixel-wise mask around each object. Hence instance segmentation gives a deeper understanding of the image.

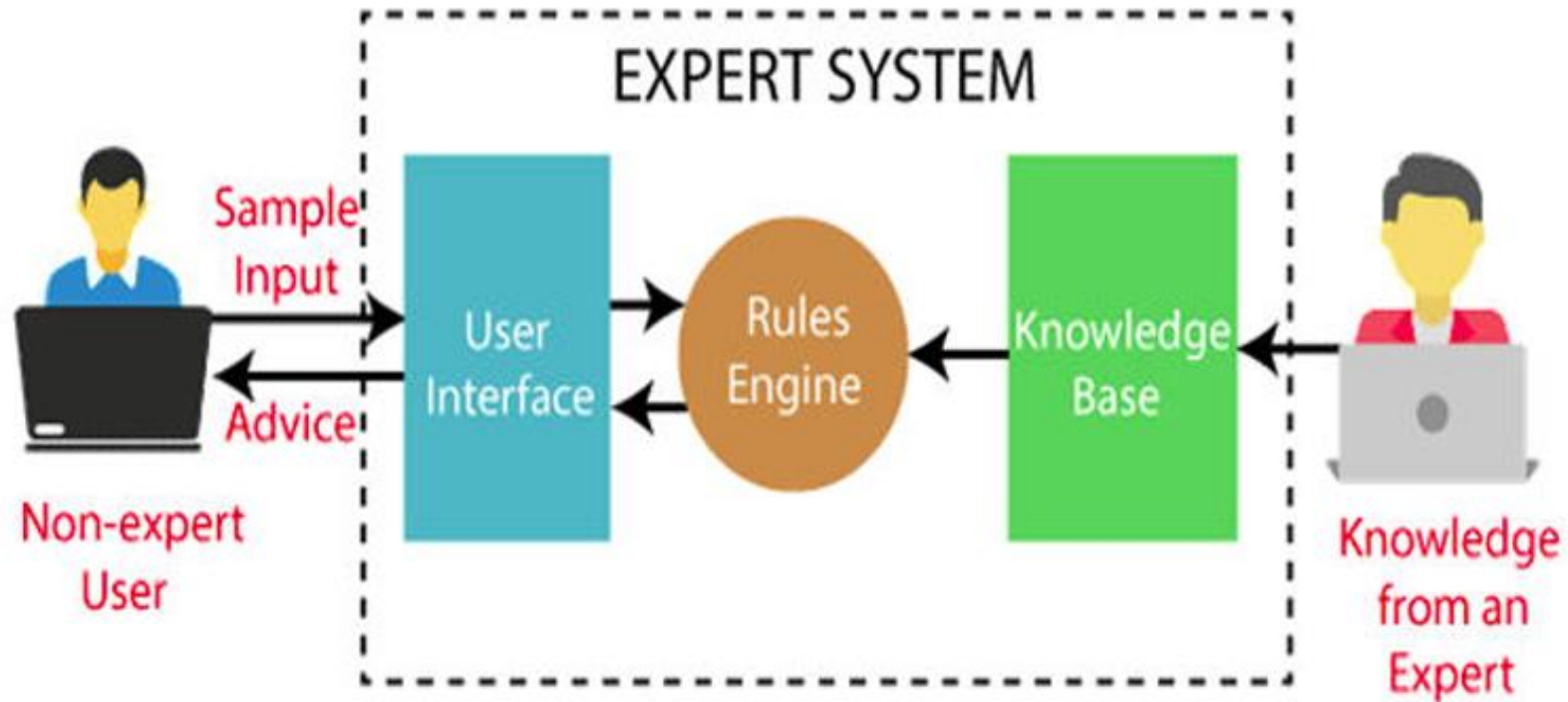
Applications of Computer Vision

- Face Recognition
- Video Surveillance
- Object Detection
- Object Recognition
- Medical Imaging Analysis
- Localization and Mapping
- Augmented Reality/Virtual Reality
- Human Expressions & Emotional Analysis
- Transforming the paperwork into digital data

Expert System

- An expert system is a computer program that is designed to solve complex problems and to provide decision-making ability like a human expert.
- It performs this by extracting knowledge from its knowledge base using the reasoning and inference rules according to the user queries.
- The expert system is a part of AI, and the first ES was developed in the year 1970, which was the first successful approach of artificial intelligence.
- It solves the most complex issue as an expert by extracting the knowledge stored in its knowledge base.

Expert System



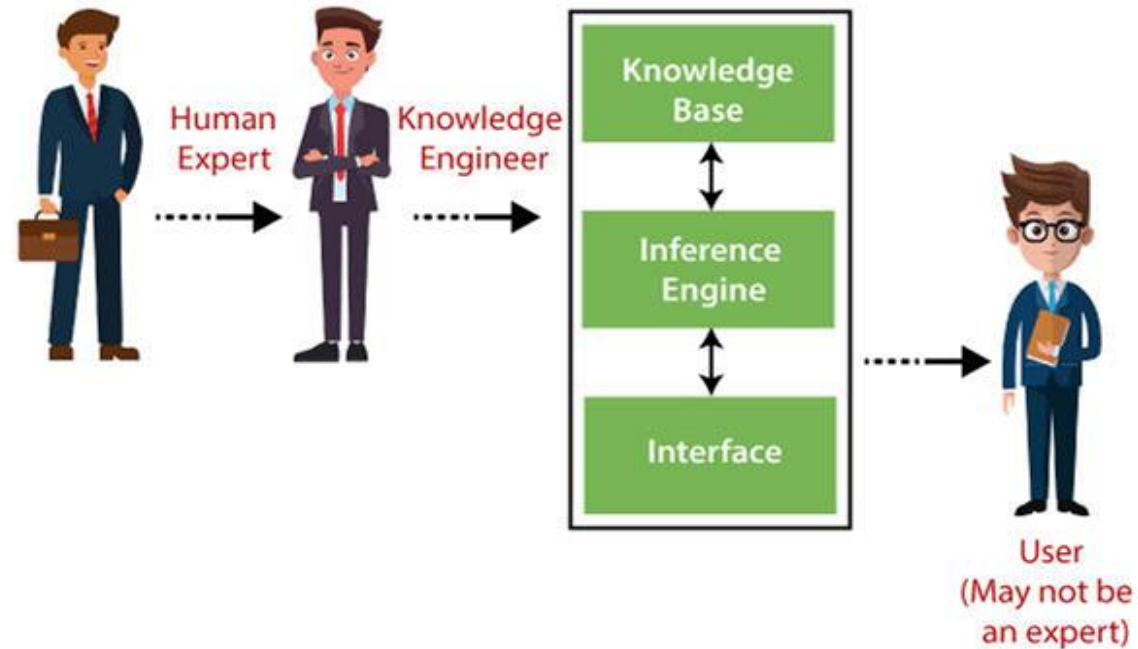
Characteristics of Expert System

- **High Performance:** The expert system provides high performance for solving any type of complex problem of a specific domain with high efficiency and accuracy.
- **Understandable:** It responds in a way that can be easily understandable by the user. It can take input in human language and provides the output in the same way.
- **Reliable:** It is much reliable for generating an efficient and accurate output.
- **Highly responsive:** ES provides the result for any complex query within a very short period of time.

Components of Expert System

- An expert system mainly consists of three components:

- User Interface
- Inference Engine
- Knowledge Base



References

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- D. W. Patterson, *Artificial Intelligence and Expert Systems*, Prentice Hall, 2001.

Chapter 8

Philosophy, Ethics and Safety of AI

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Overview

- The Limits of AI
- Can machine really think?
- The Ethics of AI

The Limits of AI

1. Technological Limits

a. Lack of General Intelligence

- **Narrow AI:** Current AI systems are specialized, excelling in specific tasks like image recognition or language translation, but they lack general intelligence comparable to human reasoning.
- **Context Limitation:** AI struggles to understand or adapt to unfamiliar scenarios beyond its training data.

b. Dependence on Data

- **Data Quality:** AI systems require vast amounts of high-quality, labeled data to function effectively. Poor or biased data leads to inaccurate outcomes.
- **Data Availability:** Some fields lack the data necessary to train robust AI models, especially in niche or emerging areas.

The Limits of AI

c. Computational Power

- **High Costs:** Training advanced models like GPT or AlphaFold requires immense computational resources, making it financially prohibitive for smaller entities.
- **Energy Consumption:** Large-scale AI models consume significant energy, raising concerns about scalability and environmental impact.

d. Limited Creativity

- AI can generate content or ideas based on patterns in data, but it cannot genuinely innovate or exhibit original thought like humans.

e. Inability to Understand Emotions

- While AI can simulate emotional responses (e.g., chatbots mimicking empathy), it doesn't genuinely feel or understand emotions, which limits its effectiveness in deeply human contexts like counseling or caregiving.

The Limits of AI

2. Ethical and Social Limits

a. Bias and Discrimination

- AI inherits biases present in its training data, leading to potentially harmful or discriminatory outcomes (e.g., biased hiring algorithms or racial profiling in facial recognition).

b. Privacy Concerns

- AI systems often require extensive personal data, raising concerns about misuse, surveillance, and lack of consent.

The Limits of AI

c. Lack of Transparency

- Many AI models, especially deep learning systems, are black boxes—their decision-making processes are difficult to interpret, reducing trust and accountability.

d. Ethical Decision-Making

- AI lacks a moral compass and struggles with nuanced ethical dilemmas (e.g., self-driving cars deciding whom to prioritize in an unavoidable accident).

The Limits of AI

3. Conceptual and Philosophical Limits

a. Lack of Common Sense

- AI doesn't possess common sense reasoning. It can fail in simple situations that require basic human intuition (e.g., understanding a joke or ambiguous context).

b. No True Consciousness

- AI, no matter how advanced, lacks awareness, self-reflection, or consciousness. It operates based on algorithms, not understanding.

The Limits of AI

c. Over-Reliance on Training

- AI models are as good as their training. They can't infer knowledge outside their trained domain or self-learn independently beyond predefined parameters.

d. Misinterpretation of Human Nuance

- AI can misinterpret language, cultural context, or sarcasm, leading to errors in communication or decision-making.

The Limits of AI

4. Practical and Deployment Challenges

a. Maintenance and Updates

- AI systems require ongoing updates to remain relevant and effective, especially as new data or situations emerge.

b. Integration Challenges

- Incorporating AI into existing workflows can be complex, requiring significant changes in infrastructure and employee training.

The Limits of AI

c. Vulnerability to Manipulation

- AI systems are susceptible to adversarial attacks, where manipulated inputs deceive the AI into making incorrect decisions (e.g., altered images fooling facial recognition).

d. Dependency Risk

- Over-reliance on AI can lead to critical failures if the AI malfunctions, such as in healthcare or aviation.

The Limits of AI

5. Societal and Policy Challenges

a. Unequal Access

- Advanced AI technologies are often expensive and controlled by a few organizations, leading to technological inequality.

b. Regulation and Governance

- Developing globally consistent regulations to govern AI use is challenging, leading to potential misuse or harmful consequences.

The Limits of AI

c. Displacement of Jobs

- Automation driven by AI could displace human workers in various industries, leading to socioeconomic challenges and inequality.

d. Ethical Concerns in Warfare

- AI weaponization, like autonomous drones or surveillance systems, raises ethical concerns about accountability and misuse

The Limits of AI

6. Long-Term and Theoretical Limits

a. Superintelligence Risks

- While speculative, concerns exist about AI surpassing human intelligence (superintelligence) and potentially acting contrary to human interests.

b. Alignment Problem

- Ensuring AI systems align with human values and goals is a complex and ongoing challenge.

c. Unpredictable Emergence

- As AI systems grow in complexity, unintended behaviors or emergent properties could arise, creating risks.

Can Machine Really Think?

1. What Does It Mean to "Think"?

- Thinking can involve various processes, such as:
 - Logical reasoning.
 - Problem-solving.
 - Decision-making.
 - Creativity.
 - Awareness and understanding.
- AI systems excel at some of these tasks but struggle with others. The distinction between human thinking and what machines do lies in consciousness, intention, and understanding.

Can Machine Really Think?

2. How Do Machines "Think"?

- Machines don't think in the way humans do. Instead, they process information using algorithms and data:
- **Pattern Recognition:** AI systems analyze patterns in data to make predictions or decisions.
- **Rules-Based Processing:** Machines follow predefined rules to solve problems (e.g., chess algorithms).
- **Learning from Data:** Through machine learning, AI systems improve their performance over time, appearing to "learn."
- For example: A machine learning model like ChatGPT generates responses based on patterns in text data it has been trained on. It doesn't "understand" the text or have an internal thought process—it simply calculates probabilities of word sequences.

Can Machine Really Think?

- **Strong AI vs. Weak AI**
- **Weak AI (Narrow AI):**
 - Machines perform specific tasks (e.g., voice recognition, medical diagnosis).
 - They simulate thinking but don't possess awareness or consciousness.
 - Example: Siri, Google Translate.
- **Strong AI (General AI):**
 - Hypothetical systems with human-like general intelligence.
 - Can understand, reason, and apply knowledge across a broad range of tasks.
 - As of now, strong AI does not exist, and machines cannot "truly think."

Can Machine Really Think?

4. Differences Between Machine and Human Thinking

a. Absence of Consciousness

- Machines lack self-awareness or subjective experiences.
- Humans think with emotions, creativity, and intuition, which are deeply tied to consciousness.

b. Lack of Intentionality

- Human thinking is often goal-driven and intentional.
- Machines only perform tasks they are programmed or trained for.

c. Understanding vs. Simulation

- Humans understand concepts and can generalize knowledge.
- Machines simulate understanding by correlating data but don't grasp meaning.

Can Machine Really Think?

5. Philosophical Perspectives

- **Alan Turing's Test (Turing Test)**

- Proposed by Alan Turing in 1950, the test suggests that if a machine's behavior is indistinguishable from a human's, it can be said to "think."
- Many argue this is a functional perspective rather than proving true thinking.

- **John Searle's Chinese Room Argument**

- Searle argued that machines follow rules to manipulate symbols but don't truly understand them. Like a person following a guide to translate Chinese, AI lacks comprehension.

- **Functionalism**

- Suggests that if machines replicate the functional processes of thinking, they may be said to "think" in a meaningful way.

Can Machine Really Think?

6. Can Machines Ever Truly Think?

- While current AI does not "think" in a human sense, there are debates about the future:
- **Yes, Machines Might Think:**
 - If consciousness can emerge from complex computations, machines may someday think.
 - Advances in neuromorphic computing (mimicking the human brain) could lead to more human-like AI.
- **No, Machines Will Never Think:**
 - Critics argue that thinking requires consciousness, emotion, and understanding, which may not be programmable.

Can Machine Really Think?

7. Practical Implications

- Whether or not machines can think, their abilities are transformative:
- Machines can outperform humans in specific cognitive tasks (e.g., analyzing vast datasets).
- Understanding the limits of machine "thinking" helps us build ethical, effective systems.

The Ethics of AI

1. Key Ethical Principles in AI

a. Fairness and Equity

- Challenge: AI systems can reflect and amplify biases present in their training data, leading to discrimination in hiring, lending, law enforcement, and more.
- Ethical Goal: Ensure AI treats all individuals and groups fairly, regardless of race, gender, socioeconomic status, or other attributes.

b. Transparency and Explainability

- Challenge: Many AI systems, especially deep learning models, operate as "black boxes," making their decision-making processes difficult to understand.
- Ethical Goal: Develop AI that is interpretable and transparent, enabling users to understand how and why decisions are made.

The Ethics of AI

c. Accountability

- Challenge: When AI causes harm (e.g., autonomous vehicles in accidents), determining responsibility can be unclear.
- Ethical Goal: Establish accountability frameworks where developers, organizations, or regulators are responsible for AI outcomes.

d. Privacy and Data Protection

- Challenge: AI often relies on large datasets, raising concerns about the misuse of personal information.
- Ethical Goal: Protect individual privacy by adhering to robust data protection standards and obtaining informed consent for data usage.

The Ethics of AI

e. Safety and Security

- Challenge: Malfunctioning or hacked AI systems can lead to catastrophic consequences in critical domains like healthcare, aviation, and defense.
- Ethical Goal: Prioritize safety in AI design and ensure systems are secure against misuse or cyberattacks.

f. Beneficence

- Challenge: AI has the potential to cause harm if misused, such as in surveillance or autonomous weaponry.
- Ethical Goal: Ensure AI is used to benefit society and improve quality of life while minimizing harm.

The Ethics of AI

2. Ethical Challenges in AI

a. Bias and Discrimination

- Example: Facial recognition systems have been found to misidentify people of certain racial groups more often than others.
- Solution: Use diverse and representative datasets, and audit AI systems for bias.

b. Job Displacement

- Example: Automation driven by AI threatens jobs in manufacturing, transportation, and service industries.
- Solution: Invest in workforce reskilling and create policies to support displaced workers.

The Ethics of AI

c. Deepfakes and Misinformation

- Example: AI-generated fake videos or news can deceive people and spread misinformation.
- Solution: Develop tools to detect and counter deepfakes, and promote digital literacy.

d. Autonomous Weaponry

- Example: AI-powered weapons raise concerns about accountability and potential misuse in warfare.
- Solution: Advocate for international agreements to regulate or ban autonomous weapons.

The Ethics of AI

e. Lack of Regulation

- Example: Rapid AI advancements outpace the creation of legal and ethical guidelines.
- Solution: Governments and organizations must collaborate to establish and enforce AI regulations.

The Ethics of AI

3. Ethical Theories and AI

- Ethical considerations in AI often draw from traditional ethical theories:
 - Utilitarianism: AI should maximize overall happiness or benefit while minimizing harm.
 - Deontology: AI must follow rules and principles (e.g., respecting privacy or avoiding harm), regardless of outcomes.
 - Virtue Ethics: AI should align with human virtues like honesty, fairness, and compassion.

The Ethics of AI

4. Ethical Frameworks and Initiatives

- Several organizations and frameworks address AI ethics:
 - Asilomar AI Principles: Focus on safety, transparency, and shared benefit.
 - EU Guidelines on Trustworthy AI: Emphasize human-centric AI, accountability, and ethical design.
 - UNESCO AI Ethics Recommendation: Advocates for inclusivity, sustainability, and diversity in AI.

The Ethics of AI

5. Ethical Dilemmas in AI

a. Surveillance vs. Privacy

- Governments and companies use AI for surveillance, which can enhance security but infringe on individual privacy.
- Question: How do we balance safety with civil liberties?

b. Profit vs. Public Good

- Companies may prioritize profit over ethical considerations, such as creating addictive algorithms for social media.
- Question: How can businesses align profit motives with ethical practices?

The Ethics of AI

c. Autonomy vs. Control

- Autonomous systems (e.g., self-driving cars) make decisions independently, raising concerns about trust and oversight.
- Question: Should humans always have the final say in critical decisions?

d. Access vs. Inequality

- Advanced AI technologies are often expensive and inaccessible to underprivileged communities, exacerbating inequality.
- Question: How can we ensure equitable access to AI benefits?

The Ethics of AI

6. The Role of Stakeholders in Ethical AI

a. Developers and Engineers

- Responsible for building unbiased, safe, and transparent AI systems.

b. Businesses

- Must adopt ethical practices, prioritize long-term benefits over short-term profits, and ensure AI aligns with societal goals.

The Ethics of AI

c. Governments and Regulators

- Should create policies and regulations that address AI's ethical and societal impacts.

d. Civil Society

- Advocacy groups, researchers, and the public play a crucial role in holding organizations accountable and shaping ethical AI use.

The Ethics of AI

7. Path Forward: Ensuring Ethical AI

- Incorporate Ethics in Design: Embed ethical principles into AI development from the start.
- Education and Awareness: Train developers, policymakers, and users about AI ethics.
- Independent Audits: Regularly review AI systems to ensure compliance with ethical standards.
- Global Collaboration: Foster international dialogue and cooperation to create unified ethical guidelines.